Order Carnivora

Perhaps artificially, carnivores are split into terrestrial forms (fissipeds: dogs, bears, raccoons, genets, mongooses, aardwolves, hyaenas, cats) and marine forms (pinnipeds: fur seals, walruses, seals). The aquatic pinnipeds generally give birth to a single, highly precocial offspring, while the terrestrial carnivores more often give birth to several somewhat altricial young. Gestational delays occur in both aquatic (all pinnipeds) and terrestrial (Ursidae, Mustelidae) forms.

Details of reproductive cyclicity (9350) and pinniped lactation (8037) are available. One journal, Carnivore, focuses on this order.

Order Carnivora, Family Canidae

Foxes, wolves, coyotes, jackels, and dogs (*Alopex* 2-9 kg; *Canis* 5-80 kg; *Chrysocyon* 20-32 kg; *Cuon* 10-20 kg; *Dusicyon* 4-13 kg; *Lycaon* 17-60 kg; *Nyctereutes* 4-11 kg; *Otocyon* 3-5 kg; *Speothos* 5-8 kg; *Urocyon* 2-7 kg; *Vulpes* 1-14 kg) are nocturnal carnivores with a worldwide distribution. Male canids often are larger than females (*Canis* 3571, 3649, 3768, 4645, 5210, 5632, 5761, 7979, 8495, 8501, 9175, 9327, 10499; cf 9945; *Cuon* 5330; *Nyctereutes* 11866; cf 4643; *Urocyon* 5210; *Vulpes* 502, 4253, 5210, 6420, 6520, 7979, 10462, 10499), although the reverse may be true for *Otocyon* (5761, 6743, 9945).

A copulatory tie is common (39, 209, 744, 2210, 2302, 3452, 3453, 3637, 4643, 5131, 5844, 6520, 7147, 10734, 11069, 11144, 11445, 11929, 11984; cf 2171, 5837, 9609). The endotheliochorial, zonary placenta implants into a bicornuate uterus (2479, 6520, 7563) and is usually eaten after birth (7147, 7158, 7710, 7712, 10462, 10499, 10734).

Neonates are furred, but blind (39, 333, 744, 1474, 2440, 3155, 3768, 4643, 5131, 5761, 6305, 9175, 10154, 10229, 10254, 10499, 11163, 11866). Their eyes open at 8-14 d (*Canis* 746, 1474, 3571, 3769, 4643, 4645, 5761, 7147, 7979, 9175, 9769, 10308, 10499, 11144, 11984; *Chrysocyon* 1206, 3005, 4297; *Nyctereutes* 562, 4643, 5130, 7979, 9611, 10499; *Vulpes* 2702, 3571, 3598, 4643, 5725, 6520, 7979, 9175, 10306, 10499; cf 3155, 3815, 5908), 8-17 d (*Speothos* 745, 909, 2734), 9 days (*Urocyon* 5210), 13-15 d (*Cuon* 745, 8798; *Dusicyon* 909, 1205; *Lycaon* 745, 2109, 11163), or 14-16 d (*Alopex* 745, 7979; cf 4643).

Females have 4 mammae (*Otocyon* 10158), 6 mammae (*Urocyon* 1474, 6305, 9885), 6-8 mammae (*Vulpes* 502, 1474, 3134, 3571, 4253, 6305, 9175, 9885, 10158), 8-10 mammae (*Canis* 744, 1474, 3571, 8613, 11144; cf 10158), or 12-16 mammae (*Cuon* 2303, 4643; *Lycaon* 331, 333, 918, 11163). Males have a baculum (1473, 9347).

Details of anatomy (261, 2123, 4295, 5317, 7449), endocrinology (*Alopex* 7447, 7450, 7452, 7453, 7462, 7463, 7981, 10083, 10084, 10085, 10086, 10087; *Canis* 395, 396, 3769, 8186, 9705a; *Lycaon* 11163; *Nyctereutes* 11068; *Urocyon* 6306; *Vulpes* 1100, 1104a, 1105, 1106, 1107, 5318, 5319, 5320, 5323, 6306, 6982, 6983, 6985, 6987, 7458, 7459, 7461, 8131, 8132), spermatogenesis (6982, 10082), conception (8311, 8313), birth (2017a), milk composition (4253, 5190, 6247), and light (3419, 3420, 3421) are available. Information on reproduction of domestic dogs is summarized in supplement 39 of the Journal of Reproduction and Fertility published in 1989.

Note that dingos, wolves, and domestic dogs are all considered *Canis lupus*.

NEONATAL MASS

Alopex lagopus	57 g	10254
	60-85 g	4643
	60-90 g	7979
Canis aureus	201-214 g; 2 d	4643
Canis latrans	154.9 g; sd10.81; n7 near term, 1 lt	5668
	177.8 g; sd6.78; n5 near term, 1 lt	5668
	200-250 g	3769
	275 g; r255-284	746
Canis lupus	300-500 g	4643
	mode 300-400 g; r300-675	3571
	350-400 g	3649
	f: 535 g	10308
	m: 547 g	10308
	m: ~1 kg	6137
	f: ~1 kg	6137
Canis mesomelas	159 g	745
Chrysocyon brachyurus	340-430 g	2578

	350 g; n1 lt	1206
	~500 g	8798
Cuon alpinus	200-350 g	10229
Dusicyon thous	120-160 g	1205
Lycaon pictus	f: 296.6 g; sd40.4; n3 dead, 1 d	11163
	m: 308 g; sd71.9; n5 dead, 1 d	11163
	350-380 g; n8 yg, 1 lt	2440
Nyctereutes procyonoides	60-80 g	10499
	60-90 g	7979
	60-110 g	4643
	60-110 g	562
	f: 90.4 g; r63-125; n11	9626
	m: 103.5 g; r50-150; n16	9626
	109.1 g; r105-115	5130
	122 g	582
Otocyon megalotis	r99.4-142.0; n5 yg, 1 lt	10158
	106.5-140.5 g; term emb	10158
Speothos venaticus	130-140 g; n1 lt, 6 yg	5256
	137.5 g; r126-154; n4	5837
	182.5 g; r175-190; n2, 1 d	5256
	~0.5 kg	2734
Urocyon cinereoargenteus	85 g; n5	11893
	~100 g	9885
	115 g	745
Vulpes bengalensis	52-65 g	39
Vulpes corsac	60-65 g	4643
Vulpes vulpes	50-150 g	4253
	70-100 g	10152
	71-120 g	10461
	80-98 g	10734
	86.9 g; r56-110	4643
	r95-106 g; n6 near term	7071
	99.91 g; sd13.28; n6 term lt, 21 emb	6305
	~100 g	9885
	100-130 g	6520
	f: 105 g; r100-110; n2	7710
	105.13 g; r101.5-110.6; n3 lt, 18 yg	10460
	114 g; n6	2702
	m: 140 g; r90-165; n5	7710
Vulpes zerda	23.75 g; r23.5-24; n2	11325
	m: 26.9 g; r23.1-29; n4	5908
	f: 28.2 g; r26.9-30.7; n3	5908
	r74-104 g; n4, 8 d	7183
NEONATAL SIZE		
Canis latrans	CR: 125 mm	522
	CR: 152.4-165.0 mm	746
Vulpes corsac	135-140 mm	4643
Vulpes velox	TL: 155 mm; r149-165; n8 term emb	2913
Vulpes vulpes	CR: 131 mm	6520
	145 mm	4643
	145 mm	7979
	est 150 mm	5941
Vulpes zerda	HB: 108.5 mm; r108-109; n2	11325
	HB: 115 mm; n1	8426
WEANING MASS		
Nyctereutes procyonoides	400-500 g	5130
	1100-1300 g; 2 mo	562
Vulpes vulpes	400-800 g	5941
	1057 g; r795-1107; n10, 28 d	2702
	~1600 g	9885
	f: 1731 g	6520
	m: 1997 g	6520
WEANING SIZE		
Vulpes vulpes	HB: 250-350 mm	5941

LITTER SIZE

Alopex lagopus	1-13	6546
	2.8 yg; r1-5; n16 dens	836
	3-12; max 18	10499
	mode 4-6; 7-8 occasional	3647
	4-12	1780
	4 yg; n1 lt	10217a
	5 yg; n1	5844
	5-6; r1-14	1474
	5.5; max 10; n4 lt	11041
	6.0; r3-12; n9 lt	5492
	6.4; r4-10; n17 lt	566
	7-10; max 21; 4-6 poor yr; max 26 emb	7979
	7.7 wean; n2030 lt	10052
	8-13 yg; max 22	4643
	8.7; r1-17; n1331 lt	5324
	8.75 yg; n2984 lt, 1401 f	10052
	9.8 yg; n2030 lt	10052
	10.0; n272 lt	5324
	11; max 22	10254
	14.0; n2 lt	10203
	20 emb; r17-23; n2 f	3647
	20; max 25	11042
Canis spp "gray jackals"	5 yg; n1 lt	5905
Canis adustus	2.0; r1-3; n2 lt	12128
	3-6	5761
	4; mode 5; r2-6	12128
	4-6	11144
	5; r3-7; n2 lt, 1 f	4180
	5.4 emb; r4-6; n5 f	10154
	7 emb	10158
	12 emb	9945
Canis aureus	1-5	5761
	2; r1-3	11965
	2.7; r2-4; n3 lt	12128
	3 emb; n1 f	8613
	3.5; r3-4; n2 lt	8495, 8501
	4-5	4645
	4-5	9175
	4-9	7979
	4.5 yg; r3-7; n6 lt	9066
	5; r3-8	4643
	6 emb ea; n2	10616
Canis latrans	1 yg; n1 lt	11645a
	2-15	3769
	3.3; r1-7; n19 lt	2109
	3.8 ps; 95%ci 1.7; n9 yearling f	10857
	3.8 emb; n26 yearling f	10856
	4.45; sd0.33	1560
	4.7 emb; n52 adult f	10856
	4.8 ps; 95%ci 0.9; n17 adult f	10857
	4.9 cl; 95%ci 1.5; n14 yearling f	10857
	mode 5-7 yg; usually 4-8; 9-12 not uncommon; 17, 19 recorded	5210
	5-10	1474
	5 emb; n1 f	522
	5.0; r3-7; n2 lt	5844
	5.0 yg; r3-6; n3 lt	3793
	5.0 yg; r2-8; n6 lt	9066
	5.2 cl; n41 yearling f	10856
	5.4 emb, ps, or yg; sd1.8; n20 lt	436a
	5.5; sd1.3; per pack; 5.3; sd1.5; per pair	747
	5.6 cl; n17, 2 yr f	5667
	5.70 yg; n1582 dens	4295

	6-7 yg; max 11	5146
	6.0; r5-7; n2 lt	5668
	6.13 emb, yg; r4-7; n13 lt, 73 emb, 7 yg	10954
	6.2 cl; n9, 3 yr f	5667
	6.23 emb; n1330 f	4295
	6.3 yg; n20 lt	2136
	6.6-8.0 cl; 7.1; r6.1-7.9	5276
	6.9 cl; 95%ci 1.3; n26 adult f	10857
	6.9 cl; n63 adult f	10856
	7 emb; n1 f	5405b,
	7.1 cl; n6, 4 yr f	5667
	8.4 cl; 6.15 ps; n23 f	3768
	9 yg; n1 den	495
Canis lupus	2.8-6.5; r1-11	7147
	max 3; n3 lt	7236
	3-5; young f; r6-8; adult f; occasionally 10-13	7979
	3-8	4643
	3-12	4645
	3.0; r3-5; n3 lt	11984
	3.4; r1-5; n5 lt	3714
	3.4 yg; r1-6; 7 lt	6239
	3.7; r1-9; 6 lt	2109
	3.8; r1-7; n15 lt	2109
	mode 4-6; r1-8	3571
	4.0 yg; se0.46; n15 lt	1663a
	4.1; mode 4; r1-8	12128
	4.3 ps; 95%ci 0.9; n7 f	3620
	4.33 yg; n316 lt	7149
	4.5; r2-10	6382
	4.5 yg; r2-7; n20 lt	3526
	4.6 emb; 95%ci 0.7; n5 f	3620
	4.7 yg; r1-8; n7 dens	502
	4.7 yg; se0.25; n48 lt	1663a
	4.75; r4-6; n4 lt	12001
	4.8; 3.1 weaned	3526
	4.8; r4-5; n6 lt	3649
	4.8; r1-8; n26 lt	1230
	4.9 yg; r3-7	8547a
	mode 5; r1-8	12128
	mode 5-8; r3-12	10499
	5 yg; n1 lt	2496
	5 ps; r3-7; n5 lt	8415
	5.0 yg; se0.41; n14 lt	1663a
	5.2 emb; r1-9; n30 f	6239
	5.4 cl; 95%ci 0.8; n9 f	3620
	5.41 emb; 6.08 cl; 2 yr old f	8964
	5.5 emb, ps; sd1.4; r2-9; n48 f	5402
	5.6; se0.67; r2-8	8695,
	5.7 emb; se0.22; n44 f	1663a
	mode 6-7; r3-14	1474
	6.4 ca; se0.6; 5.5 yg; r2-9; per pack	543
	6.4 yg; r4-9; n8 lt	10371a
	6.5 emb; 7.3 cl; adult f	8964
	6.6 ps, yg; r4-8; n9 lt	3556a
	6.6 emb; 6.8 cl	3620
	6.6 emb; n18 f	8964
	6.8 cl; n56 f	8964
	~7 yg	5146
	7 emb; n1 f	3526
	7.1 ps; n45 f	8964
Canis mesomelas	1-2	8514
	2-8	9327
	mode 3; r1-6	10158
	mode 3-4; r2-6	9945

	3-4; r2-7	11965
	3; r2-4; n4 lt	12128
	4-10	11144
	4; max 9	5761
	4.3; r1-8; n16	1230
Canis rufus	2-7	10809
	2.6; r1-4; n5 lt	12128
	4-7	1474
	6.2 emb; r2-10; n29 f	8227
Chrysocyon brachyurus	1.6; r1-2; n5 lt	1206
	1.67; r1-3; n3 lt	2210
	2; n1 lt	4297
	2.47; r1-5; n17 lt	2577, 2579
	3.0; r2-4; n3 lt	3180
	4 ea; n2 lt	6499
	4 ea; n2 lt	3003
	max 7	1648
Cuon alpinus	2 or 4; r2-6	6654
	2-6 yg	1478
	3 yg; n1 lt	4466
	3.5; r2-5	12128
	4-5	10229
	~8	5330
	8 emb; n1 f	1960, 1961
	8 emb; r7-9; n2 f	1478
	9 emb; n1 f	61
Dusicyon culpaeus	5.16 emb; n6 f	2129
Dusicyon gymnocercus	2.5; r1-4; n2 lt	12128
	3.35 emb; se0.40; r1-8; n20 f	2129
Dusicyon thous	3 yg; n1 lt	1968
	r3-6; n4 lt	1205
Dusicyon vetulus	4 yg; n1 lt	1968
Lycaon pictus	2-6	11145
	2-8	331, 333
	2-19	5761
	4 yg	1521
	4-6	5277
	6; n1 lt	11616
	6.0 yg; r1-9; n3 lt	2440
	6.5 yg; r6-7; n2 dens	334
	6.33; r4-11; n3 lt	2109
	mode 7; r2-12	2449
	mode 7-10; r2-19	10158
	7.0; r4-12; n3 lt	2171
	7.11 yg; r2-12; n18 lt	918, 1230, 1231
	7.6 yg; r6-9; n5 lt	9066
	8.1 yg; sd1.88; n20 lt	11163
	9.0; n2 lt	5221
	10.1 yg; r6-16; n15 lt not birth	3453
	11 yg; r4-16; n4 dens	9546
	16 emb; n1 f	2284
	max 16	743
Nyctereutes procyonoides	3.0 yg; r2-5; n7	8086
	mode 4-6; r4-12	5131
	4.0 yg; r2-5; n4 lt	5130
	r4.9-7.9; mode 6-7; max 15-16; n7 regions	562
	5-7	4643
	mode 5-7; max 14-19	7979
	5-8	9609
	5.0 yg; r2-8; n2 lt	2109
	5.0; sd2.3; n9 lt	11069
	5.49; farm	6739
	5.5-7.0 yg	11067
	6-8; max 19	10499

	6.33 yg; r1-10; n6 lt	9626
	7-8; r5-12	9611
	7; n1 lt	12128
	9.4; r5-14; n8 lt	11866
	10.2 emb; r7-15; n10 f	581, 582
	11 normal +2 resorbed emb	579
Otocyon megalotis	2-5	5761
	3.2 cubs; n5 lt	6743
	4 yg; n1 nest	9945
	5 yg; n1 lt	9284
	5.0 emb; r4-6; n4 f	10153, 10154, 10158
Speothos venaticus	3 yg; n1 lt	5837
	3.8 yg; r1-6; n32 lt	8689
	5.0; r1-9; n13 lt	2734, 2735
	6 yg; n1 lt	2109
	6 yg ea; n3 lt	5256
Urocyon cinereoargenteus	2 emb; n1 f	4249
	2 yg; n1 lt	9066
	3 emb; n1 f	11228a
	3 yg ea; n3 lt	7887
	3.49 ps; mode 4; r1-9; n85 f includes juveniles	9267
	3.59 ps; r1-9; n41 adult f	9267
	3.62 ps; se0.15; r2-5; n32 f	6299
	3.66 emb, ps; r1-7; n35 f	9885
	3.80 ps; n20 f	10542
	3.9 ps; se0.2; n44 f	9101
	3.96 emb; se0.24; r2-6; n24 f	6299
	4-6	190
	4-6	9164
	4.1 ps; n29 f	3767
	4.1 emb; n18 f	3767
	4.16 emb; n6 f	10542
	4.31 ps; sd1.01; n67 f	11893
	4.4 ps; r3-7; n32 f	6306
	4.42 ps; r3-5; n12	11669
	4.5 emb; r3.7; n10 f	6306
	4.90 emb; sd1.12; n40 f	11893
	5.2 cl; n30 f	3767
	5.2 cl; r2-9; n47 f	6306
	5.67 cl; r4-7; n3	11669
	5.67 cl; n3 f	11669
	4.42 ps; r3-5; n12 f	11669
Vulpes bengalensis	3 emb; n1 f	7382
	4; n1 lt	39
Vulpes chama	1 emb; n1 f	10154
	r1-4; n5 lt	1230
Vulpes corsac	est 2-6, occasionally larger	2280
	2-6	4645
	2-11	10499
	2-11; 16 rare	7979
	5.0; mode 3-6; r2-16; n100	4643
	6.1; r3-9; n28 lt	9966
	16 emb; n1 f	10499
Vulpes ferrilata	2-5 yg	7983
Vulpes macrotis	2-7	6387
	4 emb; n1 f	502
	4.0 yg; mode 4; r3-5; n5 lt	7525
	4.6 yg; r3-7; n6 f	8014
	4.55 yg; r4-5; n11 lt	2905, 2908
Vulpes rueppelli	2 yg, n1 lt	9175
Vulpes velox	3-6 yg	1871a
	3.5 emb; r3-4; n4 f	5725
	4 emb ea; n2 f	5405b,
	4.25 yg; r3-6; n4	5725

	4-7	1474
Vulpes vulpes	1-4	397
	2; n1 lt	12128
	2-5 yg	6420
	3 yg; n1 lt	8720
	mode 3-5; r3-12	4253
	r3.0-6.6; n345 f	3044
	r3.1-5.8 live emb; n136 f	197
	3.3; r2-5	12128
	3.5; r3-4; n2 lt	7710
	3.5 yg; r1-10; n384 lt	10461
	3.5-5	11358
	3.7 ps; sd1.5; mode 3; r1-9; n142 lt	9390
	3.8 emb; r1-6; n8 f	7071
	3.8 yg; r1-12; n175 lt	10461
	4-6; max 13	4643
	mode 4-8; occasionally >10	9164
	4 yg; r3-5; n2 dens	5405b$_1$
	4.0 emb; sd1.6; mode 3; r1-8; n35 f	9390
	4.05 yg; n333 lt	4153
	4.16; se0.15; sd1.52; r1-8; 7-10 d; n100 yearling f	8310
	4.25 yg; r1-5; n6 lt in den	7071
	4.29 emb; n37 f	6518
	4.3 ps; r1-8; n38 f	7071
	4.3; n436 lt	391
	4.43 yg; r3-6; n7 dens	9640
	4.5 cl; sd1.5; r2-9; 4.2 ps; sd1.6; r1-9; n75 f	9390
	4.52 yg; sd1.35; mode 4-5; r1-8; n793 lt, 361 f	5322, 5323, 5324
	4.59; se0.10; sd1.56; r1-9; 7-10 d; n254 adult f	8310
	4.6 emb; r1-10; n35 f	6306
	4.64 ps; se0.13; r1-10; n146	4395
	4.71 emb; n14 f	6518
	4.72 ps; sd1.55; n252 f	4397
	4.76; mode 5; r1-10; n108 lt	10511, 10515, 10518
	4.76 ps; sd1.52; n192 f	4397
	4.8; r3-7	466
	4.9 yg; ±0.9; n19 lt	8131
	4.92 yg; n210 lt	10606
	5 yg; r3-7; n10 dens	2441
	5.0 cl; 3.9 ps; n27 f	7071
	5.0 emb; n38 f	5940
	5.0 ps; n41 f	7832
	5.1 yg; se0.046; n1809 lt	9633
	5.1 yg; se0.3; n25 lt	9101
	5.1 ps; se0.2; n103 f	9101
	5.36 emb; se0.18; max 8; n73 f	3155
	5.37 emb, ps; r1-9; n95 f	9885
	5.4 ps; r1-12; n541 f	6306
	5.41 ps; se0.15; n114 f	3155
	r5.48-7.50 cl; r3.13-6.50 emb; mode 5; n136 f	198
	5.7 cl; n6 f	9347
	5.9 ps; n≥38	5940
	5.9 cl; r2-12; n590 f	6306
	5.99 cl; se0.16; max 12; n76 f	3155
	6 ps; n24 f	3767
	6.0 cl; n2 f	7832
	6.1 yg; ±0.3; n23 lt	8131
	6.5 emb; n2 f	7832
	6.7 emb; n7 f	3767
	6.7 emb; r3-12; n48 f	10461
	6.8 emb; se0.338; r4-13; n30 f	4882
	6.8 emb; r2-9; n34 f	10461
	7; n1 lt	10761
	7.1 ps; sd1.9; r4-12; n29 f	10461
	7.2 cl; n15 f	3767

	11 emb; n1 f	10606
	max 12 ps; max 8 live emb	197
	17 yg; n1 den	4912
Vulpes zerda	1 yg; n1 lt	8426
	1.5; r1-2; n8 lt	5908
	1-2	3637
	2.0 yg; r1-3; n2 lt, 1 f	9439
	3.0 yg; n2 lt	3598
	3-4	8223
SEXUAL MATURITY		
Alopex lagopus	f: 1st conceive: 1 yr	1780
	m: 1st sperm: 9-10 mo	4643
	m: testes: large 9 mo	10083
	m: testes: large 10 mo	1780
	f/m: 1st mate: 9 mo	745
	f/m: 9-10 mo	7979
	f/m: end 1st yr	4643
	f/m: 2nd yr	10499
Canis adustus	f: 1st mate: 9 mo	745
	m: 1st mate: 9 mo	745
Canis aureus	f: 10-11 mo	4643
	f: 11 mo	5761
	f: 1st mate: 9 mo	745
	m: 1 yr	5761
	m: 21-22 mo	4643
	m: 1st mate: 9 mo	745
	f/m: 2nd yr	7979
Canis latrans	f: 1st mate: 9 mo	745
	f: 1st mate: 10 mo	3769
	f: 1st mate: 10 mo	5666
	f: 1st mate: 1st yr	9769
	f: 1st birth: ~1 yr	1474
	m: epididymal sperm; 43-45 wk	5666
	m: 1st mate: 9 mo	745
	m: 1st mate: 20 mo	3769
Canis lupus	f: 1st estrus: <2 yr	5402
	f: 1st estrus: 342.4 d; 95%ci 15.7; n118	10148
	f: 1st mate: 12 mo	745
	f: 1st mate: 20 mo	745
	f: 1st mate: 22 mo	8415
	f: 1st birth: 2nd yr	1474
	f: 1st birth: 2nd yr	4645
	f: 1st birth: 2 yr	7236
	f: 22 mo	6382
	f: end 2nd-early 3rd yr	4643
	f: 2 yr	7979
	f/m: 1st mate: 22 mo	8964
	f/m: 1st mate: 22 mo	7147
	f/m: 1st repro: ~12 mo	7158
	m: sperm: ≥10 kg	5402
	m: 1st mate: 12 mo; dingo	745
	m: 1st mate: 20 mo	745
	m: 1st mate: 2 yr	11929
	m: 22 mo	6382
	m: 2-3 yr	7979
	m: end 2nd-4th yr	4643
	f/m: 11-24 mo	3571
Canis mesomelas	f: 1st mate: 7 mo	745
	f: 1st mate: 12-14 mo	11144
	m: 1st mate: 6 mo	745

Cuon alpinus	f: 1st mate: 12 mo	745
	m: 1st mate: 12 mo	745
Lycaon pictus	f: 1st estrus: 23 mo	11163
	f: 1st birth: 22-39 mo	3452
	f: 2 yr	9008
	m: 1st mate: 1.75-5 yr	3453
	m: 1st mate: 11 mo; n1	11163
Nyctereutes procyonoides	f: 1st mate: 10 mo	745
	f: 1st birth: 2 yr	9626
	m: 1st mate: 10 mo	745
	f/m: 1st mate: ~1 yr	9611
	f/m: 8-10 mo	4643
	f/m: 8-10 mo	562
	f/m: 9-11 mo	7979
	f/m: 9-11 mo	10499
	f/m: 10-11 to 22-23 mo	8798
Speothos venaticus	f: 1st mate: 12 mo	745
	f: 1st conceive: 10 mo	8689
	m: 1st mate: 12 mo	745
Urocyon cinereoargenteus	f: 1st mate: 12 mo	745
	f: 10 mo	3529
	f: 1st yr	11893
	m: 1st mate: 12 mo	745
	f/m: ~1 yr	11669
Vulpes corsac	f: 1st repro: 3 yr; n1	2280
	f: 9-10 mo	4643
Vulpes vulpes	f: 1st mate: 10 mo	10462
	f: 1st mate: 10 mo	8310
	f: 9-10 mo	7071
	f: 9-10 mo; mode end 2nd yr	4643
	f: 10 mo	9347
	m: spermatogenesis: 9 mo	5318
	m: 1st mate: 10 mo	745
	m: ~9 mo	9347
	f/m: 1st mate: 10 mo	6520
	f/m: 1st mate: 1 yr	4153
	f/m: after 7 mo; 1st yr	10462
	f/m: 9-10 mo	4253
	f/m: 9-11 mo	7979
	f/m: 10-12 mo	3571
	f/m: 2nd yr	10499
Vulpes zerda	f: 1st mate: 9 mo	745
	m: 1st mate: 9 mo	745
	f/m: sexual activity: 1 yr; n1 pair	9439
CYCLE LENGTH		
Alopex lagopus	estrus: 3-5 d	7453, 7462
	estrus: 4-5 d	4643
	estrus: 12-14 d	7979
Canis aureus	estrus: 3-4 d	4643
Canis latrans	proestrus: 2-3 mo; n20 f	5667
	estrus: 10.1 d; r4-15	5666
	estrus: 10.2 d; n20 f	5667
Canis lupus	estrus: 7-9 d	4535
	estrus: ~2 wk	4643
	6 mo	12001
	218 d; 95%ci 4.0	10148
Canis mesomelas	estrus: ~7 d	11144
Chrysocyon brachyurus	estrus: 5 d	8798
	estrus: 7.8 d	2577
Lycaon pictus	proestrus: 14 d; n29 cycles	11163
	estrus: 20 d; n29 cycles	11163
Nyctereutes procyonoides	proestrus: 7.6 d; sd3.5; r2-14; n35 cycles	11067, 11069
	estrus: 3.9 d; sd1.2; r2-6; n35 cycles	11067, 11069
	estrus: max 6 d	4643

	20-24 d	4643
Speothos venaticus	estrus: 4.1 d; r1-12; n22	8689
Vulpes vulpes	proestrus: ~10 d	4253
	proestrus: 14.4 d; ±3.5	8131
	proestrus: 18.8 d; ±3.0	8131
	estrus: mode 1-4 d; sometimes 5-6 d	4153
	estrus: 1-5 d	5322
	estrus: 1-6 d	8310
	estrus: 1-6 d	10461
	estrus: 2-3 d	8131
	estrus: 2-3 d	4253
	estrus: 2-4 d	10312
	estrus: 3 d; r1-5	8311
	estrus: 3-8 d	7979
	estrus: 7-9 d	4535
	10-12 d	10312
Vulpes zerda	estrus: 1-1.5 d	5908
	estrus: 1-2 d	3637
GESTATION		
Alopex lagopus	46-58 d	6546
	50 d; n1	5844
	51-52 d; r49-57	7979
	51-54 d	1474
	51-57 d	10254
	52 d	1780
	52 d	566
	52 d	4643
	52-53 d	10499
	53.66 d; mode 54; r47-62; n2030	10052
	53-54 d; n9	7447
Canis adustus	57-70 d	5761
	60 d	4180
	60-70 d	10153
	70 d	11697
Canis aureus	~60 d	4645
	60-62 d	7979
	60-63 d	8495, 8501
	60-63 d	4643
	2 mo	5761
	63 d	7712
	63 d	9175
Canis latrans	50-65 d	2088
	59.5 d; r58-61; n2	5844
	~60 d	3768
	2 mo	257
	60-63 d	1474
	60-63 d	9769
	62 d; n2	11645a
	~63 d	747
	est 63 d; r60-65	5666, 5668
	63 d; r58-65	744
	63-65 d	6387
	64 d	1355
	65 d; n1	3594
Canis lupus	~2 mo	5632
	mode 60-65 d; r56-70	3571
	60-66 d; n1	7625
	2 mo; n1	7158, 8185
	62-65 d	7979
	62-65 d	10499
	62-75 d	4643
	62 d; se3-4	11929
	62.4 d; r56-70	10308
	63 d	5146

	63 d	7712
	9 wk	1355
	9 wk	1474
	63-65 d	4645
	65.5 d; r59-73; n4	5844
	68 d	9175
	68 d; n1	11984
	2.5 mo	10698
Canis mesomelas	60-65 d	11144
	~2 mo	5761
Canis rufus	60-62 d; n1	8227
Chrysocyon brachyurus	56-64 d	6499
	~2 mo	209
	60-62, 60-65, 70 d; n3	3180
	62 d	3003
	65.5 d; r65-66; n2	2210
	r64-66 d; n4	1206
	66-67 d	4297
Cuon alpinus	60-62 d	5330
	60-62 d	10229
	60-63 d	1961
Dusicyon culpaeus	est 55-60 d	2129
Dusicyon gymnocercus	58 d	745
Dusicyon thous	56 d; r52-59; n4	1205
Lycaon pictus	2 mo	5277
	60-80 d	5761
	69 d ea; n2	2440
	69.3 d; r69-70; n3	5221
	72-73 d	1521
	72.4 d; sd1.6; n7	11163
	80 d; mode 80; r78-86; n6	2171
Nyctereutes procyonoides	59 d; r51-70	562
	59 d; r58-61	8273
	59 d; r59-70	4643
	59-64 d	7979
	69 d; r59-79; n2	9626
	~60 d	11068
	~60 d; temperate zone, ~max 79 d; north	8798
	~60 d; field data	581, 582
	60-64 d	10499
	60-65 d	5131
	61.0 d; sd2.0; r58 or 59-64; n9	11067, 11069
	61 d	9609
	61-63 d	9611
	61.7 d; r60-63; n6	8086
	~64 d	4645
Otocyon megalotis	60-70 d	10153
	70 d	11697
	75 d; n1	9284
Speothos venaticus	65 d	2109
	67 d	8689
	67-70 d	2734
	76-83 d; n1; last vs 1st mating to birth	5837
Urocyon cinereoargenteus	~51 d	1474
	~60 d	9164
	63 d	6387
Vulpes bengalensis	50-51 d	39
Vulpes chama	51-52 d	9945
Vulpes corsac	49-50, 50, 51 d; n3	2280
	50-60 d	10499
	50-60 d	4645
	52 d	4643
	est 60 d	7979

Vulpes vulpes	implant: 8-10 d	4643
	implant: 10-14 d	6520
	30-50 d	5323
	mode 51-52 d; r49-55	4253
	51 d	9347
	51 d	9164
	51.99 d; sd0.88; se0.036; mode 52; r49-55; n587	5322
	52 d	3044
	52 d; n1	7710, 7712
	52 d; r49-58	4643
	mode 52 d; r51-56; n171 f	10152
	52-53 d	8131
	52-54 d; field data	10511
	55-60 d	397
	~2 mo	2702
Vulpes zerda	50 d; n1	11325
	50.5 d; r50-51; n2	5908
	50.5 d; r50-51; n2	7952
	51 d; n1	8426
	51.0 d; r50-52; n2	9439
	62.5 d; r62-63; n2	3598
LACTATION		
Alopex lagopus	den emergence: 3 wk	7979
	den emergence: 3 wk	10499
	den emergence: 3-4 wk	4643
	solid food: 4 wk	7979
	solid food: 6 wk	4643
	nurse: 1-2.5 mo	4643
	nurse: 6 wk	10499
	nurse: 1.5-2 mo	7979
Canis adustus	wean: 6 wk	745
Canis aureus	solid food: 15-20 d	4643
	den emergence: 2-2.5 mo	4643
	wean: 50-70 d or 90 d; n2 locations	4643
Canis latrans	den emergence: 2-3 wk	744
	den emergence: 3-4 wk	747
	solid food: 12-15 d	3768, 3769
	solid food: 3 wk	744
	solid food: 5 wk; n1 yg	11645a
	solid food: < 6 wk	4398
	wean: 5-7 wk	744
	wean: 6-7 wk	747
	wean: 2-4 mo	3768, 3769
Canis lupus	den emergence: ≤3 wk	1871a
	den emergence: 21 d	7147
	den emergence: 3 wk	4643
	den emergence: 3 wk	7979
	den emergence: 3 wk	9638
	solid food: 3 wk	6137
	solid food: ~23 d	7147
	solid food: 3-4 wk	4643
	nurse: 4-6 wk	7979
	wean: 1.5 mo	4643
	nurse: 34-51 d	6137
	wean: 5-6 wk	10499
	wean: 5-8 wk	7147, 7148
	wean: 38-39 d	9638
	nurse: 8-10 wk	3714
	wean: 4 mo	11984
	catch own food: 7-8 mo	8793
Canis mesomelas	den emergence: 14 d	11144
	den emergence: 3 wk	7412
	solid food: 2-3 wk	11144
	wean: 8-9 wk	7412

Chrysocyon brachyurus	wean: 17 wk	3003
	wean: 4-7 mo	1206
Cuon alpinus	solid food: <4 wk	5330
	solid food: 2 mo	10229
	wean: 58 d	5330
Dusicyon thous	solid food: 20 d	1205
	solid food: 30 d	848
	wean: 90 d	1205
Lycaon pictus	den emergence: 7 d	11253
	den emergence: 3.4 wk	11163
	solid food: 14 d	10158
	wean: 3 mo	10158
Nyctereutes procyonoides	den emergence: 25-30 d	4643
	solid food: 18-20 d or 25-30 d	7979
	solid food: 21-30 d	562, 4643
	wean: 30 d	5130, 5131
	wean: 30-40 d	11452
	wean: 45-60 d	562, 4643, 10499
	wean: 10 wk	9609
	nurse: 80 d	5130
Otocyon megalotis	solid food: 1 mo	5761
	wean: shortly after 1 mo	5761
	wean: 4 wk	10158
Speothos venaticus	wean: 4 wk	745
	nurse: 4-5 mo	2735
Urocyon cinereoargenteus	solid food: est 2-3 wk	5210
	solid food: ~5 wk	9164
	wean: 12 wk	745
Vulpes corsac	solid food: 28 d	4643
Vulpes velox	wean: 6-7 wk	5725
Vulpes vulpes	den emergence: 3-4 wk	4643
	den emergence: 24 d	4253
	den emergence: 6 wk	6520
	den emergence: ~6 wk	7979
	den emergence: 1.5-2 mo	397
	solid food: 20 d	4253
	solid food: 3-4 wk	2702
	solid food: 4 wk	6686
	solid food: 4 wk	3571
	solid food: 1 mo	10461
	solid food: 6 wk	9164
	solid food: 8 wk	6520
	wean: ~5 wk	6686
	wean: ~6 wk	3815
	wean: 1.5 mo	4643
	wean: 1.5 mo	7979
	wean: 6-8 wk	6520
	wean: 7-8 wk	3571
	wean: 8-10 wk	9885
Vulpes zerda	wean: 61-70 d	5908
INTERLITTER INTERVAL		
Alopex lagopus	1 yr	7979
Canis aureus	1 yr	5761
Canis latrans	1 yr	3769
	1 yr	744
Canis lupus	2 lt in 1 yr; n1	11083
	1 yr per pack	8415
Canis mesomelas	0.5-1 yr	11144
Cuon alpinus	1 yr	4643
Dusicyon culpaeus	1 yr	2129
Dusicyon thous	8 mo; n1	1205
Lycaon pictus	11 mo; r9-12 mo; n5	2171
	11.7; se0.6; n9	11163
	12-14 mo; min 6 mo if litter dies	3452

Nyctereutes procyonoides	1 yr	562
Otocyon megalotis	max 2 lt in 1 yr	5761
Speothos venaticus	249 d; r152-302; n7	8689
Vulpes bengalensis	1 yr	9175
Vulpes corsac	1 yr	10499
	1 yr	4643
Vulpes vulpes	1 yr	7071
	1 yr	10761
	1 yr	9347
	1 yr	4253
	1 yr	6520
Vulpes zerda	1 yr; yg live	5908
SEASONALITY		
Alopex lagopus	spermatogenesis: Jan-July; Canada, AK	1780
	testes: large Feb-Apr, small Aug-Oct; farm	10084, 10086
	estrus: Feb-Mar; USSR	7979
	mate: Feb-May; USSR	4643
	mate: Mar-Apr; n AK	1780
	mate: Mar-Apr; captive	7453, 7462, 10086, 10087
	mate: Apr-early May; w Siberia; later e Siberia	10499
	mate: Apr-May; Siberia	1780
	mate: end Apr-early May; Finnish Lapland, Finland	5492
	birth: Mar-May; USSR	7979
	birth: Apr-June; n Canada	1474
	birth: end Apr-early July; USSR	4643
	birth: May-June; Commander I, Aleutians	566
	birth: May-June; captive	10083
	birth: May-July; N America	1780
	birth: June; n1; Finnish Lapland, Finland	5492
	birth: peak June; AK	836
Canis adustus	mate: yr rd, peak Sept-Nov; Transvaal, S Africa	10153
	birth: end Mar-mid June; n11; zoo	12128
	birth: June-July, Sept; Uganda	5761
	birth: Aug-Jan; S Africa	10158
	birth: Dec-Apr; Angola	11697
	birth: June; S Africa	11144
Canis aureus	mate: mid Jan-Feb; USSR	7979
	mate: mid Jan-Apr; Europe, w USSR	4645
	mate: Feb-early Mar; USSR	4643
	mate: spring-summer; Pakistan	9175
	birth: end Mar-early May; USSR	4643
	birth: Apr-early May; Europe, w USSR	4645
	birth: Apr-mid May; USSR	7979
	birth: Dec-Apr; Tanzania	5761
	lact f: Mar-May; Nepal	7382
	yg found: yr rd; Bangladesh	8613
	wean: June; USSR	7979
	wean: mid July-early Aug; USSR	4643
Canis latrans	spermatogenesis: mid Oct-mid Mar; captive	5666
	mobile sperm: Dec-May; captive	4033
	testes: large Nov-Mar; AR	3793
	estrus: Jan-Mar; WY	1871a
	estrus: Mar; captive	5844
	estrus: Mar; captive	5666
	mate: Jan-Feb; WY	747
	mate: Jan-Feb; N America	1474
	mate: Jan-Feb; San Patricia Co, TX	257
	mate: Jan-early Mar; KS	3768
	mate: Jan-mid Mar; ID	6249
	mate: peak Feb; Canada	9769
	mate: Feb; Jasper ALB, BC, Canada	2088
	mate: Feb; WI	5210
	ovulate: Feb-Mar; KS	3768
	preg f: Jan-May; OR	4295

	preg f: Jan-June; AZ	4295
	preg f: Feb-Apr; WY	4295
	preg f: Feb-May; ID	4295
	preg f: Feb-May; NM	4295
	preg f: Feb-May; NV	4295
	preg f: Feb-May; WA	4295
	preg f: Feb-June; CA	4295
	preg f: Feb-June; n1; SD	4295
	preg f: Feb-June; n2; CO	4295
	preg f: Feb-July; n1; UT	4295
	preg f: Mar; n1; NE	5405b$_1$
	preg f: Mar-May; MT	4295
	preg f: Mar-May; TX	4295
	preg f: mid Mar-Apr; ID	6249
	preg f: Apr; n2; Coahuila, Mexico	522
	preg f: end Apr-May; Black Hills, SD, WY	10954
	birth: Mar-Apr; NM	502
	birth: Mar-July; Chiapas, Mexico	3633
	birth: end Mar-May; KS	3768
	birth: Apr; WI	5210
	birth: Apr; San Patricia Co, TX	257
	birth: Apr-May; N America	1474
	birth: end Apr; MAN, Canada	2136
	birth: May; WY	747
Canis lupus	testes: large Mar, small Jan; Russia	3649
	testosterone: low July-Jan; captive	1663
	estrus: Jan-Feb; captive	396
	estrus: Jan-Apr; WY	1871a
	mate: Jan; n Tibet	8740
	mate: Jan-Mar; Canada	1474
	mate: Feb-Mar; captive	12103
	mate: Feb-Mar; MN, AK	8185
	mate: Feb-Mar; WI	5210
	mate: Feb-Mar; w US	5146
	mate: Feb-Mar; central, n USSR	7979
	mate: Feb-Apr, peak Mar; AK	8964
	mate: end Feb-?; MI	7147
	mate: Mar; AK	7625
	mate: end Mar-early Apr; AK	5632
	mate: Apr-May; n central Australia	1663a
	mate: Apr-June; captive	1663
	mate: Dec; n1; zoo, India	2496
	mate: Dec-Mar, peak Feb; zoo	11929
	mate: Dec-Mar; south 1-1.5 mo before north; USSR	4643
	mate: end Dec-early Jan; USSR	4645
	mate: end Dec-Feb; s USSR	7979
	mate: end Dec-Mar; Germany	3571
	mate: end Dec-Apr; USSR	10499
	birth: Feb-Mar; USSR	4645
	birth: end Feb-mid Apr; s USSR	7979
	birth: Apr-May; central, n USSR	7979
	birth: Mar-May; Germany	3571
	birth: Mar-Sept; n25; zoo	1230
	birth: Mar-Sept, peak June-Aug; VIC, Australia	5402
	birth: Apr-May; Canada	1474
	birth: Apr-May; MN, AK	8185
	birth: Apr-May; WI	5210
	birth: Apr-early June; USSR	4643
	birth: early May; AK	7625
	birth: peak May; Russia	3649
	birth: May; AK	8415
	birth: May-June; NWT, Canada	6137
	birth: mid May; n1; zoo	9638
	birth: mid May-early June; captive	6382

	birth: end May-early June; AK	5632
	birth: June-July; n central Australia	1663a
	birth: June-Aug; captive	1663
Canis mesomelas	mate: Apr-Aug, peak May-July; S Africa	11144
	mate: mid-Aug-mid Nov; zoo	1230
	birth: June-Aug 79%; Natal, S Africa	9327
	birth: July-Nov, peak Oct; S Africa	11144
	birth: July-Sept; Kenya, Uganda, n Tanzania	5761
	birth: Aug-Oct; ne S Africa	3145
	yg seen: July-Oct; S Africa	8514
Canis rufus	mate: Jan-Feb; se TX, sw LA	9134
	preg f: Mar-mid May; AR	8227
	preg f: Mar-mid May; TX	8227
	preg f: mid Mar-mid Apr; OK	8227
	birth: Mar-Apr; se TX, sw LA	9134
	birth: Apr-May; s central US	1474
Chrysocyon brachyurus	estrus: Oct-Jan; captive	1206
	mate: Apr; Geraes State, Brazil	2579
	mate: end Oct-mid Nov; zoo	209
	birth: early Jan; zoo	209
	birth: Jan-Feb; n2; zoo	3003
	birth: Aug; Geraes State, Brazil	2579
Cuon alpinus	mate: Jan-Feb; zoo, China	4643
	mate: Feb; zoo	10229
	mate: Sept-Jan; India	1961
	birth: Jan-Feb; Nepal	7382
	birth: Jan-Mar; India	988a, 6654
	birth: Apr; zoo, China	4643
	birth: Nov-Dec; India	61
	birth: Nov-Mar; India	2303
	yg seen: Dec; Karnataka, India	5330
Dusicyon culpaeus	estrus: Aug-Oct; Argentina	2129
	birth: Oct-Dec; Argentina	2129
Dusicyon gymnocercus	testes: increase mid June-Sept; Argentina	2129
	preg f: end Aug-mid Dec; Argentina	2129
	birth: end Oct-mid Dec; Argentina	2129
Dusicyon thous	birth: peak Jan-Feb; Venezuela	7983
Lycaon pictus	estrus: end Jan-early May; S Africa	11163
	mate: Mar-May; Africa	5775
	mate: Mar-Sept, peak Apr-June; S Africa	11145
	birth: Jan, May, Nov; zoo	2171
	birth: Mar; Zaire	11253
	birth: Mar-July; zoo	1230
	birth: peak Mar-July; zoo	1230
	birth: Mar-Sept, peak Apr; Namibia	9945
	birth: Apr; n1; Natal	11616
	birth: Apr-July; S Africa	11163
	birth: Apr-Nov, none July, Sept; zoo	918
	birth: peak May-June; ne S Africa	3145
	birth: May-June; S Africa	9008
	birth: May-July; n Zimbabwe	331, 333, 334
	birth: yr rd, peak Mar-June, none Sept; Serengeti, Tanzania	3452
	yg seen: Mar, Sept; S Africa	8514
	yg seen: June-July, Sept; Africa	5775
Nyctereutes procyonoides	testes: large Dec-Feb, small May-Nov; se Europe	581, 582
	mate: early Feb-Apr; USSR	4643
	mate: Feb-?; captive	9611
	mate: Feb-Apr; USSR	562
	mate: Feb-Mar; captive	11069
	mate: mid Feb-mid Mar; se Europe	581, 582
	mate: end Feb; farm	9609
	mate: Mar; USSR	4645
	mate: peak Mar, some Feb; USSR	10499
	mate: Dec-Mar; zoo, Germany	9626

	ovulate: Mar; Japan	5131
	birth: Mar-Apr; Japan	11451a
	birth: Apr, June; captive	2109
	birth: Apr, peak May-June; USSR	4643
	birth: Apr-May; USSR	562
	birth: Apr-June; n10; captive, Japan	8086
	birth: mid Apr-early May; n6; zoo, Germany	9626
	birth: end Apr-mid May; se Europe	581, 582
	birth: May; USSR	4645
Otocyon megalotis	mate: Nov-Apr; w, sw Zimbabwe	10153
	birth: end Aug-Oct; sw Kenya	6743
	birth: Sept-Nov; S Africa	10158
	birth: Oct; Tanzania	5761
	birth: Mar; ne Uganda	5761
	birth: Nov; n1; sw Africa	9945
	birth: Dec-Apr; Angola	11697
Speothos venaticus	birth: Oct; n1; captive	2109
	birth: yr rd; zoo	8689
Urocyon cinereoargenteus	testes: large Feb; MD	11669
	testes: large Nov-Feb; e US	11669
	mate: Jan-Apr; WY	1871a
	mate: mid Jan-mid May; NY	9885
	mate: peak end Jan-early Feb; s GA, n FL	11893
	mate: end Jan-May, Mar 52.5%; NY	6306
	mate: peak Feb; AL	10542
	mate: Feb-Mar; US	1474
	mate: mid Feb-Mar; WI	5210, 9101
	mate: end winter; Mexico	6387
	preg f: Jan-Mar; s GA, n FL	11893
	preg f: Feb-Mar; IL	6299
	preg f: Mar; n1; CA	11228a
	birth: Mar-Apr; Mexico	6387
	birth: Apr-early May; PA	9164
	birth: Apr-May; US	1474
	birth: early spring; w US	5146
	lact f: May; n3; NM	502
Vulpes bengalensis	mate: Nov-Jan; India	988a
	birth: Feb-Apr; Pakistan	9175
	birth: Feb-Apr; India	988a
Vulpes chama	birth: mid Sept-mid Oct; Transvaal, S Africa	8965
	birth: mid Sept-mid Oct; n5; zoo	1230
Vulpes corsac	estrus: Jan-Feb; USSR	7979
	mate: Jan-Feb; USSR	4643
	mate: end Jan-Feb; n Tibet	8740
	mate: Feb; s USSR	4645
	mate: Feb; w Siberia, Transbaikal, Russia	10499
	mate: end Feb-Mar; Altai, Russia	10499
	birth: Mar; s USSR	4645
	birth: Mar-Apr; USSR	7979
	birth: mid Mar-mid May, peak Apr; USSR	4643
Vulpes fernilata	birth: Apr-May; Nepal	7983
Vulpes macrotis	mate: Jan; UT	2905
	mate: Dec-Feb; CA	7525
	preg f: Feb; n1; AZ	502
	birth: Feb; Mexico	6387
	birth: Feb-Mar; CA	7525
	birth: Feb-Apr; sw US, nw Mexico	1474
	birth: Mar-Apr; UT	2905
	birth: mid Mar-mid Apr; n6; NE	8014
Vulpes rueppelli	lact f: Mar; n1; Pakistan	9175
Vulpes velox	preg f: Apr; n2; OK	5405b,
	birth: Feb-Apr; central US	1474

Vulpes vulpes	sperm: yr rd, peak May-Aug; n434; NSW, Australia	9390
	spermatogenesis: Apr-Sept; ACT, Australia	7071
	spermatogenesis: Oct-Jan; captive	10312
	spermatogenesis: Nov-Feb, peak 3rd wk Jan; IL, IA	10461
	spermatogenesis: Nov-Mar, peak Jan; UK	6520
	spermatogenesis: end Nov-Mar; NY	954
	testes: large Jan-Feb, small Sept; UK	2123
	testes: large Oct-Mar; England, UK	2124
	testes: large Nov-Feb, small May-Aug; Ireland, UK	3155
	testes: large Dec-Jan, small July-Aug; England, UK	6521
	testes: large Dec-Feb, small July; Wales, England, UK	2124
	testes: large Dec-Feb; USSR	4643
	testosterone: high Jan-Mar; captive	6983
	epididymal sperm; Nov-Apr; Ireland, UK	3815
	estrogen: peaks Dec, Apr; lab	7461
	estrus: Jan; captive	7710
	estrus: Jan-Feb; NY	6306
	estrus: Jan-Feb; zoo	397
	estrus: Jan-May, peak Feb; captive	8310
	mate: Jan; captive	6686
	mate: Jan-Feb; w US	5146
	mate: Jan-Feb; Ireland, UK	3815
	mate: Jan-Feb; central Europe	4253
	mate: Jan-Feb; USSR	4643
	mate: Jan-mid Feb; Ireland, UK	3155
	mate: Jan-Mar; farm	5322
	mate: Jan-Mar; captive	8131
	mate: Jan-Apr; Scotland, UK	5940
	mate: Jan-Apr; USSR	4645
	mate: mid Jan-mid Feb; Ireland, UK	3156
	mate: mid Jan-mid Feb; Germany	10511
	mate: mid Jan-Feb; WI	9101
	mate: mid Jan-early Apr; farm	5323
	mate: mid Jan-3rd wk Apr, end Jan-early Feb 88.3%; NY	6306
	mate: end Jan; UK	9347
	mate: end Jan-Feb; s USSR	7979
	mate: end Jan-Mar; NY	954
	mate: Feb; farm	8310
	mate: Feb-Apr; USSR	10499
	mate: Feb-Apr; captive	10052
	mate: end Feb-early Apr; n Europe	4253
	mate: Mar; central USSR	7979
	mate: Oct-May; captive	10926
	mate: Dec-Jan; USSR	466
	mate: Dec-Jan; s Europe	4253
	mate: Dec-early Feb; PA	9164
	mate: Dec-Mar; WY	1871a
	mate: mid Dec-early Feb; IA, IL	10461
	mate: mid Dec-mid Feb; captive	5319
	mate: mid Dec-Mar, peak end Jan-Feb; NY	5323, 9885
	mate: end Dec-mid Mar; Germany	3571
	ovulate: Jan-Feb; England, UK	2123
	ovulate: Jan-Mar; Ireland, UK	3155
	ovulate: Feb-July, peak Mar; Sweden	3044
	CL present: Jan-Mar; Ireland, UK	3815
	conceive: Dec-Apr; IL, IA	10461
	preg f: peak Jan-Feb; se England, UK	6520
	preg f: Feb-Mar; Wales, UK	6520, 6521
	preg f: Feb-Mar; NY	6306
	preg f: peak Mar-July, low Nov-Feb; Luxemburg, Belgium	818
	preg f: peak Aug; NSW, Australia	9390
	birth: Feb; captive	7710
	birth: Feb-May; Germany	3571
	birth: mid Feb-May; Pyrenees, Spain	11255

	birth: Mar; captive	6686
	birth: Mar; Ireland, UK	3155
	birth: Mar; USSR	466
	birth: Mar-Apr; central US	1474
	birth: Mar-Apr; Europe, mid east	6420
	birth: Mar-Apr; Nepal	7382
	birth: Mar-Apr; USSR	4643, 4645
	birth: Mar-Apr; zoo	397
	birth: Mar-Apr; farm	8310
	birth: Mar-May; Scotland, UK	10761
	birth: Mar-May; farm	5322
	birth: mid Mar; SD	261a
	birth: mid Mar; lab	5318
	birth: mid-end Mar; France	391
	birth: mid Mar-mid Apr; Germany	10511
	birth: peak mid Mar-mid Apr; ND	9501
	birth: mid Mar-Apr; WI	5210
	birth: end Mar-early Apr; UK	9347
	birth: end Mar-early Apr; IL, WI	2060
	birth: Apr-early May; Scotland, UK	2702
	birth: ~Apr-May; USSR	7979
	birth: Apr-May; n9; zoo	12128
	lact f: Mar-mid May; Ireland, UK	3155
Vulpes zerda	mate: Jan-Feb; n African, Arabian desert	3637
	mate: Feb-Apr; captive	9439
	birth: mid Mar-Apr; n African, Arabian desert	3637
	birth: Apr-June; captive	9439

Order Carnivora, Family Ursidae

Bears (*Ailuropoda* 75-160 kg; *Helarctos* 30-65 kg; *Melursus* 55-145 kg; *Tremarctos* 60-175 kg; *Ursus* 50-800 kg) are often diurnal, solitary, terrestrial omnivores from Europe, Asia, and the New World. Males are often larger than females (3571, 4383, 4643, 6469, 8495, 8501, 8623, 8895, 8900, 9175, 9549, 10414, 10434, 10499). Copulation may trigger ovulation (*Ursus* 3060, 11802) and implantation can be delayed (8799, 11802, 11952). The endotheliochorial, labyrinthine, discoid or zonary placenta with villous, circumperipheral hematomae implants into a bicornuate or bipartite uterus (3060, 7278, 7563, 8895, 8968, 9707, 10532, 11802). Placentophagia is recorded (*Ursus* 2276).

Neonates are blind, but usually have some fur (1002, 2109, 2475, 4643, 4851, 6387, 8895, 9175, 9308, 10579, 11207). Eyes open at 3 wk (*Melurus* 2802, 8495, 8501), 3-6 wk (*Ursus* 507, 1505, 2416, 4375, 4704, 5970, 6469, 7234, 7277, 7979, 9600, 9624, 10308, 10357, 10499), or 40-48 d (*Ailuropoda* 5314, 9549). Females have 2-6 mammae (3058, 4745, 9549, 11207). Males have a baculum (1473, 2314, 3059) and prepenial testes (*Ailuropoda* 2314). Details of anatomy (3059, 4641, 5961), endocrinology (*Ailuropoda* 1098, 1763, 4861, 7471, 7622, 9903a; *Ursus* 3407, 4599, 7103, 8213, 10927a, 10928), and milk composition (512, 4852, 5292, 8896, 9233) are available.

NEONATAL MASS

Ailuropoda melanoleuca	90-130 g	5314
	100 g	325
	104 g; r60-130	9549
Helarctos malayanus	325 g; n1	5121
Ursus americanus	200-340 g	1474, 5210
	225-400 g	5435
	~227 g	5942
	~250 g	6387
	250-280 g; n1	1505
	250-400 g	4850
	252-336 g	507
	f: 292 g; n1	9600
	300 g	10308
	m: 301 g; n1	9600

	310.5 g; r265-362; n4	9069
	316 g; se9; n9, <3 d	4599
	<453.6 g	7234
	<454 g	6823
Ursus arctos	280-780 g	1474
	298 g; r286-310; n2 near term	8116
	300-625 g	10308
	~450 g	2060
	500 g	7979
	501-510 g	4643
	f: 629 g; n1	10579
Ursus maritimus	500 g	10434
	540-660 g	11328
	566 g; r520-610; n5 dead	9739
	~600 g; r400-700	8895
	603 g; r453-753; n2	2109
	666 g; r652-680; n2	4704
	675 g; 2 d	10357
	680 g; 1 d	7277
	697.5 g; r675-720	3179
	~700 g	4375
	755 g; r650-840; 3 d	4643
	~1 kg	10433
Ursus thibetanus	305 g; n1	9064
	300-450 g	4643, 7979, 10499
	710 g; n2; a few d after birth	10499
NEONATAL SIZE		
Ailuropoda melanoleuca	15-17 cm	5314, 9549
Helarctos malayanus	19 cm; n1	5121
Tremarctos ornatus	22.5 cm; n1	9308
Ursus americanus	~15-20 cm	5942
	<15.2 cm	7234
	20 cm	2088
Ursus arctos	20-25 cm	2060
	23-23.5 cm	4643
	23-28 cm	10499
	25 cm	3571
Ursus maritimus	~25 cm	4375
	25 cm; r24-26; n2	2109
	28-32 cm; 3 d	4643
	~30 cm	7979
	f: 34 cm; n1 nose to tail	1002
Ursus thibetanus	26.5 cm; n2; a few d after birth	10499
WEANING MASS		
Ailuropoda melanoleuca	22 kg; 6 mo; from graph	2314
Ursus arctos	26 kg; n1 7 mo cub self-sufficient	5350
Ursus maritimus	5.78 kg; r2.8-10.25	9739
	den emergence: 10-15 kg; 1 yr: 45-80 kg; 2 yr: 70-140 kg	2475
LITTER SIZE		
Ailuropoda melanoleuca	1.5	5855
	1.56 born; mode 2; r1-3; n55 lt, 24 f	11574
	1.7; r1-2; n37 lt	9549
	2; n1 lt	4861
Helarctos malayanus	1	9064
	1 ea; n2 lt	5121
	1 yg ea; n2 lt	9066
	1 yg ea; n3 lt, 1 f	7044
	1.1; r1-2; n18 lt	2277, 2279, 2281, 2283
Melursus ursinus	1 yg ea; n2 lt	9066
	1-2 yg; n5 lt	5220
	1.3; r1-2; n3 lt	9052
	1.6; r1-2; n24 lt	6250
	2	7652
	2; n1 lt	1230

	2; r1-2	8495, 8501
	2; sometimes 3	2802
	2; occasionally 3	6654
Tremarctos ornatus	1	9308
	1.4; r1-3	2286
	1.5 yg; r1-2; n2 lt	3711
	2 yg; 1 lt	1019, 1020
Ursus americanus	1-4	7652
	1.5; r1-2; n2 lt	9600
	1.7	5437
	1.73 yg; r1-3; n30 lt	4493
	1.77 ps; r1-2; n9	8623
	mode 2	502
	mode 2; r1-4	2088
	mode 2; r1-4	5942
	mode 2; r2-4	6387
	mode 2-3; r2-4	7234
	2-4 cl	11802
	2 yg; n1	4851
	2 yg ea; n3 lt	9066
	2.05 yg; r1-4; n219 lt	3060
	2.1 cl; r1-3; n11	8623
	2.2 yg; n5	4599
	2.2; r1-4; n10 lt	4383
	2.26; n15 lt, 2 f	508
	2.4 cl; r1-3; n12 f	3060
	2.4 yg; n38 lt	10260
	2.4 yg; r1-4; n47	5943
	2.4 yg; max 6; n284 lt	5210
	2.5 emb; r2-3; n2 f	2545
	2.5; r2-3; n4 lt	3407
	2.6 emb, yg; n19 lt	10414
	2.7, 2.8 cl; max 4; 2 samples	1986
	2.75 cl	5961
	2.9; r1-5; n68 lt	205
	mode 3	10839
	3.2 ps; r1-4; n13 f	3060
	3.5 yg; r3-4; n4 lt, 2 f	9319
	5 yg; n1 f; anecdote	6950
	6 yg; n1 f	9320
Ursus arctos	1-2, occasionally more	7979
	1-5	5824
	1.5 yg; r1-2; n6 f	10929
	1.6 yg; n22 f	8288, 8289
	1.64 new ca; n14 f	10929
	1.69; mode 2; r1-3; n13 lt	6469
	1.7 yg, 1.8 yearlings; n65 lt	6917
	1.76 ps; n16 f	10929
	1.8; r1-3	3986
	1.9 cl; n9 f	8289
	1.98 yg; r1-3; n168 f	7615
	2	4645
	2; r1-4	1474
	2; n1 lt	10350
	2; n3 lt	1230
	mode 2; 3 rare	6387
	mode 2; r1-3	11330
	mode 2; r1-4	2060
	mode 2; r1-5	4643
	mode 2-3; r2-5	3571
	rarely > 2	12049
	~2; r1-3; 4 rare; n46 lt	8791
	2.05; mode 2; r1-4; n213 lt	2611, 2612
	2.23 yg; se0.06; r1-4; n98 f	4641

	2.26 yg; mode 2; r1-3; n31 lt, ~145 d	7092
	2.3; mode 2; r1-4; n18 lt	6469
	2.4 yg; r2-3; n8 lt	2276
	2.53 ps; n15 tracts	4641
	2.55	2106
	2.8 born; r2-3; n4 lt	542
	3 emb	7980
	3 emb; r2-4; n2 f	8116
	3; n1 lt	10352
	5 yg; n1 f	10499
	6 emb; n1 f	10499
	6 yg; n1 f	11697a
Ursus maritimus	r1-3; n4 lt	76
	1 yg; yearling; 2 yg; adult; 3,4 rare	7979
	1; n1	5514
	1 ea; n2 lt	2109
	1.4; r1-2; n14 lt	9739
	1.43; mode 1; r1-2; n14 lt	11327
	1.46; n13 lt	11328
	1.5; r1-2; n4 lt	5970
	1.52; r1-3; n54 lt	8899
	est 1.6 yg	5436
	1.64 yg < 1 yr; n36 f	3568
	1.69; r1-3; n65 lt	11049
	1.7; n10 lt, 3 f	9233
	1.8; r1-3; n6 lt	2916
	1.8; r1-2; mode 2; n10 lt	12128
	1.8; r1-2; n13 lt	9733
	1.81 yg emerge; r1-4; n129 f	6220
	1.82; n34 lt	10433
	1.9 yg; r1-3; n118 lt	8898, 10434
	mode 2	12057
	mode 2; r1-4	4375
	mode 2; r1-4	4643
	2 yg; n1 lt	9066
Ursus thibetanus	mode 1-2; r1-3	7979
	1 yg; n1 f	7382
	1 ea; n2 lt	1230
	2; 1,3 rare	10499
	mode 2	6653, 6654
	mode 2; r1-2	9175
	mode 2; 3 rare	1310
	2 yg; n1 lt	9064
SEXUAL MATURITY		
Ailuropoda melanoleuca	f/m: 4-10 yr; native reports	9884a
	f/m: 5.5-6.5 yr	9549
	f/m: 6-7 yr	1816
Ursus americanus	f: 1st cl: 3.5 yr	3060
	f: 1st estrus: 4.5 yr	5437
	f: 1st mate: 38% 2.5 yr; 100% breed 3.5 yr	5961
	f: 1st mate: 3 yr	2088
	f: 1st mate: 3.4 yr	8623
	f: 1st conceive: 2.5 yr; n1	1986
	f: 1st conceive: 3.5 yr	6387
	f: 1st birth: 3.5-4.8 yr; n4 groups, 23 f	7091
	f: 1st birth: 5.7 yr; r5-8; n18	5943
	f: 1st lt raised: 6.5-7.51 yr	5435, 5437
	m: spermatids in testes: 3.5 yr	3060
	f/m: 1st mate: 3.5 yr	1474
	f/m: 3.5 yr	5942
Ursus arctos	f: ovarian histology: 1-4 yr	10927a
	f: 1st estrus: 4.5 yr; r3.5-5.5	542
	f: 1st conceive: 3 yr 4 mo	4641
	f: 1st conceive: 6.5-8.5 yr	8288, 8289

	f: 1st birth: 2 yr; n1	2611, 2612
	f: 1st birth: 3 yr	6509
	f: 1st birth: 5th yr	10499
	f: 1st birth: 6 yr; n5	7092
	f: 6-7 yr	12049
	m: testes histology: 2-4 yr	10927a
	f/m: 1st mate: 3 yr	1474
	f/m: 1st mate: 4-5 yr	6387
	f/m: 2.5-4 yr	3571
	f/m: 3rd yr	7979
Ursus maritimus	f: 1st mate: 4 yr	10433
	f: 1st mate: 50% 4 yr; rest 5-6	8895
	f: 1st birth: 4-5 yr	8898, 8899
	f: 1st birth: 5 yr	3568
	f: 1st birth: 6-7 yr	2475, 5435
	m: 1st sperm: 3 yr	2475
	f/m: 3-4 yr	10499
	f/m: 3-4 yr wild; 5 yr; zoo	7979
	f/m: 5 yr	11327
Ursus thibetanus	f: 1st birth: 3 yr	6509
	f/m: 3rd yr	7979
CYCLE LENGTH		
Ailuropoda melanoleuca	estrus: 1 d	4860
	estrus: mode 10 d; r4-14	1816
	estrus: 12-25 d; peak receptivity 2-7 d	9549
Helarctos malayanus	estrus: a few d	2279
Ursus americanus	estrus: a few d	9227
Ursus arctos	estrus: 27 d	7615
	4-18 d; interestrous interval	7615
	16 d	11330
	24 d; r18-30; n3 cycles, 1 f	10350
	max 27 d	2106
Ursus maritimus	estrus: 3 d	11327
GESTATION		
Ailuropoda melanoleuca	~135 d; r97-161; n19	9549
	16 wk	7471
	140 d; ±8; r122-163	1816, 11952
	159 d; n1	4861
	2-5 mo	5314
Helarctos malayanus	95.7 d; r95-96; n3	2277, 2279, 2283
	96-97 d	10352
	214 d; r174-240; n3	7044
Melursus ursinus	est 169-176 d	5220
	7 ? mo	7652
	7 mo	7974
	~7 mo	2802, 8501
Tremarctos ornatus	194-208 d	3711
	215 d; r175-240; n6	1019, 1020
	7 mo 3 wk-8.5 mo; n1	2286
	8 mo; n1	9308
Ursus americanus	implant: 6 mo	11802
	~6-7 mo	5435
	200-217 d	10308
	7 ? mo	7652
	~7 mo; field data	3060
	~7 mo	6387
	210-253	3407
	7 mo 1 wk	1355
	215 d; r205-226; n9	507, 4851
	~220 d; r210-234; n1	4851
	~225 d	5210
	8 mo with delay	11802

Ursus arctos	implant: 1-2 mo	3571
	implant: 62.5 d	6469
	implant: r18-33 wk; n20	2612
	postimplantation: 7 mo	3571
	min 50 d	2106
	115-136 d; n1; 1st, last mating to birth	10352
	180-240 d	10308
	6-9 mo	3571
	est 189-251 d	8289
	196 d; r179-213	10350
	r197-288 d; n20 with delay	2611, 2612
	~7 mo	4582
	~7 mo	4645, 7979, 10499
	7-8 mo	8741
	7-8 mo	7652
	227 d; r185-251; n8	4643
	236 d	1355
	237 d	6469
	est 8 mo	4641
	8 mo; r209-272 d	11330
	8 mo	3986
	9 mo	2276
Ursus maritimus	6-7 mo	5435
	201-303 d	11328
	207 d; r201-214; n3	3179
	226 d; r195-253; n4	10357
	230-250 d	4643
	mode 235-254 d; r228-303	11327
	~8 mo	10499
	~8 mo; n13	9733
	8 mo; r230-250 d	7979
	246 d; r245-248	5970
	246 d; r228-262; n4	76
	10-11 mo	12057
Ursus thibetanus	~6 mo	6654
LACTATION		
Ailuropoda melanoleuca	solid food: 5 mo	5314
	independent foraging: 3-4 mo	1816
	nurse: 6 mo	5314
	wean: 46 wk	9549
Helarctos malayanus	wean: ~3 mo	2281
Melursus ursinus	wean: 2-3 mo	2802
Tremarctos ornatus	solid food: 89 d; n1 lt	1020
Ursus americanus	independent: 5.5 mo	3057
	wean: 6-7 mo	1474
	mode 1st yr; max 18-22 mo	5435
Ursus arctos	den emergence: 3-4 mo	10499
	independent: 7 mo; n1	5350
	wean: 1.5 yr	2276
	wean: 1-2 yr	8288, 8289
	wean: 4 mo	10499
	wean: ~5 mo	7979
	wean: 6 mo	4643
Ursus maritimus	den emergence: 45 d	7979
	den emergence: 2 mo	4643
	solid food: 12 wk	2416
	solid food: 6 mo	10499
	wean: 52 d	10357
	wean: 73.4 d; n19	9739
	wean: 5-8 mo	6342
	wean: 6-8 mo	4643
	wean: 9-21 mo	4375
	wean: ~15 mo	7979
	nurse: max 15 mo	10499

	wean: 20+ mo	11327
	wean: 24-28 mo	2475, 8899
Ursus thibetanus	den emergence: ~2 mo	7979
	den emergence: ~2-2.5 mo	10499
	wean: 3-3.5 mo	7979
	wean: 3-3.5 mo	10499
INTERLITTER INTERVAL		
Ailuropoda melanoleuca	1 yr; yg die	9549
	2 yr; yg live	9549
Melursus ursinus	3 yr	2802
Ursus americanus	1 yr; yg die before June	205
	2 yr; yg live	205
	2 yr	6387
	2 yr	2088
	2-3 yr	5435
Ursus arctos	1 yr; yg die	10499
	2 yr; yg live	10499
	2 yr	2106
	2 yr	3571
	2 yr	4643
	2-3 yr	6387
	2-3 yr	1474
	2.67 yr; n9	7092
	3 yr	8289
	3 yr	12049
	mode 3 yr	542
Ursus maritimus	1 yr	5970
	1 yr	9733
	2 yr	1474
	> 2 yr	11327
	2-3 yr	5435
	2.9 yr; 40% 2 yr; r1-5; n66 intervals	8895, 8898, 8899, 10433
	3 yr	4643
Ursus thibetanus	1 yr	6654
SEASONALITY		
Ailuropoda melanoleuca	estrus: Mar-June; zoo	5855
	mate: mid Mar-mid May; captive	9549
	mate: end Apr-May; zoo	11952
	mate: May; zoo	1098
	birth: Jan; w China	9884a
	birth: Sept; zoo	11952
Helarctos malayanus	birth: Jan; n1; zoo	10352
	birth: yr rd; zoo	2281
Melursus ursinus	mate: June; India	7652
	mate: yr rd, peak 1st half of yr; Sri Lanka	8495, 8501
	birth: June; n1; captive	1230
	birth: Sept-Jan, one exception June-July; Nepal	6250
	birth: yr rd, more yg seen Aug-Sept; Sri Lanka	7974
	birth: cold weather; India	2802
Tremarctos ornatus	estrus: May-June; zoo	9308
	mate: Aug; zoo	2286
	birth: Jan, Mar; n2; zoo	1020
	birth: mid Jan; zoo	9308
	birth: end Dec-mid Mar, peak Jan; zoo	2286
Ursus americanus	spermatogenesis: Apr-mid Sept; MI	3060
	estrus: May-Aug; Rocky Mt, N America	5437
	mate: Jan-Feb; BC, Canada	2088
	mate: May-June, peak early June; N America	5435
	mate: May-June; zoo	3407
	mate: May-Aug, peak June-July; WA	8623
	mate: June-early July; WI	5210
	mate: June-July; US	11207
	mate: June-July; captive	507
	mate: June-July; PA	205

	mate: mid-end June; AK	4850
	mate: mid June-early July; MN	9227
	mate: mid June-mid July; MI	3060
	mate: end June-early Aug; N America	7652
	mate: early summer; Sierra Madre Occidental, Mexico	6387
	ovulate: May-mid June; Adirondacks, NY	11802
	implant: Nov; US	11207
	implant: mid Nov-early Dec; PA	5961
	implant: early Dec; Adirondacks, NY	11802
	implant: Dec-?; Augusta Co, VA	10414
	birth: early Jan; MI	3060
	birth: Jan; US	11207
	birth: Jan; Adirondacks, NY	7234
	birth: Jan; n32 lt; PA	205, 206
	birth: Jan; Sierra Madre Occidental, Mexico	6387
	birth: Jan-early Feb; WI	5210
	birth: Jan-Feb; BC, Canada	2088
	birth: Jan-Feb; WA	8623
	birth: Jan-Feb; WY	1871a
	birth: mid Jan; ME	10260
	birth: end Jan; n9; captive	507
	birth: end Jan-Feb; Adirondacks, NY	11802
	birth: end Nov-Feb, peak Jan-Feb; N America	5435
	birth: Dec-Mar, peak mid Jan; AK	4850
	yg appear: May; Coahuila, Mexico	522
	wean: Oct; AK	8623
Ursus arctos	sperm: end May-Aug; YUK, Canada	8289
	testes: large May, small Oct-Nov; AK	3059
	estrus: Apr-July; zoo	2612
	estrus: Apr-Aug; Europe	8741
	mate: Apr-May; zoo	2276
	mate: Apr-June; Alps, Europe	3571
	mate: May; Djunegarsky Altau, USSR	3986
	mate: May-early June; captive, Japan	10928
	mate: May-June; n, ne European USSR	4645
	mate: May-June; Canada	7615
	mate: May-June; ne Siberia	5824
	mate: peak May-June; AK	542
	mate: May-July; US	1474
	mate: May-July, some to Nov; AK	3059
	mate: May-Aug; Poland, Rumania	6469
	mate: mid May-mid July, peak early-mid June; WY	2106
	mate: mid May-mid July; WY	1871a
	mate: mid May-mid July; Tschita, USSR	4643
	mate: end May-June; s central AK	542
	mate: June-July; USSR	7979
	mate: mid June-mid Aug; Europe, central Asia	7652
	implant: Nov?; zoo	2612
	birth: early-mid Jan; USSR	4643
	birth: Jan; US	1474
	birth: Jan; St Petersburg, Russia	7980
	birth: Jan; zoo	2611
	birth: Jan; zoo	11330
	birth: Jan-Feb; Cerro Campana, Mexico	6387
	birth: Jan-Feb; zoo	2276
	birth: Jan-Mar; Altai Mt, USSR	10499
	birth: mid Jan; n, ne European USSR	4645
	birth: mid-end Jan; Djunegarsky Altau, USSR	3986
	birth: end Jan-early Feb; IL, WI	2060
	birth: June-July; upper Yang-tse-kiang, China	3034
	birth: July; few data; zoo	1230
	birth: Dec-Feb; Europe	8741
	birth: Dec/Jan-Feb; Alps, Europe	3571
	birth: mid Dec-early Mar, peak Jan; zoo	2611, 2612

Ursus maritimus	spermatogenesis: Feb-May poss June; Arctic	2475
	mate: mid Feb-mid May; zoo	11327
	mate: Mar; arctic	12057
	mate: Mar-Apr; zoo	5970
	mate: Mar-Apr; USSR	7979
	mate: Mar-Apr; zoo	9733
	mate: Apr; n Canada	4375
	mate: Apr-May; zoo	3179
	mate: Apr/May-?; N America	5435
	birth: ~Jan-Feb; N America	5435
	birth: Jan-Feb+; USSR	7979
	birth: Nov-Dec; zoo, USSR	7979
	birth: Jan-Apr; arctic	4643
	birth: Jan-May; arctic	12057
	birth: Feb-Mar; Finland, Greenland	2594
	birth: end Jan-May; Siberia	10499
	birth: Nov; few data; zoo	2109
	birth: Nov; zoo	11327, 11328
	birth: Nov-early Dec; n10; zoo	12128
	birth: Nov-Dec; zoo	9733
	birth: Nov-Jan; n Canada	11049
	birth: mid Nov-Dec; zoo	2594
	birth: end Nov-early Dec; n Canada	4375
	birth: end Nov-Jan; se Baffin I, Canada	10433, 10434
	birth: Dec; zoo	3179
	birth: Dec; zoo	5970
	birth: end Dec-Jan; MAN, Canada	8899
	wean: July; circumpolar	4375
Ursus thibetanus	den emergence: Apr-mid May; USSR	4643
	mate: June-July; Primocsky, USSR	1310
	mate: June-July; Siberia	10499
	mate: mid June-mid Aug; USSR	4643
	mate: Oct; Pakistan	9175
	mate: fall; Nepal, s Tibet	6654
	birth: mid Jan-mid Feb; USSR	4643
	birth: Feb; Pakistan	9175
	birth: May; few data; zoo	1230
	birth: end Dec-Mar, peak Jan-Feb; USSR	7979
	birth: spring; Nepal, s Tibet	6653, 6654
	birth: end Dec-Mar, peak mid-Feb; Siberia	10499

Order Carnivora, Family Procyonidae

Raccoons and their allies (*Ailurus* 3-6 kg; *Bassaricyon* 940-1700 g; *Bassariscus* 700-1400 g; *Nasua* 3-6 kg; *Nasuella*, *Potos* 1.4-4.5 kg; *Procyon* 2-12 kg) are semiarboreal to arboreal, generally nocturnal (except *Nasua*) carnivores or omnivores primarily from the New World, although red pandas, *Ailurus*, inhabit the Himalayas. Female *Bassaricyon* are larger than males, but the reverse is recorded for *Nasua* and *Procyon* (5580, 7979, 8628, 9155b).

Ovulation may be induced by copulation (*Potos* 8798) or by the presence of males (*Procyon* 7534; cf 9467). Long copulations (0.5-1 h) suggest a copulatory lock (8625, 9174, 9549, 10297; cf 3870, 5580, 8628). Transuterine migration occurs (6514a). Latitudinal variation in gestation length suggests delayed implantation in *Ailurus* (9174). The 9-15 g (for the smaller genera), deciduate, zonary or discoidal, labyrinthine, hemoendothelial or endotheliochorial placenta with hemophagous organs has an antimesometrial implantation into a bicornuate or bipartite uterus (932, 2125, 2126, 7563, 8628, 8632, 9467, 9707, 11483) and is eaten after birth (8624, 8628, 8632, 10877). A postpartum estrus is not common, but may occur at 10-16 d in *Procyon* (8626, 8628, 9467).

Neonates are furred, but blind at birth (2109, 7624, 8624, 8625, 9174). *Ailurus* neonates open their eyes either at 17-18 d (9170, 9549) or 36-41 d (3074, 9051). Eye-opening in *Bassaricyon* is at 21-25 d (8628), in *Bassariscus* is at 21-34 d (1474, 5146, 8633, 9105, 10877), in *Nasua* is at 4-11 d (5580), in *Potos* is at 1-3 wk (900, 8629, 8798), and in *Procyon* is at 14-23 d (1474, 3571, 4271b, 7979). Procyonids have 2 (*Potos* 8798), 4

(*Ailurus* 9549; *Bassariscus* 9105), or 6 (*Bassariscus* 1474; *Procyon* 3571) mammae. Males have a baculum (1473, 6558, 7464, 8624, 9467, 11483). Details of anatomy (9467), birth (9173), photoperiod (962, 963), and reproductive energetics (3796) are available.

NEONATAL MASS		
Ailurus fulgens	f: 73 g; n1	626
	m: 95 g; n1	626
	110-130 g	9173
	118 g	9174
Bassaricyon spp	m: 51 g; n1 some tail eaten	8625
	52.5 g; r49.6-60.2; n8	8628
Bassariscus astutus	14-40 g	8633
	25-30 g	1871a
	37.5 g; sd0.7; r37-38; n2, 2 d	10877
Nasua nasua	100-180 g	5580
Potos flavus	125-200 g	8798
	150 g; n1	900
	173.2 g; r148.7-191.4; n4	8624
	191.4 g; n1	1904
	203 g; r175.4-248; n8	8629
Procyon lotor	75 g	4271b
	80 g	7979
	85 g	1474, 5210
NEONATAL SIZE		
Ailurus fulgens	f: 190 mm; n1	626
	m: 220 mm; n1	626
Bassaricyon spp	HB: 119.3 mm; r112-127; n6	8628
Bassariscus astutus	110.5 g; sd0.7; r110-111; n2, 2 d	10877
Nasua nasua	155-165 mm	5580
Potos flavus	HB: 180.6 mm; r168-208; n4	8624
	HB: 185.5 mm; r165-197; n8	8629
Procyon lotor	CR: ~95 mm; term emb	6513
WEANING MASS		
Bassariscus astutus	258.7 g; sd78.3; r168.3-304.5; n3, 40 d	10877
Procyon lotor	600-900 g; 40-50 d	4271b
	1.4-1.8 kg; 10-12 wk	5210
WEANING SIZE		
Bassariscus astutus	220.7 mm; sd19.3; r198-234; n3, 40 d	10877
LITTER SIZE		
Ailurus fulgens	1.5 yg; r1-2; n2 lt	7618
	1.67; r1-2; n3 lt	12128
	1.7 yg; r1-4; n74	9170, 9174
	mode 2 yg	7577, 7578
	2 yg	2592
	2 yg; n1 lt	5176
	2 yg; n1 lt	11415
	2 yg; n1 lt	626
	2 ea; n3 lt	2285
	2.0 yg; mode 2; r1-4; n39 lt	9165
	2.5 yg; r2-3; n2 lt	9066
Bassaricyon spp	1 yg ea; n12 lt	8625, 8628
Bassariscus astutus	2-4	10721
	2.5 yg; r1-4; n2 f	9105
	3 yg; n1 lt	2109
	3.5 emb; r3-4; n2 f	502
Bassariscus sumichrasti	mode 2	3633
Nasua spp	4.0 yg; r3-5; n3 lt	9066
Nasua nasua	2 ea; n4 lt	12128
	3.5; r3-4; n2 lt	5580
	4 emb; 1 f	4228
	4 emb; n1 f	7624
	4-6	1474
	5 yg ea; n2 f	7118

Potos flavus	1 yg ea; n2 lt	900
	1 emb; n1 f	2348
	1 yg ea; n17 lt	8624, 8629
	1.5 emb; r1-2; n2 f	4228
Procyon cancrivorus	2.5; r2-3; n2 lt	12128
Procyon lotor	1.83 yg; n6 lt	9588
	2-6; max 8	7979
	2; r1-3	12128
	2 emb; n1 f	6298
	2.2 yg; r1-6; n55	2109
	2.3; n11 lt	9155a
	2.46 recent ps	5332a
	2.5 yg; r2-3; n2 lt	9066
	2.5; n101 lt	5332a
	2.6; n10 lt	9155a
	2.6 implantations; se0.1; r1-4; n95	3248
	2.62 emb; se0.11; n21 f	5332a
	2.7; n45 lt	9155a
	2.7; n88 lt	9155a
	2.7 ps; se0.1; r1-6; n115	3248
	2.75 cl; se0.35; n4 f	935a
	2.8 emb; r2-4; n5 f	11892
	2.8; n20 lt	9155a
	2.8; n31 lt	9155a
	2.8 ps; mode 3 75%; n20 f	7975a
	2.9; n12 lt	9155a
	2.9; n25 lt	9155a
	2.87 emb; se0.26; n9 lt	935a
	2.89 ps; se0.21; n26 f	935a
	2.9 ps; r2-5; n26 f	10138
	2.93 ps; r2-5; n14 f	12092
	3 ps; n1 f	7498a
	3 emb; n1 f	9588
	3.15 yg; r2-4; n13 lt	10523a
	3.2 emb; se0.18; r2-5; n17 f	7078
	3.23 ps; sd0.837; r1-5; n74 lt	2808
	3.30 emb; sd0.675; r2-4; n10 f	2808
	3.3; n11 lt	9155a
	3.33 cl; sd0.072; r2-4; n15 f	2808
	3.33; n24 lt	963
	3.4 ps; n69	5466a
	3.5 yg, 4.0 ps; n8 lt	9463
	3.5 yg; r3-4; n2 lt	11892
	3.56 ps; se0.03; n1421 f	9466
	3.58 ps; mode 3; r1-6; est n287	3530
	3.6; n217 lt	9155a
	3.63 ps; r2-4; n19 f	11892
	3.66 ps; mode 3; r1-10; est n200	3530
	3.7; n23 lt	9155a
	3.7; n212 lt	9155a
	3.8; n14 lt	9155a
	3.83 emb; r3-5; n6 f	1525
	4-5 cl	7534
	4 ps; n1 f	10523a
	4.0 ps; r3-5; n2 f	6298
	4 yg; r3-7; n10 dens	10523a
	4.25 emb; r4-5; n4 f	2125
	4.6 ps; r2-7; n5 f	7149a
	4.6 yg; n8 lt	10297
	4.67 yg; r3-6; n3 nests	3723
	4.8 emb, yg; mode 4; r3-7; n25	3528
	4.8; mode 4; r1-5	1230
	4.8; n52 lt	9155a
	mode 5-6; r2-8	3571

SEXUAL MATURITY

Ailurus fulgens	f: 1st conceive: 18 mo	9172
	f: 1st birth: 24-26 mo; n7	9174
	f/m: 18-20 mo; n7	9174
Bassaricyon spp	f: 2 yr	8628
	m: 1.5 yr	8628
Bassariscus astutus	f: 1st birth: ~1 yr	8632
	m: testes descend: 16 wk	8633
Potos flavus	f/m: 1st mate: 3 yr	8626
	f/m: 18-30 mo	8798
Procyon lotor	f: 1st mate: 1 yr 9.3%	2808
	f: 1st mate: 1 yr	9467
	f: 1st preg: yearling	3528
	m: sperm, testes wt: 1 yr	9467
	m: sperm: 1.5 yr	2808
	m: 1st mate: 2 yr	2088

CYCLE LENGTH

Ailurus fulgens	estrus: 18-24 h	9172
	26-44 d	9173
Bassaricyon spp	estrus: 1-3 d; possibly 4	8628
	23.9 d; r19-40; n22 cycles, 3 f	8628
Bassariscus astutus	estrus: 24-36 h	8633
Potos flavus	vulvar swelling: 2 d; r1-3; n4 f	8629
	estrus: 17.6 d; r12-24	8629
	62.8 d; r46-92; n43 cycles, 4 f	8629
Procyon lotor	cornified epithelium: 3.4 d; partially isolated f	7534
	cornified epithelium: 14.3 d; n6 isolated f	7534
	r80-140 d; n5 f	9467

GESTATION

Ailurus fulgens	112 d	2592
	125-132, 155? d; via f behavior	3074
	131 d; sd9.27; r114-145; n9	9172, 9174
	131.5 d; r131-132; n2	7618
	132 d; n1	9051
	134.2 d; sd14.7; r112-158; n17 [literature compilation]	9173
Bassaricyon spp	73.25 d; r72-75; n12	8625, 8628
Bassariscus astutus	51.92 d; r51.33-53.58; n9	8632
	~56 d	1871a
Nasua spp	71 d; n1	3594
Nasua nasua	67-73 d	4582
	10-11 wk; field data	5580
	77 d	1355
Potos flavus	115.5 d; r111.5-118; n2 f; 8 lt	8629
	119-120 d	8626
Procyon lotor	60-63 d	3571
	63 d	4271b
	63 d	9467
	~9 wk	1474, 2088, 5146, 6387, 7979
	65 d; n1	3594
	69 d	1355

LACTATION

Ailurus fulgens	den emergence: 3-4 mo	9174
	solid food: 90 d	9173
	solid food: 125-135 d	9170, 9174
	wean: 4-5 mo	9172, 9174
Bassariscus astutus	solid food: 3 wk	6387
	solid food: 30-40 d	10877
	wean: ~35-50 d	10877
Nasua nasua	den emergence: 3-5 wk	5582
	nurse: 4 mo	5580
Potos flavus	solid food: 35 d	8798
	solid food: 8 wk	8624
	nurse: 2.5-4 mo	8626, 8629

	wean: 16-18 wk	8798
Procyon lotor	den emergence: 6-8 wk	2109
	solid food: ~1 mo	7979
	solid food: 40 d	4271b
	solid food: 9 wk	7481
	wean: 10-12 wk	5210
	wean: 13-17 wk	7480
	nurse: 16 wk	7481
LITTERS/YEAR		
Ailurus fulgens	1	9174
Bassariscus astutus	1	1474
Potos flavus	1	8624
Procyon lotor	1	10297
	1	7234
	max 2/season; only 1 raised	9467
SEASONALITY		
Ailurus fulgens	mate: Jan; zoo	9051
	mate: Jan-Feb; zoo	3074
	mate: Jan-mid Mar; n hemisphere zoos	9172, 9173
	mate: July-Aug; s hemisphere zoos	9173
	birth: May-Aug, peak June 79%; n199; n hemisphere zoos	9173
	birth: June; Himalayas, Asia	9173
	birth: June; zoo	9051
	birth: June; zoo	7578
	birth: June; zoo	3074
	birth: June-Aug, peak June; n39; zoo	9165
	birth: mid June-mid July; zoo	9172
	birth: Dec-Mar; n11; s hemisphere zoos	9173
Bassaricyon spp	birth: Jan-Mar, May-June, Sept, Dec; n12; zoo	8628
Bassariscus astutus	preg f: mid Apr-May; TX	10721
	preg f: May; n1; NM	502
	birth: Apr-July; WY	1871a
	birth: May-June; sw US	1474
	birth: May-June; w US	5146
	birth: June; n1; zoo	2109
	lact f: July; n1; Durango, Mexico	523a
Bassariscus sumichrasti	birth: Mar; Yucatan, Mexico	3633
Nasua nasua	mate: end Jan-mid Mar; Barro Colorado I, Panama	5580
	birth: Mar-June; n1; Chiapas, Mexico	6387
	birth: Apr-June; Barro Colorado I, Panama	5580
	birth: June-July; n2; nw Chihuahua, Mexico	6387
	birth: est July; sw US	1474
	birth: July; n1; Sinaloa, Mexico	6387
	birth: July-Sept; n1; Guerrero, Mexico	6387
Potos flavus	preg f: Apr; Panama	3398
	preg f: Sept; Suriname	3398
	preg f: Dec-Jan; n2; Veracruz, Mexico	4228
	birth: Jan-Feb, Aug-Sept; captive (literature compilation)	3398
	birth: est yr rd; zoo	8624, 8629
Procyon lotor	no sperm: July-Oct; IL	9467
	sperm: Oct-Apr; MD	2808
	testes large: peak Feb-Mar; AL	5332a
	estrus: Feb; Europe	7979
	mate: Jan-early Apr, peak mid Feb; IL	9467
	mate: peak end Jan-early Feb; NY	4271b
	mate: peak end Jan-mid Mar; WI	5210
	mate: end Jan-Mar; lab	963
	mate: peak Feb; PA	9164
	mate: Feb-Mar; n US; earlier; s US	1474
	mate: Feb-Mar; Germany	3571
	mate: Feb-June, peak Feb; MD	2808
	mate: peak Mar; GA, FL	7078
	mate: Dec-June, peak Feb; KS	10297
	mate: mid Dec-early Apr; IL	9467

mate: yr rd; zoo	1230
conceive: Feb-mid June; n18; AL	5332a
preg f: Apr-Sept; n9; s FL	935a
birth: Jan-Apr; IL	9467
birth: Mar; MI	3723
birth: Mar-May; MI	6587
birth: Mar-July; zoo	2109
birth: Apr-early May; KS	10297
birth: peak Apr-May; WI	5210
birth: Apr-May; IL, WI	2060
birth: Apr-May; Europe	3571
birth: Apr-May; ND	3528
birth: Apr-June, Oct; n6; zoo	12128
birth: Apr-Aug, peak Apr; MD	2808
birth: peak mid Apr; IL	9467
birth: mid Apr; Adirondacks, NY	7234
birth: end Apr-early May; PA	9164
birth: end Apr-early May; Europe	7979
birth: peak May; GA, FL	7078
birth: May-Aug; AL	6587
birth: Aug; n1; WV	786
birth: est Sept; as 3 mo animals killed Dec; IN	6363
birth: Dec-Apr; w US	5146
lact f: June; n1; SD	261a
lact f: July; NE	5405b,

Order Carnivora, Family Mustelidae

Mustelids (*Aonyx* 1-34 kg; *Arctonyx* 7-14 kg; *Conepatus* 0.9-4.5 kg; *Eira* 4-6 kg; *Enhydra* 15-45 kg; *Galictis* 1 kg; *Gulo* 7-35 kg; *Ictonyx* 420-1400 g; *Lutra* 3-16 kg; *Lyncodon* 300-350 mm; *Martes* 0.6-5.5 kg; *Meles* 6.5-34 kg; *Mellivora* 7-14 kg; *Melogale* 1-3 kg; *Mephitis* 700-6300 g; *Mustela* 25-1600 g; *Mydaus* 1.3-3.6 kg; *Poecilictis* 200-250 g; *Poecilogale* 115-400 g; *Pteronura* 22-34 kg; *Spilogale* 200-1000 g; *Taxidea* 4-12 kg; *Vormela* 370-715 g) are a diverse group of terrestrial to semiaquatic, primarily carnivorous inhabitants of the New World, Eurasia, and Africa. Males are often larger than females (190, 1304, 2068, 2421, 3076, 3797, 3802, 4225, 4269, 4285, 4576, 5210, 5583, 5631, 5678, 5681, 5761, 6927, 7195, 7250, 7599, 8495, 8501, 8636, 8715, 9327, 10499; cf 8637).

Ovulation is triggered by lengthy coitus (ovulation: 834, 958, 1632, 1633, 2422, 2687, 2689, 3026, 3028, 4299, 4300, 4340, 6156, 7655, 7975, 8276, 9391, 10189, 11277, 11381, 11382; cf 7126; duration of coitus: 958, 2687, 4299, 4300, 4340, 4391, 4458, 4643, 5681, 6089, 6725, 8276, 8715, 9326, 9610 8627, 10499). In *Martes* development may be slow (5966). Implantation may also be delayed (1578, 11945; *Martes* 487, 1580, 1586, 3030, 5438, 6385; *Meles* 1102, 1103, 1104, 1572, 1573, 1574, 1575, 1576, 1577, 1579, 1581, 1584, 1585, 2822, 4416, 4429, 6160, 6167, 7457, 7460, 7975, 9502; *Mephitis* 11383; *Mustela* 489, 2255, 2421, 3028, 3406, 4340, 5325, 5326, 6906, 7132, 7134, 7135, 7637, 7638, 7640, 7643, 7643a, 7644, 8522, 9271, 9460, 9880, 9881, 10469, 11946; *Taxidea* 3245). Some *Spilogale* subspecies have delayed implantation, while others do not (840, 3019, 3185, 3407a, 3408, 3410, 7126, 7127, 7129, 7129a, 7130, 7132a, 7133, 7140, 7141, 8969, 8970, 10002, 10003). Precocious implantation in *Mustela vison* is induced experimentally by a) transfer from natural daylight to long photoperiods (6907); b) administration of prolaction which stimulates secretion of progesterone by the corpus luteum (6905, 7641, 7642, 8226a); c) administration of pimozide which stimulates prolactin secretion by the hypophysis (7639); and d) the administration of progestagens (2018, 5264; cf 4303, 7638).

The labyrinthine, endotheliochorial, zonary or bidiscoid placenta has a mesometrial or antimesometrial, superficial implantation into a bipartite or bicornuate uterus (1412, 2126, 2789, 3006, 3026, 4261, 4340, 4576, 7126, 7563, 7655, 8303, 8715, 10000, 10003, 10004, 10532, 11277, 11857, 11950) and may be eaten after birth (2421, 4269, 4689, 7655, 7709, 8627, 9822). A postpartum estrus occurs (1104, 1474, 1572, 1573, 1577, 1578, 2088, 4225, 4285, 4299, 4416, 4643, 6047, 6559, 7979; cf 1578, 1582, 6559, 7975, 9028).

Most mustelids are blind and naked or sparsely furred at birth (334, 576, 1240, 3416, 5146, 5210, 5761, 6385, 6387, 6559, 7234, 8276, 9325, 11277; cf 9610). Their eyes open at 20-30 d (*Conepatus* 8798), 3-5 wk (*Mephitis* 10330, 11277), ~4 wk (*Aonyx* 5761; *Mellivora* 296, 6559), 4-6 wk (*Meles* 3571, 4643, 6559, 7813,

7975, 7979, 10499; *Mustela* 2862, 3571, 4269, 4458, 4576, 4689, 7598, 7599, 8769, 9344, 10499, 10571, 11321; cf 1304, 4645), 4-7 wk (*Martes* 410, 1240, 2068, 3571, 4225, 7137, 10499), 41-48 d (*Arctonyx* 8232; *Ictonyx* 9326, 9327, 10158), 51-54 d (*Poecilogale* 9326, 9327, 10158), or 62 d (*Eira* 8631). Other neonates have fur, but do not open their eyes until 4 wk (*Pteronura* 465, 2812; *Spilogale* 2100, 10758) or 4-5 wk (*Gulo* 1474, 4643, 6046; *Lutra* 5242, 6927, 8318, 10499; cf 1941, 7979, 9175; *Taxidea* 5210, 6559, 7250). *Enhydra* neonates are fully developed with fur and open eyes (565, 5678, 5681, 8303, 9589, 10499).

Females have 4 mammae (*Aonyx* 10158; *Ictonyx* 10158; *Martes* 1474, 2068, 3571, 9175; cf 1474; *Mellivora* 6559, 9945, 10158), 5 mammae (*Lyncodon* 8618), 6 mammae (*Arctonyx* 6559; *Conepatus* 502; *Meles* 3571; *Mydaus* 6559), 4-6 mammae (*Lutra* 1474, 3571, 4643, 5277, 9175, 10158; *Poecilogale* 332, 4744, 9325, 10158), 6-10 mammae (*Mustela* 502, 1474, 3571, 4269, 4643, 5210, 7655), 8 mammae (*Spilogale* 502, 1474; *Taxidea* 502, 1119, 1474), or 10-15 mammae (*Mephitis* 502, 1474, 3134, 6387). Males have a baculum (636, 1473, 2068, 4576, 4643, 4776, 7128, 7979, 10499, 11277).

For *Mustela*, details of anatomy (1517, 2963, 3026, 5945, 6734, 6849, 11497), endocrinology (161, 664, 699, 995, 996, 1049, 1053, 1054, 1413, 1632, 1633, 2018, 2424, 2425, 2686, 2687, 2690, 2691, 2692, 3083, 3084, 3409, 3473, 4143, 4144, 4145, 4527, 4775, 5769, 5839, 6905, 6912, 7085, 7113, 7136, 7137, 7142, 7448, 7451, 7645, 7798, 7799, 8966, 9127, 9393, 9394, 9395, 9879, 10010, 10011, 10228, 10855, 11716a, 11961, 11962), spermatogenesis (1049, 1129, 5317, 5971, 7799, 8338, 9123, 10815a, 10815b, 11439), semen (9958), mating (6540, 7316a), conception (59, 1694, 4300), fertility (3777, 7635, 7636, 10553, 10556, 10557), sex differentiation (662, 3083, 3085, 6883, 10854), temperature (7654), light (454, 665, 959, 960, 961, 1050, 1051, 1646, 2688, 2762, 4302, 4651, 4652, 4653, 4777, 4911, 5046, 5977, 6312, 6851, 6906, 6907, 9391, 9392, 9393, 9394, 9396, 10800, 10812), milk composition (1999, 7580, 9635), and reproductive energetics (3172, 11448a) are available.

For other genera details of anatomy (*Enhydra* 10000; *Lutra* 9584; *Meles* 1108; *Taxidea* 11950), endocrinology (*Gulo* 7138; *Martes* 444, 445, 488, 834, 1101, 1586, 9955; *Meles* 100, 446, 447, 448, 1103, 1587, 3264, 6981, 6983, 6985, 6986, 6987, 7457, 7460; *Mephitis* 4194, 11277; *Spilogale* 7133, 8970; *Taxidea* 3245), oogenesis (1517), ovulation (4051), copulation (4391), milk composition (5297), light (*Martes* 756, 3030; *Meles* 1574, 1576, 6984), and reproductive energetics (4381, 8716) are available. One journal, Lutra, focuses on mustelids.

NEONATAL MASS

Aonyx cinerea	m: 57 g; n1	6393
Arctonyx collaris	58 g	8232
Eira barbara	m: 77.95 g; r74.4-81.5; n2	8631
	f: 85.05 g; r78.0-92.1; n2	8631
Enhydra lutris	1300 g	1474
	1500 g; r1400-1600	4643, 7979
	m: 1750 g; r1200-2500; n4	5677, 5678, 5681
	f: 1960 g; r1020-2830; n6	5677, 5678, 5681
	2000 g; n1	565
	2000 g; r1600-2500; n3	9589
	2000 g; r1700-2400; n3	7478
	3000 g; n1	3617
Gulo gulo	m: 73 g; near term	4643
	f: 83 g; near term	4643
	84 g; n3	7171
	88.1 g; r84.5-94.1; n3, 1 d	9916a
	107.7 g; n4; term emb	548a
Ictonyx striatus	10.0-15.0 g	10158
Lutra canadensis	132 g; n1, near term	4286
	r140-145 g; n4 near term	4739
Martes americana	28 g; n1	1240
	30-40 g	5210
Martes martes	30 g	9610
Martes pennanti	f: 28.9 g; r26.0-30.7; n3, n1 lt	6385
	m: 30.0 g; n1	6385
	m: 39.0 g; n1, 3 d	8715
	f: 41.2 g; n1, 3 d	8715

Martes zibellina	30 g; r25-34; (34 g singletons; 25 g ltsz >1)	4643
	30 g	9610
	30-35 g	7979, 8276
Meles meles	50-132 g	6559
	75 g	10499
	75 g; r63-84	4643, 7979
	89.5 g; r85-94; n2	11255
Mellivora capensis	210 g; n1	6559
Mephitis mephitis	32-35 g	11384
	33.4 g; r26.7-41.4; n12	11277
Mustela erminea	1.7 g; n6, 1 d	4269
	3 g; r2.7-4.2	5741
Mustela eversmanni	4-6 g	10499
	4.5 g	4643
	~8.4 g	7137
Mustela frenata	3.1 g; n6, 1 lt, 1 d	4269
Mustela lutreola	6.5 g	4643
	f: 7.6 g	12050
	m: 8.4 g	12050
Mustela nivalis	1.42 g; r1.09-1.70; n8	4576
	1.5 g	4458
	3.0 g; r2.0-4.0; n14, 1 d	2863
	4.5 g; n6 (1 m, 5 f)	2862
Mustela putorius	7 g	4643
	9-10 g	3571
	~10 g	9344
	10 g ea; n2	4689
Mustela vison	6.2 g; r5.9-6.5; n2	10571
	6-11 g	4643
	9-11 g	10499
Poecilogale albinucha	4 g	10158
Pteronura brasiliensis	m: 189 g; r161-213; n3	465
	f: 219.5 g; r209-230; n2	465
Spilogale putorius	9.5 g; n1 (alcohol)	2100
Spilogale pygmaea	6.9 g	10758
Taxidea taxus	93.5 g; r89.5-97.5; n2	6559
NEONATAL SIZE		
Eira barbara	f: 137 mm; n1	8631
	m: 137 mm; r134-140; n2	8631
Enhydra lutris	380 mm	4643
	500-550 mm	7979
	500-560 mm	4643
Gulo gulo	CR: 112.5 mm; n4 term emb	548a
	CR: 121 mm [text says cm]; n3	7171
	122-125 mm; a few days before birth	4643
	HB: 157.3 mm; r155-160; n3, 1 d	9916a
Lutra canadensis	CR: 136 mm; n4 near term	4739
	~270 mm	1871a
Martes zibellina	110-120 mm	8276
	110-120 mm	4643
	110-120 mm	7979
Meles meles	max 130 mm	10499
Mellivora capensis	100 mm	296
Mephitis mephitis	94.2 mm; n5, 1 lt	11277
Mustela eversmanni	65-70 mm	4643
	65-70 mm	10499
	80 mm	11321
Mustela frenata	52 mm; n1, 1 d	4269
	~56 mm	1871a
Mustela lutreola	TL: f: 71.9 mm	12050
	TL: m: 73.9 mm	12050
Mustela nivalis	42 mm; r38-43; n8	4576
	HB: 50 mm; n2	2863
	60 mm; n1, 4 d	2862

Mustela putorius	53 mm; r51-55	4689
	70 mm	4643
Mustela vison	50-90 mm	10499
Pteronura brasiliensis	m: 327 mm; r310-335; n3	465
	f: 330 mm; r325-335; n2	465
Spilogale pygmaea	HB: 63 mm	10758
WEANING MASS		
Eira barbara	1290 g; r1134-1146; n2, 60 d	8630
	2125 g; r2100-2150; n2, 90 d	8630
Enhydra lutris	est 6000-15000 g	3617
	10000-13600 g	5681
	f: est 15200 g	7478
	m: est 16200 g	7478
Martes americana	327 g; n3	1240
	435-480 g	6821
Martes martes	680 g	9610
Martes zibellina	600 g	9610
Mustela erminea	f: 28.2 g; n1, 35 d	4269
	m: 32.5 g; r30-34.7; n4, 35 d	4269
Mustela frenata	f: 30.5 g; 4 wk	4269
	m: 38.7 g; 4 wk	4269
Mustela nivalis	20 g; 3 wk; from graph	4576
	f: ~50 g; 8 wk; from graph	4576
	m: ~60 g; 8 wk; from graph	4576
Taxidea taxus	half grown	6559
WEANING SIZE		
Martes zibellina	390-400 mm; 2 mo	4643
LITTER SIZE		
Aonyx capensis	mode 2 yg	11507b
	mode 2-3	9327
	2-3	9945
	2-5	5761
	max 4	10153
Aonyx cinerea	1.25; r1-2; n4 lt, 1 f	6393
Arctonyx collaris	3 yg; r1-4; n3 lt	6559
	3 yg; r2-4; n2 lt	8232
Conepatus leuconotus	2-4	1474
Conepatus mesoleucus	mode 3-5; r2-10	8798
	3-6	190
Eira barbara	1.75; mode 2; r1-2; n4 lt	8627, 8631
Enhydra lutris	1 yg	7634
	1; est max twins	5246
	1 yg; 2 emb observed once	565
	1; 2 emb seen; never 2 yg	5681
	1 emb; n1 f	8303
	1 emb ea; n11 f	10004
	1 ea; n278 lt	5677, 5678, 10000
	2% twin emb, 4% twin ovulations	3098
Gulo gulo	mode 2-3; r1-5	10499
	2; r2-3; n7	4643
	2.00 blastocysts; se0.19; n18 f	6505
	2.2; r2-3; n13 lt	6046
	2.2; r2-3; n15 lt	8793a
	2.5; r1-4; n38 lt	8793a
	2.5 yg; r1-4; n58 lt	8793a
	2.6; r2-3	8792
	2.60 emb; se0.24; n5 f	6505
	2.6; r1-4; n45 lt	8793a
	2.78 cl; se0.18; n14 f	6505
	2.9; mode 2-3; r1-4; n33 lt	12091
	3 yg; n1 lt	7171
	3-4	11949
	3.2 emb; sd0.8d; n23 f	548a
	3.3 ps; sd1.2; n18 f	548a

	3.5 yg; r1-6	1871a
	3.9 cl; sd1.0; n23 f	548a
	4; n1 lt	4201
	5 emb; n1 f	8793a
Ictonyx striatus	1-3	9326, 9327
	1-3 wild; 3-5 captive	10153, 10158
	2-3 yg	4354a
Lutra canadensis	2; r1-5	1474
	2-4 yg; r1-6	1871a
	2.22 emb; r1-3; n9	4286
	2.68 emb; ±0.71	4739
	2.875 emb; r2-4; n8 f	11763
	2.92 cl; ±0.79; n60 f; 1.9 blastocysts; n7 f	4739
	mode 3; r1-5	2088
	3 yg; n2 nests	11763
	3.0 yg; r2-4; n3 lt	2109
	3.43 yg; r2-4; n7 lt	6442
	4 ?; young with f, but some mixing of pups	7195
Lutra felina	mode 2 or 4-5	10809
Lutra longicaudis	4 emb; n1 f	4228
Lutra lutra	r1-2 weaned; n13 lt	5286
	1.5; mode 1; r1-3; n4 lt; n6	8318
	1.78 yg; sd0.57; mode 2; r1-3	6075
	2; n2 lt	12128
	2.0 yg; sd0.6; r1-3; n19 f	4569
	2.1 emb, implantations; sd0.7; r1-3; n17 f	4569
	2.1; mode 2; r1-5; n160 lt	6927
	2.2 cl; sd0.8; r1-3; n14 f	4569
	2.3; mode 2; r1-4; n70 lt	10513, 10514
	2.8; r2-5; n48 lt	11206
Lutra maculicollis	1-2 yg; n2	9327
	mode 2; r1-3	10158
	2-3	5277
Lutra perspicillata	5; n1	9175
Martes americana	1-3 cl	6863
	2-3 or 4 yg; n3 lt	11402
	2.2 yg; r1-3; n5 lt	1240
	2.6 yg; r1-4; n20	6821
	2.85 yg; r1-5	1869
	3 yg ea; n3 lt	410, 411
	3.02 cl; r2-5	11949
Martes flavigula	2-3	4643
	2-3; sometimes 5	10499
	2-3	9175
	max 5	7979
Martes foina	2-3	9175
	2-6	4645
	3-4; r1-8	7979
	mode 3-4; max 8	10499
	3-5	7652
	mode 3-5; r2-7	3571
	3-7; 2 occasional	4643
Martes martes	2-5	3571
	2; n1 lt	1942
	3; r1-4	9610
	3; r2-4	4643
	3-4; max 8	7979
	mode 3-4; r2-7	4643
	3-5	7652
	3-5	4645
	mode 3-5; r2-8	10499
	3.8; r3-5	4643
	5; r3-7	4643

Martes pennanti	1-5	2060
	mode 2-3; r1-5	2088
	mode 3; most 2-5; r1-6	8714, 8715
	2 yg; max 5; r1-4	5146
	2.66; r1-4; n2 lt + blastocysts counts of 7 f	4285
	2.7 yg; mode 3; r1-4; n26 lt	4225
	2.72 cl; mode 3; r2-4; n23 f	2852
	3-4	11949
	3-5 cl	3031
	3.28 cl; n44 f	2068, 11953
	3.45 emb; mode 3; r3-4; n11 lt	2068, 11953
	3.5 cl; mode 3; n16 f	6385
	3.67 cl; ±0.26; n12 f	5631
	4; n1	8716
Martes zibellina	mode 3-4; r1-10	8276
	2.8-3.0; across years	9610
	3.2 emb; n153 f	7987
	3.2; r1-5; n116 lt	756
	3.4 emb; n14 f	675
	3.49 cl	834
Meles meles	1-5	1572
	2 yg; n1 lt	11255
	2; mode 2; r1-4	12128
	2.23 cl; r2-11; n43 lt	4429
	3-4 cl; r1-10	89
	3.6; r2-7; n16 lt	11205
	4.7 cl; 3.1 emb	11444
Mellivora capensis	1-4	5761
	2	9175
	2	10158
	2	9945
	2-4	11253
	2; n1 lt	1230
	mode 2; r1-4; n3 lt	6559
Melogale personata	2 emb; n1	5403
	3 yg	6369
Mephitis macroura	3 emb; n1 f	2348
	5 emb; n1 f	502
	5 emb; n1 f	1474
	5; r3-8	6387
Mephitis mephitis	2.5 yg; se0.4; n6 dens	966
	3 yg; n1 f	9822
	3.3 yg; r2-5; n3 lt	9066
	3.5; r3-4; n2 lt	9821
	4.28 yg; sd2.23; r1-10; n118 lt, 67 f	11382
	5-19 cl	11381
	5 yg; n1 lt	576
	5.2 yg; se0.8; n9 dens	966
	5.7 yg; r4-7; n3 lt	502
	6 yg ea; n2 f	10330
	6; r2-14	9164
	6.0 ps; se0.5; n15 f	966
	6.0 emb, ps; se0.33; n3-10; n25	3557
	6.3 cl; n116 f	9644
	6.45 cl, emb, yg; n11 f	498
	6.7 ps; n274 f	9644
	6.8 emb; n87 f	9644
	7.15 ps; se0.28; r4-11; n39 f	11277
	7.33 emb; r5-9; 6.33 live emb; r0-5; n21 f	11277
	8.0; r5-9; n3 lt	8245
Mustela altaica	2-8	10499
	5.5; adult f; 3.6; young f	4643
	7-8	7979

Mustela erminea	3-7	3571
	3-9 emb	2418
	4-8	1474
	mode 4-8; r2-13	4645
	mode 4-8; r2-18	4643
	4-13	7652
	4-13	5146
	4; n1 lt	8769
	4 emb; n1 f	502
	5; n1 lt	7598
	5-10 weaned	9175
	7 yg; n1 f	4269
	7 blastocysts; n1 lt	4283
	8-9; r3-14; max 18	7979
	9.7; r9-10; n3 lt	10512
	9.7 cl; 8.8 emb; r6-13	5744
	11 emb; n1 f	4073
	max 19 ova	5741
Mustela eversmanni	4-14; mode 8-9	10499
	6.8 yg; r4-10; primiparous f	7137
	8-11; max 18	4645
	9.5; r3-18	4643
	13.25; n4 lt	11321
	8-11; max 18	7979
Mustela frenata	3-9 cl	11946
	4 yg; n1 f	4224
	6-9 yg	11948
	mode 6; r3-8	6387
	6.5 yg; r6-7; n2 lt	4269
	9 yg; n1 nest	576
Mustela lutreola	mode 4-5; r2-7	3571, 4643, 4645, 7979, 10499
Mustela nigripes	2.4 wean; sd1.7; r1-6; n62	11716
	3.0 yg; sd1.4; r1-6; n62	11716
	3.3 yg; sd0.89; mode 3; r1-5; n68 lt; over 4 yr	3416
	3.33 yg; r1-5; n3 f	9876
	3.43 yg; n21 f	327, 328
	3.5; r1-5; n11 f	4802
	4; r3-5	1867
Mustela nivalis	4; n1 lt	1304
	4.4; r4-5; n8 lt	1010
	4.5; r3-6; n4 lt	4458
	4.7 yg; r2-7; n3 lt	2863
	4.7; mode 6; r1-6; n9 lt	4576
	5.0; mode 4; r3-10; n7 lt	4227
	5.2 emb; sd1.25; r4-8; 4.9 live; sd1.49; n14 f	5277a
	5.5 emb, 5.0 alive; n2 f	580
	5.6 emb; n12 f	5740
	6; n1 lt	2862
	6; n1 lt	4457
	6; r4-12	10512
	6.75 yg; r5-9; n4 lt	8636
	7.1; r4-11	2422
	9; n1 lt	8637
Mustela nudipes	4; n1 lt	7161
Mustela putorius	5-11 cl	1633
	5-13	4299
	5.7; r2-12; n13 lt	1230
	5.83; r2-8; n6 lt	7652, 7654
	7; n1 lt	4689
	8 implantations; se1.5; n4 lt, 20 d postcoitus	7637
	8-9	6156
	9	9028
	9.3; r7-12; n3 lt	7709

	9.86 cl; mode 7-8; r6-10	7655
	13.5; se1.3	7135
Mustela sibirica	2-12	7979, 10499
	7; r4-10	4643, 4645
Mustela vison	5 blastocysts; n1 f	161
	4.0 implantations; r1-7; n2 f	161
	1.20-4.32; 3 wk; across strains	2778
	2.29-4.70; across strains	4911
	3.04; n84 lt	3473
	3.8; r3-6; n10 lt	3735
	3.87 yg; se0.6; n15 young f	8522
	4.2; mated before Mar 21	5133
	4.25 yg; n697	1166
	4.31-5.38; n295 lt	5325
	4.40; se0.5; r3-6; n5 adult	2018
	4.48 yg; n>600 lt	3026
	4.62; n2032 lt	5954
	4.74; se0.17; n158	7636
	4.8 yg; ±2.6; n10 f, 10 lt	5264
	5.16 mated after Mar 21	5133
	5.20; r1-8; n5 young f	2018
	5.3 yg; se0.9; n10 adult f	8522
	5.34; se0.09	7638
	5.47; se0.09	7638
	5.5; se0.1	7639
	5.7 emb; r5-7; n3 f	161
	6 emb; n1 f	1290
	6.17; r4-12; n30 lt	57
	9.8 cl; n50	3026
	max 17 rare	7979, 10499
Poecilictis libyca	1-3 yg	4354a
	2 yg; n1 lt	8427
Poecilogale albinucha	1-3	9326, 9327
	2	11253
	2 yg	4744
	2-3 yg	10153, 10158
Pteronura brasiliensis	1 yg; n1 lt	10891
	r1-3 yg; n19 sightings	2812
	2.75 yg; r2-5; n4 lt	465
Spilogale putorius	1-4 blastocysts	10003
	5.3; r4-7; n3 lt	2100, 2101
	5.5	7126
Spilogale pygmaea	4.0; r2,6 yg; n2 lt	10758
Taxidea taxus	mode 2-3; r1-7	6559
	2; max 5	5146
	2-5	1474
	2 emb; n1 f	4249
	3 emb; n1 f	1119
	4-5 yg	502
Vormela peregusna	4-8	7979
	4-8	10499
	4-8	4645
	4.3; r3-8; n13	4643
SEXUAL MATURITY		
Aonyx congica	f/m: ~1 yr	7983
Conepatus mesoleucus	f/m: 1st mate: ~10-11 mo	8798
Eira barbara	f: 1st estrus: 23 mo; n1	8630
	m: testes: descend 6 mo	5583
Enhydra lutris	f: 1st mate: 2 yr	7979
	m: 5-6 yg	3098
	f/m: 2 yr	4643
Gulo gulo	f: 1st mate: 1 yr	11949
	f: 12-15 mo	1871a
	f: 2 yr	6505

	f: 4 yr	6046
	m: >3 yr	6505
	f/m: 1-2 yr	6048
	f/m: 2 yr	10499
Ictonyx striatus	f: 1st birth: 10 mo	9326
	m: testes: scrotal 16-20 wk	9326
	m: 1st mate: 22 mo	9326
Lutra canadensis	f: 1st birth: 2 yr; n1 f	6443
	f/m: 1st mate: 24 mo	7195
	f/m: 2nd yr	857
	f/m: ~2 yr; max 3 yr	8798
Lutra lutra	f: 1 yr	10499
	m: 1st sire; 18 mo	11502
	m: scrotal sac length: 20-22 mo; n1	5287
	m: 2 yr	10499
	f/m: 1st mate: 3 yr (some earlier)	6927
	f/m: 2 yr; wild; 1 yr	4643
	f/m: 2-3 yr, some 1 yr	3571
Martes americana	f: 1st mate: 2 yr	5146
	f: 1st mate: 2 yr	11949
	f: 1st mate: not as yearling	5438
	f: 1st birth: 1st yr	1474
	m: 1st mate: 1 yr	5438
	f/m: 1st mate: 15 mo	6821
Martes foina	f: 1st birth: 1 yr	4645
	f/m: ~2 yr	3571
Martes martes	f: 1st mate: 456 d; n1	6047
	m: 1st mate: 14 mo	6047
	f/m: 2 yr, 3 mo	7979
	f/m: 2-3 yr	3571
Martes pennanti	f: 1st mate: 1 yr	2068
	f: 1st mate: 1 yr; n1	4225
	f: 1st birth: 2 yr	2068
	f: 1st birth: 2 yr	2088
	m: sperm: 1 yr	2068
	m: 2 yr	8714
	f/m: 1st mate: 1 yr	11949
	f/m: 1st mate: 1 yr	6385
	f/m: 1 yr	4285
	f/m: 1-2 yr	1871a
Martes zibellina	f: 1st mate: 18 mo (20%)	9610
	f: 1st preg: 26-33% 1 yr	4643
	f/m: 1st mate: 15 mo	9955
	f/m: ~15 mo	8276
	f/m: 15-16 mo	7979
	f/m: 15-18 mo	488
Meles meles	f: 1st mate: 12-17 mo	4416
	f: 1st ovulate: 13 mo	89
	f: 1st birth: 1 yr	4645
	m: spermatogenesis: 1 yr	89
	m: spermatogenesis: 2 yr	10419
	m: spermatogenesis: > 2 yr	4416, 4429
	m: 1st mate: 1 yr	4645
	m: ~15 mo	3988
	f/m: 1 yr	7979
	f/m: 1 yr	10499
	f/m: ~1.5 yr	3571
	f/m: 2 yr	4643
Mephitis mephitis	f: 1st mate: 1 yr	11277
	f/m: 10 mo	1871a
Mustela erminea	f: 1st mate: 17-75 d; n29	10752
	f: 1st mate: 6-8 wk; before weaning	11564
	f: 1st mate: 3 mo; n1	7598
	f: conceive: 5-6 wk	7599

	f: 1st preg: 3 mo	4283
	f: 3 mo	10512
	f: 3-4 mo	1871a
	m: spermatogenesis: 1 yr	4144, 11564
	m: 1 yr	4283
Mustela eversmanni	f: 1st mate: 10 mo	7137
	f: 1st birth: 1 yr	7137
	f/m: ~9 mo	10499
	f/m: ~10 mo	7979
	f/m: 10 mo; completely developed 2 yr	4643
	f/m: 1 yr	4645
Mustela frenata	f: 1st mate: 3-4 mo	11945, 11948
	m: 1st mate: 1 yr	11947, 11948
Mustela lutreola	f: 9-10 mo	12050
	f/m: ~9 mo	3571
	f/m: 1 yr	7979
	f/m: next spring	4645
	f/m: yr after birth	4643
Mustela nigripes	f/m: 1 yr	3416
	f/m: 1 yr	11716
Mustela nivalis	f: 1st mate: 16 wk	4576
	m: testes: ~4 mo	4776
	m: 1st mate: 32 wk	4576
	m: 1st mate: 1 yr	11564
	f/m: 3-4 mo	1871a
	f/m: 3-9 mo	3571
Mustela putorius	f: vulvar edema: 30-50 wk	9394
	f: 1st mate: 6-9 mo	9344
	m: spermatogenesis: 8 mo	5167
	f/m: 1st mate: 1 yr	9028
	f/m: ~9 mo	3571
	f/m: ~1 yr	4643
	f/m: 1 yr	7979
Mustela vison	m: spermatogenesis: 8-9 mo	1129
	f/m: 10 mo	3026
Poecilogale albinucha	m: 1st mate: 33 mo	9326
Spilogale putorius	f: 10 mo	7126
	f/m: 5 mo	1871a
Taxidea taxus	f: 1st mate: 4-5 mo	7250
	f: 6-9 mo	6559
	m: sperm: 15 mo	11951
	m: 1st mate: 16-17 mo	7250
	m: >1 yr	6559
	f/m: 1st mate: ~1 yr	11950
CYCLE LENGTH		
Aonyx cinerea	24-30 d	2813
	28 d; n1 f	6393
Eira barbara	estrus: 12.1 d; r5-20; n20, 3 f	6395, 8627
Lutra lutra	estrus: 14 d	11502
	24-28 d	2813
	36 d; via estradiol peaks	10918
	40-45 d	11502
Lutra perspicillata	estrus: 14 d	9175
	~1 mo	2813
Martes martes	estrus: 3-4 d; n34, 24 f	9610
	estrus: ~10 d	6047
	3-7 d; r3-18; n34 cycles, 24 f	9610
Martes zibellina	estrus: 1-2 d; rarely 3	8276
	estrus: 1-3 d	4643
	estrus: 2 d; n116 f	9610
	7-20 d	4643
	mode 9-12 d; r7-17; n177 cycles, 116 f	9610
	10-12 d	8276
Mephitis mephitis	estrus: 1 wk	4194

Mustela erminea	4 wk	4643
Mustela eversmanni	estrus: 41 d; ± 27	11716
Mustela lutreola	estrus: 5 d	7559
	metestrus: 10 d; r1-32	12050
	diestrus: 17 d; r3-41; n4	12050
Mustela nigripes	estrus: 32-42 d; n1 f	11716
	proestrus: 2-3 wk; r8-31 d	11716
Mustela putorius	estrus: 3-5 d	7979
	estrus: 4-5 mo	6156
	estrus: max 17 wk	4299
Mustela vison	estrus: mode 9 d; r1-40	7979
	estrus: ~3 wk	3026
	estrus: 1 mo; n some f	4643
	7-10 d; interestrus interval	1871a
GESTATION		
Aonyx capensis	63 d	11697
	~63 d	5761
	~63 d	10153
Aonyx cinerea	60-64 d	2813
Arctonyx collaris	postimplantation: 6 wk	8232
	5-9.5 mo	8232
Conepatus leuconotus	~2 mo	1474
Conepatus mesoleucus	42 d	8798
Eira barbara	63-67 d; n≥7	8627, 8630, 8631
	~70 d; n1	11235a
Enhydra lutris	6.5-9 mo	3098
	est 8 mo with delay	5678, 5681
	est 8-9 mo	10499
	238 d; n1; r8-9 mo	4643
Gulo gulo	active development: 30-40 d	1871a
	4 mo	7652
	215 d; n1	7429a
	~250 d with delay	1871a
	272 d; n1	7171
Ictonyx striatus	36 d ea; n3	9326
Lutra canadensis	active development: ~50 d	1871a
	max 43 d; n1	2088
	245-365 d	2813
	9 mo 18 d; n1	6443
	9.5-10 mo	1474
	9 mo 18 d-12 mo 15 d	6442
	est 12 mo with 9 mo delay	4286
Lutra felina	60-120 d	10809
Lutra lutra	~59 d; r53-100; n5	8318
	60-62 d	2813
	61 d; n1	1941
	61-63 d	3571, 4643
	61-74 d	11502
	61-87 d; 1st vs last mating to birth	1355
	63 d	10513
	9 wk	7652
	9-10 wk	10499
Lutra maculicollis	2 mo	8760
	2 mo	5277
Lutra perspicillata	61-63 d	9175
	62 d	2813
Martes americana	implant: est 60-102 d	411
	implant: ≥6 mo	11402
	implant: 7-8.5 mo	5438
	implant: 248-250 d	11949
	postimplantation: ~27 d	5438
	postimplantation: 28-29 d	11949
	220-230 d	1240
	244 d	410

	8.5-9 mo	3030
	259-275 d	411
	259-276 d; with delay	11949
	268.5 d; r261-276; n2	6821
	9 mo	8312
	~9 mo with delay	11402
Martes foina	implant: 2-3 mo	3571
	implant: 240 d	1586
	postimplantation: 28-30 d	1586
	postimplantation: 5.5-7.5 mo	3571
	236-274 d with delay	7979, 10499
	236-275 d with delay	4643
	est 8-9 mo	7652
	8.5-9.5 mo with delay	3571
	268-270 d with delay	1586
Martes martes	implant: 5-7.5 mo	3571
	implant: ~240 d	1582
	postimplantation: 27-28 d	4643
	postimplantation: 30 d	1582
	postimplantation: 2-3 mo	3571
	postimplantation: est 2-3 mo	9610
	r94-106 d; n3	1942
	230-270 d	7979
	230-275 d with delay	10499
	236-274 d with delay	4643, 4645
	8-9 mo	7652
	8-9.5 mo with delay	3571
	264-274 d; n1	6047
	270 d with delay	1582
	267-291 d	9610
Martes pennanti	implant: 9 mo	2088
	implant: 10 mo	4285
	postimplantation: 1 mo	2088
	postimplantation: 2 mo	4285
	9 mo	3031
	327-358 d	11949
	11.5 mo	8715
	350 d; r211-370	2088
	352 d; r338-358; n15	4225
	~1 yr	4285
	1 yr	2068
Martes zibellina	implant: 5 mo	4643
	implant: 7-9 mo	7979
	postimplantation: 27-28 d	4643
	postimplantation: 25-40 d	7979
	postimplantation: est 2-3 mo	9610
	236-315 d; repeated matings	8276
	~8 mo	7652
	245-298 d with delay	4643
	250-300 d; 1 mating	8276
	253-297 d with delay	7979
	267-291 d	9610
	~11 mo	756
Meles meles	implant: 2-10 mo	6559
	implant: 3-9 mo	4416, 4429
	implant: 4 mo	3264
	implant: 4-5 mo	3571
	implant: 6-10 mo	11444
	implant: 300-320 d	1575, 1584
	implant: 10 mo	1577
	postimplantation: 45 d	1104, 1572, 1575
	postimplantation: 6 wk	6559
	postimplantation: 2 mo	1577
	postimplantation: 2-3 mo	3571

	postimplantation: 7-8 wk	3264
	5 mo-~1 yr	4416, 4429
	6 mo	3264
	6 mo	7652
	min 6 mo; 2 mo	7813
	6-8 mo	3571
	200-350 d	4645
	271-284 d; summer mating	4643
	271-284 d; mate mid-July	7979
	10-11 mo	11444
	12 mo	1577
	12 mo; with delay	1104, 1572, 1575
	~1 yr; spring mating	4643
	~1 yr; r339-376 d; non-July matings	7979
	14-15 mo	10419
	14-15 mo; winter mating	4643
Mellivora capensis	180 d	12059
	~180 d	9175
	6 mo	11697
	est 6 mo	6559
	~6 mo	5761
Mephitis macroura	~8 wk	6387
Mephitis mephitis	implant: ~19 d	11381
	59-77 d	11383
	59-77 d; 61.5 d 2nd preg of yr	11382
	61 d; r59-62 d; n4	8245
	61-67 d	9822
	62 d	11674
	63 d	1355
	63.2 d; sd2.2; r60-69; n26 mate on or after mid Mar	11382
	63.3 d; r62-66; n3	11277
	64, 66-75 d; n2	9821
	69.1 d; sd4.1; r63-77; n33 mate before mid Mar	11382
Mustela altaica	30-40 d	4643
	~40 d	7979
	~40 d	10499
Mustela erminea	implant: ~5 wk; longer 2nd mating	3571
	implant: est 5-6 mo	7979
	implant: 9-10 mo	2421
	active development: 21-28 d	1871a
	postimplantation: ~6 wk	3571
	postimplantation: est 2-3 mo	7979
	4 wk	9175
	42 d	11564
	6 wk or 9-10 mo	4643
	8-9 wk, mate Mar; 8-9 mo; other matings	11497
	77 d; n1 capture to birth	4269
	10-11 wk	7652
	10 wk, mate Feb-Apr; 9-10 mo other matings	10512
	10.5-11 wk or 9-10 mo 2nd mating	3571
	7 mo; adult f	7598
	8 mo; young f	7598
	r240-393 d; n35	10752
	8.5-10 mo	1474
	8-9 mo	5744
	est 9-10 mo	7979
	270 d with delay	1871a
	280 d; r>71-378 d; n16	7599
	~11 mo	4645
	1 yr; with delay	2421
Mustela eversmanni	implant: <13 d	7137
	implant: 14 d; with 7-8 d delay	4643
	postimplantation: 29 d	7137
	36-40 d	7979

	36-43 d with delay	4643
	38-41 d	10499
	40.8 d; ±1.2; r39-43; n9	7137
	40-42 d	4645
Mustela frenata	implant: 8 mo	11949
	active development: 21-28 d	1871a
	postimplantation: 23-24 d; r21-28	11944, 11948, 11949
	min 70 d; n1; capture to birth	4269
	≥131 d; n1	4224
	270 d with delay	1871a
	279 d; r205-337; n18	11944, 11945, 11946, 11948
Mustela lutreola	35-42 d	7979
	39-48 d; sometimes 71-76 d	3571
	40-43 d	12050
	42-46 d	4643
	6-8 wk	4645
	43-72 d	10499
Mustela nigripes	42.7 d; sd0.7; r42-45; n62	11716
Mustela nivalis	5 wk	1304
	5 wk	10512
	5 wk	3467
	35; mode 35; r34-36; n9	4576
	35.1 d; r33-36	1010
	35.7 d; r35-36; n3	2863
	36 d; r35-37; n4	4457, 4458
	5-6 wk; after main mating; 8-10 wk; rest of yr	3571
	~42 d	11564
	min 7 wk	4645
	~50 d	8223
Mustela putorius	implant: 11-13 d	7655
	implant: 12 d	7135, 7142
	implant: 13 d	11962
	implant: 12-13 d	3018
	40-43 d	3571
	~40 d	7979
	40 d 17 h; r39 d 5 h-42.0 d; n5	7654, 7655
	41 d	9028
	41 d	4689
	42 d; mode 42; r41-43; n21	4299, 4527
	42 d; sd3	6156
Mustela sibirica	28-30 d	10499
	28-30 d or 2 mo	7979
	~30 d	4645
	35-42 d	4643
Mustela vison	implant: 1-46 d	7979, 10499
	implant: 9-20 d	7448
	implant: 14-16 d	7641
	implant: 2-3 wk; max 1 mo	4643
	implant: 16-20 d	4340
	implant: 21-23 d	3028
	postimplantation: 25-26 d	4643
	postimplantation: 28-30 d	3028
	postimplantation: 29-35 d	7979
	postimplantation: ~31-33 d	8312
	postimplantation: 40 d	7448
	36-75 d with delay	7979, 10499
	38-85 d	5146
	42 d; r39-76	1474
	46.4-56.8 d	2779
	48.0-51.5; across treatments	4911
	7-8 wk	7652
	49.05 d; se0.19; n374	7638
	49.66 d; sd4.89; r39-76; n924	3026
	50.95 d; r39-76; n747	1166

	51 d	3028
	51 d; sd5.3; r40-76; n233	8312
	51.3 d; r40-65	4643
	51.7 d; sd5.2; r40-71; n459	345a
	52.2 d; ±5.1; n10 f	5264
	53.3 d; se0.3; n390; last mating to birth	7638, 7639
	56.2 d; se1.8; adult	8522
	60 d	7448
	2 mo; n2	161
	61.8 d; se1.5; young f	8522
	75 d; n1	10571
Poecilictis libyca	37 d; n1	8427
Poecilogale albinucha	31.75 d; mode 31; r31-33; n4	9326
Pteronura brasiliensis	66.7 d; r65-70; n3	465
	70 d; n1	10891
Spilogale putorius	delay: est 14-16 d	7126
	implant: 180-230 d	7130, 7133
	postimplantation: 28-31 d	7130, 7133
	33 d	10003
	est 50-60 d	7126
	120 d; capture to birth	2026a
	210-266 d	7130, 7133
	218 d; r203-234; n7	7129
Spilogale pygmaea	43-51 d; ~47 d after mating; n1	10758
Taxidea taxus	implant: 6-7 mo	7250
	postimplantation: 5 wk	6559
	postimplantation: 6 wk	4287
	postimplantation: 2-3 mo	7250
	6.5 mo	11950
	9-10 mo	7250
Vormela peregusna	est 8 wk	4643
	~8 wk	4645
	~2 mo	7979
LACTATION		
Arctonyx collaris	solid food: 85 d	8232
Conepatus mesoleucus	den emergence: ~35 d; r6-8 wk	8798
Eira barbara	solid food: 62-64 d	8631
	wean: 75-100 d	8631
Enhydra lutris	wean: ~2 mo	7979
	dependent: 169 d; r76-333; n23	7478
	dependent: 6-8 mo	3098
	independent: 7 mo	7979
Gulo gulo	solid food: the middle of lactation period	10499
	wean: 2.5-3 mo	10499
	wean: 3 mo	4643
Ictonyx striatus	solid food: ~33 d	9327
	solid food: 32 d	9326
	wean: ~8 wk	9326
Lutra canadensis	solid food: ~8 wk	8798
	solid food: 63-75 d	1871a
	wean: ~4 mo	8798
Lutra lutra	solid food: 2 mo	6927
	wean: 21 d	8318
	wean: ~3 mo	10499
Lutra maculicollis	den emergence: ~2 mo	5761
Lutra perspicillata	wean: 6 wk	9175
Martes americana	solid food: 6 wk	1869
	weight decrease: 6 wk	1240
Martes foina	solid food: 6 wk	3571
	wean: 40-45 d	4643
	wean: ~3 mo	3571
Martes martes	solid food: 6 wk	3571
	solid food: 6 wk	4643
	den emergence: 44 d	9610

	wean: 8 wk	9610
	wean: ~3 mo	3571
	wean: 6-7 wk	10499
	wean: 1.5 mo	7979
Martes pennanti	den emergence: 9-10 wk	2088
	wean: begins 8-10 wk	8714
	wean: ~10 wk	1871a
	nurse: 114 d	2068
	effective predation: 4 mo	8714
Martes zibellina	den emergence: 44 d	9610
	solid food: 50-52 d	4643
	wean: 8 wk	9610
Meles meles	den emergence: 6-9 wk	4643
	solid food: 2.5-3 mo	10499
	wean: 2.5 mo	7979
	wean: 3 mo	4643
	independent: 150 d	7975
	wean: 2-2.5 mo	3571
	wean: 75 d	7975
	wean: 12-18 wk	6559
	wean: 4 mo	4429
Mellivora capensis	solid food: 1 wk	11253
	den emergence: 6 wk	11253
Mephitis mephitis	solid food: 36 d	10330
	wean: ~2 mo; max 4	11277
Mustela altaica	independent: 2 mo	7979
	wean: 34-41 d	4643
	wean: ≤8 wk	11384
	wean: ~2 mo	10499
Mustela erminea	solid food: 3 wk	4269
	solid food: 4-5 wk	5741
	solid food: 35 d	8769
	wean: 4 wk	3571
	wean: 5 wk	4269
	wean: ~5 wk	9175
	wean: 7-12 wk	5741
	den emergence: 2 mo	7979
Mustela eversmanni	solid food: 20 d	11321
	solid food: 28-34 d	4643
	solid food: 6 wk	4645
	solid food: 2 mo; ~2.5 mo	10499
	nurse: >40 d	11321
	wean: ~6 wk	4645
	wean: 1.5-2 mo; 1.5 mo	7979
Mustela frenata	solid food: 3 wk	4269
	nurse: 5 wk	11948
	wean: 36 d	4269
Mustela lutreola	solid food: 20-25 d	4643
	wean: 4-5 wk	3571
	prey capture: 52-62 d	12050
	wean: ~2 mo	4645
	wean: 2-2.5 mo	4643
	wean: 10 wk	7979
Mustela nivalis	solid food: 2.5 wk	2863
	solid food: 18 d	4576
	den emergence: 28 d	2862
	wean: 2-9 wk	3467
	wean: 24 d	4457, 4458
	wean: 28-35 d	1871a
	wean: 42-56 d	4576
	wean: 6-8 wk	11564
	nurse: 45 d	1304
	wean: ~12 wk	2862

Mustela putorius	nurse: ~10 wk	4299
	wean: 6-8 wk	9344
	wean: < 3 mo	10512
	independent: 2 mo	7979
	wean: 4-5 wk	3571
Mustela sibirica	wean: 2 mo	4643
	wean: 2 mo	10499
Mustela vison	solid food: 20-25 d	4643
	wean: 2-2.5 mo	4643
Poecilogale albinucha	solid food: 35 d	9326, 9327
	wean: ~11 wk	9326
Pteronura brasiliensis	wean: ~4 mo	2812
Spilogale putorius	solid food ≤36 d	1871a
	solid food: ~40 d	2100
	wean: 50 d	1474
Taxidea taxus	wean: near 8 wk	5210
INTERLITTER INTERVAL		
Enhydra lutris	1 yr	10499
	2+ yr	5678
Gulo gulo	1 yr	6048
	2-3 yr	1474
Lutra canadensis	1 yr	7195
	1 yr	7234
Lutra felina	1 yr	10809
Lutra lutra	1 yr	9175
Lutra maculicollis	1 yr	5761
Martes americana	1 yr	1474
Martes foina	1 yr	3571, 4645, 7979, 10499
	1 yr	1586
Martes martes	1 yr	3571, 4645, 7979
Martes pennanti	1 yr	2068
	1 yr	1474, 2088
Meles meles	1 yr	4429
	1 yr	4645
	at times 2 yr?	3571
Mephitis mephitis	2.5 mo; n2 f	8245
	1 yr	1474
	1 yr	502
Mustela altaica	1 yr	10499
Mustela erminea	1 yr	4145
	1 yr	1474, 3571, 7979, 9175
Mustela eversmanni	1 yr	4643, 4645, 7979
	mode 1; 2 if 1st dies	10499
Mustela frenata	1 yr	11947, 11948
Mustela lutreola	1 yr	3571, 4645, 7979
Mustela nivalis	35.3 d; r34-38	3467
	1 yr	4645
Mustela mustela?	~1 yr	2421
Mustela putorius	20 wk	9344
Mustela vison	1 yr	4645, 5146
Vormela peregusna	1 yr	4645
LITTERS/YEAR		
Conepatus mesoleucus	sometimes 2	8798
Gulo gulo	0.5	4643
	1/2-3 yr	1474
Ictonyx striatus	mode 1/season	9327
Lutra lutra	1; r1-2 (?)	3571
Meles meles	0.5-1	3571
Mellivora capensis	max 2	6559
	2	11253
	2	9175
Mephitis mephitis	2	8245
Mustela nivalis	1	4645
	1; r1-2	3571

	> 1	1474
	1-2	2422
	3	3467
Mustela putorius	1	7979, 10499
	1; 2 if 1st lt removed	6156
	1; r1-2	3571
	2	9344
	2; r0-2	4299
Poecilogale albinucha	mode 1/season	9327
	2	11253

SEASONALITY

Aonyx capensis	mate: July-Aug; Zambia	10153
	mate: Mar; Zimbabwe	10153
	birth: yr rd?; S Africa	10158
	yg emerge: Mar-Apr; central, s Africa	5761
Aonyx cinerea	repro: not seasonal; captive	2813
Conepatus leuconotus	mate: Feb; sw US, n Mexico	1474
	birth: Apr-May; sw US, n Mexico	1474
Eira barbara	mate: Jan-Apr, June, Oct; zoo	8627
Enhydra lutris	spermatogenesis: yr rd; w US	5678
	mate: peak Mar-Apr; USSR	7979
	mate: peak Mar-May; USSR	10499
	mate: Mar-May, July-Aug; USSR	4643
	mate: peak spring; Commander I, far e Russia	565
	mate: yr rd, peak fall; w US	5678
	unimplanted blastocysts: yr rd; AK	10000
	birth: Feb-May, Oct-Nov; w US	5678
	birth: est June; w US	1474
	birth: yr rd; CA	3273
	birth: yr rd, peak May; AK	7478
	birth: yr rd, peak spring; USSR	4643
	lact f: yr rd; USSR	10499
Gulo gulo	testes: regressed Aug-Jan; captive, WA	7138
	spermatogenesis: end Apr-May; USSR	4643
	spermatogenesis: end June; AK	4643
	mate: Apr-May; Finland, Sweden	6046, 6048
	mate: Apr-Aug; AK	1474
	mate: Apr-Oct; AK	11955
	mate: early June; n1; central AK	6049
	mate: June, Aug; n2; AK	6725
	mate: June-Sept; USSR	4643
	mate: Apr-Oct; AK	4643
	mate: Sept-Oct; w Siberia	10499
	mate: Dec-Jan; n Europe, n Asia, n N America	7652
	implant: Jan; AK	11955
	birth: end Jan-Mar; Finland	8793a
	birth: Feb-Mar; Finland, Sweden	6046, 6048
	birth: Feb-Mar; Sweden	4201
	birth: Feb-Apr; AK	1474
	birth: end Feb-early Mar; USSR	4643
	birth: Mar-Apr; AK	11955
	birth: end Mar; Sweden	12091
	birth: Apr-early May; zoo	4643
	birth: Apr-June; w US	5146
	birth: end winter-mid spring; w Siberia	10499
Ictonyx striatus	testes: large Sept-Apr, small May-Aug; captive	9326
	birth: Sept-Dec; captive	9326
	lact f: Feb, June, Sept, Oct; few data; Kenya, Tanzania	5761
Lutra canadensis	estrus: Oct-May; captive	2813
	mate: est Jan-Mar; NC	11763
	mate: Apr; ID	7195
	mate: winter-early spring; MN	6442
	birth: Jan-May; WI	5210
	birth: Mar; ID	7195

	birth: est Mar-Apr; NC	11763
	birth: Apr-May; US	1474
	birth: mid Apr; NY	7234
	birth: Nov-May, peak Mar-Apr; WY	1871a
Lutra lutra	mate: Jan-Mar; Iraq	10767
	mate: end Jan-Mar; Germany	10513
	mate: Feb-Apr; USSR	4643, 7979, 10499
	mate: July-Aug; captive	1941
	mate: peak winter; s Sweden	3075
	mate: yr rd, peaks Feb, July; Germany	3571
	birth: Mar-Oct; Shetland, UK	6075
	birth: Apr-May, sometimes June, rare Dec; USSR	7979
	birth: Apr-May; Germany	10513
	birth: Apr-May; USSR	10499
	birth: end Apr-early May; Petschora, USSR	4643
	birth: June-early July; Volga delta, USSR	4643
	birth: May; Eur USSR	4645
	birth: mid Oct; captive	1941
	birth: peak spring; lower Saxony, Germany	9063
	birth: spring-early summer; Pakistan	9175
	birth: yr rd, low Mar-Apr; nw Scotland, UK	6075
	birth: yr rd, peak Apr-June; Germany	3571
	birth: yr rd; Aberdeenshire, UK	5287
	birth: yr rd; Sri Lanka	8495, 8501
	birth: yr rd; n240; England, UK	6927
Lutra maculicollis	mate: July; Tanzania	8760
	birth: Sept; Tanzania	8760
Lutra perspicillata	birth: end Aug; n1; Pakistan	9175
Lutra provocax	mate: est yr rd; Chile	10809
Martes americana	mate: June-Aug, peak July; WI	5210
	mate: mid June-Aug; ALB, Canada	1869
	mate: end June-July; AK	1869
	mate: July-Aug; nw MT	5438
	mate: July-Aug; captive	8312
	mate: July-Aug; captive, ID	411
	mate: July-Aug; MT, ID	11949
	mate: mid July-early Sept; captive	6821
	mate: mid July-Sept; NY	3029
	mate: end July-early Aug; n N America	1474
	mate: end July-mid Aug; captive	411
	mate: Aug; captive	410
	mate: Aug; zoo	1240
	unimplanted blastocysts: Nov-Jan; ID	6863
	implant: Feb-Apr; nw MT	5438
	implant: after Feb 19; no location given	3030
	birth: Mar-Apr; MT, ID	11949
	birth: Mar-Apr; zoo	1240
	birth: end Mar-mid May, peak Apr; WI	5210
	birth: Apr; captive	8312
	birth: Apr; captive, ID	411
	birth: Apr; n N America	1474, 5146
	birth: mid Apr; captive	410
	birth: end Apr-early May; IL, WI	2060
Martes flavigula	mate: June-July, perhaps Mar; USSR	10499
	birth: spring; USSR	10499
	birth: spring-early summer; Pakistan	9175
Martes foina	testes: small Nov-Jan, large May; captive	444, 445
	mate: June-July; Belgorod, Ukraine	4643
	mate: June-Aug; s, central Europe	7652
	mate: July; USSR	7979
	mate: July; France	1586
	mate: July-Aug; Germany	3571
	mate: ~Aug; Eur USSR	4645
	mate: mid-summer; USSR	10499

	mate: end summer; Pakistan	9175
	diapause: until Mar; Crimea, USSR	5966
	birth: early May; Europe, w USSR	4645
	birth: Mar-May; Germany	3571
	birth: end Mar-Apr; USSR	4643
	birth: end Mar-early Apr; France	1586
	birth: early spring; Baluchistan, Pakistan	9175
Martes martes	sexual quiescence: Sept-Mar; Europe, w USSR	4643
	mate: Jan; n1 pr; captive	1942
	mate: June-Aug; n, central Europe	7652
	mate: end June-early Aug, peak July; captive	9610
	mate: end June-early Aug, peak end July-early Aug; Europe, w USSR	4643
	mate: end June-Aug; Germany	3571
	mate: July; Austria	6047
	mate: July; captive	1578, 1580
	mate: July-Aug; Europe, w USSR	4645
	mate: summer; USSR	7979
	mate: summer; USSR	10499
	ovulate: July; captive	1580
	implant: Mar; captive	1578
	implant: Mar; Europe, w USSR	4643
	birth: Mar-May; Germany	3571
	birth: mid Mar-Apr; captive	1578, 1580
	birth: Apr; Austria	6047
	birth: Apr; n1; captive	1942
	birth: Apr-early May; Europe, w USSR	4645
	birth: mid-end Apr; captive	9610
Martes pennanti	mate: Mar?; indirect evidence; Canada	2068
	mate: end Mar-early May; WI	5210
	mate: end Mar-mid May; BC, Canada	2088
	mate: early Apr; ONT, Canada	8715
	mate: Apr; captive, CA	4225
	mate: Apr-May; Canada	3031
	implant: Feb; NY	2852
	implant: end Feb; ONT, Canada	8715
	implant: after Feb; Canada	3031
	birth: Feb-May; WY	1871a
	birth: end Feb-early Apr; Canada	2068
	birth: Mar-Apr; captive, CA	4225
	birth: est mid-end Mar; NY	2852
	birth: mid Mar-mid Apr; WI	5210
	birth: mid Mar-Apr; BC, Canada	2088
	birth: end Mar; ONT, Canada	8715
	birth: end Mar-early Apr; MAN, Canada	6385
	birth: end Mar-early Apr; n N America	1474
	birth: end Mar-early Apr; w US	5146
	birth: end Apr-early May; IL, WI	2060
	birth: ~early May; NY	7234
Martes zibellina	spermatogenesis: mid May-July, USSR	4643
	mate: end June-mid Aug, peak mid July; captive	9610
	mate: June-July; USSR	488
	mate: June-July; 30-35 d; USSR	7979
	mate: June-July; captive	756
	mate: June-July; captive	834
	mate: mid June-mid Aug; Asia	8276
	mate: July-Aug; n, e Siberia	7652
	implant: end Feb-early Mar; USSR	4643
	implant: Mar; delayed; Asia	8276
	birth: end Mar-early May; USSR	4643
	birth: Apr; Asia	8276
	birth: Apr; captive	9955
	birth: Apr-May; USSR	7979
	birth: mid-end Apr; captive	9610

	birth: May; captive	756
Meles meles	epididymal sperm: yr rd, peak Jan-Feb; Switzerland	3988
	fertile m: yr rd; Sweden	89
	spermatogenesis: spring-summer; USSR	7979
	spermatogenesis: yr rd; France	448
	spermatogenesis: yr rd, peak June-Feb; France	449
	spermatogenesis: yr rd; France	447
	testes: large Jan-Feb; France	449
	testes: large Jan-Feb; Switzerland	3988
	testes: large Feb, low July; France	9027
	testes: large Feb-Mar, small Oct-Nov; Sweden	89
	testes: large early Aug; Germany	3522
	testes: regressed Oct-Dec; Germany, Austria	10419
	testosterone: high Feb-Apr; captive	6983
	mate: Jan-Feb; France	1572, 1577, 1578
	mate: Feb-Mar; adult; Apr-June; yearlings; Switzerland	11444
	mate: Feb-Apr; w France	6986
	mate: late Mar-May, July-Sept; Somerset, England, UK	4429
	mate: May; n1; Sweden	6564
	mate: early Aug-early Nov; Europe, w, n Asia	7652
	mate: Apr-June; s Sweden	7975
	mate: (late Apr) June-Aug; Germany	3571
	mate: June; zoo	7813
	mate: July-Aug; Germany	3522
	mate: July-Aug?; Germany	3263
	mate: summer, Oct; Germany	3264
	ovulate: Feb, June, Sept-Oct; sw England, UK	4416
	ovulate: July-Aug; Germany	3522
	conceive: Feb; lab	1104
	implant: Nov-Jan; sw England, UK	4416
	implant: mid Nov-Jan; Germany	3263, 3264
	implant: after Nov; Germany	3522
	implant: early-mid Dec; sw France	1572, 1573
	implant: Dec-Jan; Switzerland	11444
	implant: ~mid-end Dec; Bordeaux, France	1577, 1578, 1582
	implant: early Jan; s Sweden	7975
	implant: Jan; lab	1104
	birth: mid Jan-mid Mar; sw England, UK	4416
	birth: mid Jan-mid Apr; Germany	3264
	birth: end Jan-Mar; Switzerland	11444
	birth: end Jan-early Apr; Germany	3571
	birth: Feb; sw France	1572, 1573, 1577
	birth: Feb-mid Mar; Germany, Austria	10419
	birth: Feb-Mar; n11; zoo	12128
	birth: Feb-Mar; s Sweden	7975
	birth: Feb-Mar; Germany	3522
	birth: mid Feb-mid Mar; Sweden	6564
	birth: Mar; zoo	7813
	birth: Mar-Apr; Europe, w USSR	4645
	birth: Mar-Apr; Lebanon	6420
	birth: Mar-Apr; USSR	7979
	birth: est Mar-Apr; Siberia, USSR	10499
	birth: end Mar-Apr; n7; zoo	4643
Mellivora capensis	birth: Feb; n1; zoo	1230
	birth: twice/yr, rainy periods; Zaire	11253
	birth: yr rd?; S Africa	10158
Melogale personata	birth: just before rainy season; Thailand	6369
Mephitis macroura	mate: end winter; Mexico	6387
	birth: May-June; n Mexico	1474
Mephitis mephitis	spermatogenesis: end Dec-Aug; NY, NJ	4194
	testes: large Mar, small Aug; n IL	11277
	estrus: end Feb-early Mar; NY, NJ	4194
	mate: Feb-Mar; IL	11277
	mate: mid Feb-mid Apr; captive	11382, 11383

	mate: early Mar; OR	11674
	mate: Mar, May; captive	8245
	mate: Apr-May; TX	2192
	birth: early May; US	1474
	birth: May; ALB, Canada	966
	birth: May; OR	11674
	birth: May-early June; WY	1871a
	birth: May, July; captive	8245
	birth: May-mid June; WI	5210
	birth: May-June; ALB, SAS, Canada	9644
	birth: May-July; captive	11382
	birth: peak May; PA	9164
	birth: end May; IL	11277
Mustela altaica	spermatogenesis: end Dec-mid Jan; USSR	4643
	mate: mid Jan-early Apr; Kazakhstan	4643
	mate: Feb-Mar; Kazakhstan	7979
	mate: Feb-Mar; Kazakhstan, Altai, w Sayan, central Russia	10499
	birth: early Apr-mid June; USSR	4643
	yg found: early May; USSR	7979
Mustela erminea	spermatogenesis: Apr-Aug; lab	4144
	testes: large mid Feb-Aug, decline Sept; lab	4144
	testes: peak Apr; Netherlands	11190
	sperm in testes: Mar; Germany	11497
	ovarian wt: peak Nov; Netherlands	11190
	ovulate: Mar, July, Aug; Germany	11497
	mate: Feb-Mar; n, w Europe, w Asia	7652
	mate: Feb-Mar, June-July; Germany	3571
	mate: Feb-mid June; n Kazakhstan	4643
	mate: Feb-Oct, peak May-July; captive	7599
	mate: mid Feb-early Apr, mid June-early Aug; Germany	10512
	mate: May-July; lab	4144
	mate: June-July; n2; lab	7598
	mate: July-Aug; WI	5210
	mate: Nov-Dec; New Zealand	5744
	mate: spring-summer; Europe, w USSR	4645
	mate: summer; Pakistan	9175
	implant: after Feb; MT	11945
	implant: Mar; captive	4145
	implant: Aug-Oct; New Zealand	5744
	birth: mid Jan-mid Feb; n2; lab	7598
	birth: end Feb-mid May; Europe, w USSR	4645
	birth: Mar-Apr; Germany	10512
	birth: Mar-May; May-July; 2 regions; USSR	4643
	birth: Apr; captive	4145
	birth: Apr-May; ne, nw N America	1474
	birth: Apr-May; w US	5146
	birth: Apr-June; Germany	3571
	birth: mid Apr-early May; WI	5210
	birth: May; n1; captive	4073
	birth: Sept-Oct; New Zealand	5743, 5744
	birth: spring; Pakistan	9175
	birth: spring; USSR	7979
Mustela eversmanni	testes: large end Dec-mid May, peak end Feb; captive	7137
	estrus: Feb-mid Mar; Kazakhstan	9772
	estrus: Feb-Mar; USSR	7979
	mate: mid Feb-mid Mar; USSR	10499
	mate: end Feb-mid Mar; Kazakhstan	9772
	mate: Mar; captive	4643
	mate: Mar-?; Europe, w USSR	4645
	mate: end Mar-Apr; captive	7137
	birth: mid Apr-May; USSR	10499
	birth: end Apr-early May; USSR	7979
	birth: end Apr-May; Europe, w USSR	4645
	birth: mid May; captive	7137

Mustela frenata	testes: large Apr-July; n US	11947
	mate: Mar-Apr; Yucatan, Mexico	3633
	mate: June-Aug; captive	11946
	mate: July-Aug; MT	11945
	mate: mid summer; MN	11948
	implant: after Jan; MT	11945
	birth: Apr; BC, Canada	2088
	birth: peak Apr; MN	11948
	birth: Apr-May; MT	11945
	birth: mid Apr-mid May; WI	5210
	birth: May-June; Yucatan, Mexico	3633
Mustela lutreola	mate: Feb-?; Europe, w USSR	4645
	mate: Feb-Apr; USSR	7979
	mate: Feb-Aug; Germany	3571
	mate: Mar-Apr; USSR	10499
	mate: Apr; n1; zoo, Russia	4643
	mate: Apr-May; captive	7559
	birth: Apr-?; Europe, w USSR	4645
	birth: Apr-May; Germany	3571
	birth: Apr-May; USSR	7979
	birth: Apr-May; Europe, w USSR	10499
	birth: June; n1; zoo, Russia	4643
Mustela nigripes	mate: mid Mar-early Apr; WY	1871a
	mate: end Mar-early May; captive	11716
	birth: June; midcentral US	1474
	birth: spring; midcentral US	3416
Mustela nivalis	anestrus: Nov-Mar; England, Scotland, UK	5740
	estrus: Mar-Apr; USSR	7979
	mate: Feb-Mar; Germany	10512
	mate: Mar-summer; Europe, w USSR	4645
	mate: Apr; captive	2862
	mate: mid Apr; captive	1304
	mate: June-July; Sept-Oct; location not stated	11564
	mate: mid Oct; lab	4458
	mate: yr rd; Europe, n Africa, n Asia	7652
	mate: yr rd, peak Feb-Apr; Germany	3571
	mate: yr rd; Germany	8637
	mate: yr rd; USSR	4643
	preg f: Mar, Aug; n2; Romania	580
	preg f: Mar-Aug; England, UK	2422
	preg f: Apr-June; England, Scotland, UK	5740
	preg f: Apr-Sept; Poland	5277a
	birth: Apr-Aug; zoo	1010
	birth: Apr-Sept; England, Wales, UK	4776
	birth: early May; NY	7234
	birth: mid May; captive	1304
	birth: end June; captive	2862
	birth: July-Aug; Nov-Dec; location not stated	11564
	birth: mid Nov; lab	4458
	birth: yr rd; n N America	1474
	birth: est yr rd; America	4227
	lact f: Apr-July, Oct; England, Scotland, UK	5740
Mustela putorius	spermatogenesis: Mar-July; captive	5167
	testes: large Feb-Oct; lab	958
	testes: large Dec-Apr; captive	1049
	testes: large Mar-May; UK	11439
	testes: large Apr-May; captive	163
	anestrus: Aug-Apr; lab	4299
	estrus: Feb-May; Germany	7652, 7655
	estrus: Feb-Aug; captive England, UK	4299
	estrus: Mar-Apr; USSR	7979
	estrus: Mar-Aug; lab	9344
	estrus: mid Mar-end May; captive	10800
	estrus: June-Aug; captive	961

	mate: mid Feb-June, peak Apr-May; zoo	4643
	mate: end Feb-early June; Germany	3571
	mate: early Mar; lab	664
	mate: Mar-?; Europe, w USSR	4645
	mate: Mar-May; captive	9028
	mate: Mar-Aug; captive	6851
	mate: Mar-Aug; Sweden; captive	6156
	mate: end Mar-mid Apr; Germany	10512
	mate: end Mar-early May; USSR	10499
	mate: Apr; n1; lab	4689
	ovulate: Mar-Aug; lab	958
	birth: Jan, Mar-Apr, Aug, Nov-Dec; n13; zoo	1230
	birth: Apr; captive	961
	birth: Apr-July, rarely Aug; Germany	3571
	birth: May; captive	9028
	birth: May; Europe, w USSR	4645
	birth: May; n1; lab	4689
	birth: ~May-June; USSR	7979
	birth: May-June; Germany	10512
	birth: mid June-mid July; zoo	4643
Mustela sibirica	estrus: end winter-early summer; USSR	7979
	mate: end Feb-Mar; USSR	10499
	mate: spring; Europe, w USSR	4645
	birth: Apr; w Siberia, USSR	10499
	birth: end Apr-early May; sw Transbaikalia, Siberia, Russia	10499
	birth: May; Nepal	7382
	lact f: Aug; n1; Nepal	7382
Mustela vison	epididymal sperm: Feb-mid Apr; captive	5971
	seminiferous tubule lumen: present Dec-Mar, absent Apr-Nov; captive	8338
	spermatogenesis: Nov-Mar; captive	4911
	testes: large Dec-May; captive	1129
	testes: large mid Jan-mid Apr; captive	5971
	testes: large Dec-Mar, peak Feb; captive	1049
	testes: large Mar, small May; captive	636
	testes: large Mar; captive, NY	8522
	testes: recrudesce Oct-Dec; captive	1050
	testosterone: peak Jan; captive	10553
	estrus Feb-Mar; captive	5046
	estrus: end Feb-Mar; captive, NY	8522
	mate: Jan-Mar; N America	1474
	mate: Feb-Mar; n10; lab	161
	mate: Feb-Mar; USSR	7979
	mate: Feb-Apr peak mid Mar; ne US	3028
	mate: Feb-Apr; USSR	10499
	mate: end Feb-early Mar; N America	7652
	mate: end Feb-early Mar; PA	9164
	mate: end Feb-early Apr, peak mid Mar; USSR	4643
	mate: end Feb-early Apr; captive	3026
	mate: Mar; captive	1129
	mate: Mar; captive	5325
	mate: Mar; captive	2018
	mate: Mar; lab	2778, 2779
	mate: Mar; lab	6907
	mate: Mar-?; Europe, w USSR	4645
	mate: mid Mar; SAS, Canada	7644
	mate: end Mar-early Apr; captive	4340
	ovulate: Feb-Apr; ne US	3028
	preg f: Mar; n1; NC	1290
	preg f: Mar-Apr; captive	4911
	implant: after Mar 21; captive	9272
	implant: mid Apr; captive	3026
	birth: Apr; n4; lab	161
	birth: Apr-May; WI	5210

	birth: Apr-May; N America	1474
	birth: Apr-May; USSR	10499
	birth: mid Apr; PA	9164
	birth: end Apr-early May; captive	3026
	birth: end Apr-early May; captive	4340
	birth: end Apr-mid May; WY	1871a
	birth: early May; captive	8522
	birth: early May; NY	7234
	birth: May; captive	2778
	birth: May; captive	5046
	birth: May; lab	6907
	birth: May; USSR	4643
Poecilogale albinucha	testes: large Sept-Apr, small May-Aug; S Africa	9326
	mate: Oct-Dec; Zambia	10153
	mate: spring-summer; Natal, S Africa	9327
	birth: Sept-Apr; captive	9326
	birth: Nov-Mar; S Africa	10158
Pteronura brasiliensis	mate: end Dec; zoo	465
	birth: mid Mar; zoo	465
	birth: Apr-Oct; Surinam	2812
Spilogale putorius	testes: regress end Apr-May; e US	7126
	estrus cycles: Sept-Jan; captive	4051
	mate: Apr; e US	7126
	mate: end Sept-early Oct; w US	7130, 7133
	mate: ~early Oct; lab	7129
	implant: early Apr; OR	7140
	implant: Apr-early May; w US	7130, 7133
	implant: end Apr-May; lab	10003
	birth: May-mid June; WI	5210
	birth: May-June; WY	1871a
	birth: May-June; US	1474
	birth: mid May; lab	7133
	birth: end May-June; e US	7126
Spilogale pygmaea	testes: scrotal Dec-May; Pacific coast, Mexico	6570
	birth: July-Aug; Pacific coast, Mexico	6570
Taxidea taxus	testes: large May-end Aug; SD	11951
	mate: July-Aug; America	11951
	mate: end July-mid Aug; SD, MT	11950
	ovulate: Aug; SD, MT	11950
	conceive: July-Aug; ID	7250
	implant: Feb; ID	7250
	implant: Feb; SD, MT	11950
	implant: mid Feb; KS	4287
	birth: Feb-May; w US	1474
	birth: Feb-June; US	6559
	birth: Mar-early Apr; WY	1871a
	birth: Mar-Apr; ID	7250
	birth: ~early Apr; SD, MT	11951
	birth: end Apr-early June; WI	5210
	wean: May-June; SD, MT	11950
Vormela spp	birth: Mar-Apr; Pakistan	9175
Vormela peregusna	mate: Mar; Europe, w USSR	4645
	birth: Feb-early Mar; Kazakhstan; later; n Caucasus Mountains	4643
	birth: Feb-Mar; Kazakhstan, central Asia	10499
	birth: est end Feb; Balkhash, Kazakhstan	7979
	birth: May; Europe, w USSR	4645

Order Carnivora, Family Viverridae

Viverrids (*Arctictis* 9-22 kg; *Arctogalidia* 2-2.5 kg; *Chrotogale* 51-64 mm; *Civettictis* 7-20 kg; *Cryptoprocta* 7-12 kg; *Cynogale* 3-5 kg; *Eupleres* 2-4.6 kg; *Fossa* 1.4-2.6 kg; *Genetta* 1-3.5 kg; *Hemigalus* 745 g-3 kg; *Macrogalidia* 4-6 kg; *Nandinia* 1.5-5 kg; *Osbornictis* 1.5 kg; *Paguma* 3.5-5 kg; *Paradoxurus* 1.5-5.5 kg; *Poiana* 650 g; *Prionodon* 600-800 g; *Viverra* 2-11 kg; *Viverricula* 1.8-4 kg) are semiarboreal, solitary, nocturnal carnivores or omnivores from Eurasia, Africa, and Madagascar. Males are often larger than females (120, 122, 8495, 8501). The endothelial, zonary placenta has a mesometrial implantation into a duplex or bipartite uterus (1597, 6563, 7418, 7563) and is eaten after birth (120, 9216). The furred neonates open their eyes at 0-4 d *Arctictis*, *Civettictis*, *Eupleres*, *Fossa* (119, 120, 121, 349, 3135, 3713, 5761, 6125, 6753, 6754, 10154, 11589); 8-11 d *Arctogalidia*, *Genetta*, *Hemigalus*, *Paradoxurus*, *Viverra* (42, 47, 652, 2301, 5761, 6603, 8615, 9216, 11326); or 12-25 d *Cryptoprocta*, *Paguma*, *Prionodon* (116, 120, 122, 179, 5929, 6603). Males have a baculum (1596). Females have 2-6 mammae (119, 331, 6446a, 7161, 10158). *Cryptoprocta* females have a large, erectile clitoris with spines and an os clitoridis (1596, 6563). Details of anatomy (5274, 5601, 10459) and the hormonal control of male scent glands (10230) are available.

NEONATAL MASS		
Arctictis binturong	307.4 g; n9	11591
	310 g; r270-350; n2	6171a
	319 g; r283.7-340.5; n4	349
	333 g; n1	6125
Civettictis civetta	162 g; n1, 6 d	3135
	f: 319 g; n1 near term	10154
	m: 324 g; n3 near term	10154
	717 g; r680-792; n3, 4 d	6754
Cryptoprocta ferox	100 g	116, 120
Eupleres goudotii	150 g	120
Fossa fossa	65-70 g	119, 120
Genetta spp	61-82 g	11329
Genetta genetta	60-85 g	9216
	83 g; r77-88; n3	11326
Hemigalus derbyanus	125 g; n2	6603
Paradoxurus hermaphroditus	83 g; r69-102	47
	94.3 g; r92-98	42
Prionodon linsang	40 g; n2	6603
NEONATAL SIZE		
Arctictis binturong	HB: 21.5 cm; n1	6125
Civettictis civetta	f: 14 cm; n1	10154
	m: 14.8 cm; n3	10154
	m: HB: 20.35 cm; n1, 1.5 d	6746
Eupleres goudotii	15 cm	120
Genetta genetta	HB: 14.6 cm; r14-15; n3	11326
WEANING MASS		
Civettictis civetta	500 g; r440-530; n3, 17 d	3135
	763 g; r680-810; n3, 33 d	3135
	2.94-3.62 kg; n5, 73-80 d	6754
LITTER SIZE		
Arctictis binturong	1.98; se0.1; mode 2; r1-6; n148 lt	11591
	2 yg; n1 lt	3707, 3708, 3713
	2; r1-3; n3	6125
	2.3; r1-3; mode 2; n3 lt	8046
	2.5; r2-3; n2 lt	349
	3 yg; n1 lt	11974
	3 yg; n1 lt	2109
	3 yg; n2 lt	1443
	4; n1 lt	9634
Arctogalidia trivirgata	2 emb; n1 f	8619
	3 yg; n1 lt	652
Civettictis civetta	1 yg; n1 f	185
	2.0 yg; r1-3; n2 lt	12128
	2.2 yg; r1-4; n22 lt, 5 f	6745, 6746, 6753, 6754

	2.4 yg; r2-3; n5 lt, 1 f	3135
	2.5 yg; r2-3; n2 lt	6612
	2.5; r2-3; n2 lt	647
	3.0; n4 lt	1230
	4 emb; n1 f	10153, 10154
Cryptoprocta ferox	3.0; r2-4; n2 lt	116, 122
Cynogale bennettii	2	556
	2-3 emb	7983
Eupleres goudotii	1 yg	120, 121
Fossa fossa	1; n1 lt	11589
	1 ea; n5 lt	119, 120
Genetta spp	2-3 yg	11329
Genetta felina	1 yg; n1 lt	12128
	2.5; r2-3; n2 lt	9945
Genetta genetta	2 emb; n1 f	8965
	2 emb; n1 f	4924
	2 ea; n2 lt	10714
	2.14; r2-3; n7 lt	9216
	2.25 yg; r1-3; n4 lt	1149
	2.34; mode 2; r1-3; n35 lt	471
	2.6 emb; r2-4; n11 f	10158
	2.75; r1-3; n8 lt, 1 f	11326
	2.8; n5 lt	10154
	max 5	10153
Genetta maculata	2 emb; 1 f	4924
	2 yg; 1 f	185
	2-3 yg	4354a
Genetta servelina	1-2	5277
Genetta tigrina	2 yg ea; n4 lt	10714
	2 yg; r1-3; n2 lt	12128
	2.9; max 5; n10	10158
	3 yg; n1	9327
	3 emb; n2 f	8965
Hemigalus derbyanus	2 yg; n1 f	6603
Nandinia binotata	1 yg; n1 lt	12128
	mode 2-3 yg	4354a
	2; max 4	5761
	2 emb; n1 f	10153, 10158
Osbornictis piscivora	1 emb; n1 f	4434
Paguma larvata	1.71; r1-3; n7 lt, n1 f	2109
	2; n1 lt	179
	2 yg ea; n3 lt	12128
	3; n1 lt	8615
	3.0 yg; r1-4; n7 lt	10873
Paradoxurus hermaphroditus	2-3 yg	557
	3 yg; n1 lt	38, 42, 47
	mode 3	8495
	3-4 yg	8501
	5 yg; n1 f	2538
Paradoxurus zeylonensis	2-3 yg	8501
Poiana richardsoni	2	647
Prionodon linsang	2 yg; n1 f	6603
	2.5 emb; r2-3; n2 f	6446a
Prionodon pardicolor	2 yg	988a
	mode 2	8619
Viverra tangalunga	2 emb ea; n2 f	2313
Viverra zibetha	3	2301
Viverricula indica	2-4	120
	3-5 yg	8495, 8501
	3-5	9175
	4-5 yg	988a
	4 emb; n1 f	8906

SEXUAL MATURITY		
Arctictis binturong	f: 1st mate: 30.4 mo; se3.7; r12.9-47.9; n10	11591
	f: 1st conceive: 30 mo; se5.6; r13-47.9; n6	11591
	m: 1st mate: 27.7 mo; se3.2; r20.5-37.8; n5	11591
Civettictis civetta	f: 1st birth: 14 mo	3135
	f: 1st birth: 1 yr 308 d; r1 yr 210 d-1 yr 357 d	6753
	f: 1 yr	3135
	m: 1st mate: 7 mo	3135
Genetta spp	f: 4 yr; n1 f	11329
Nandinia binotata	f/m: 3rd yr	1742a
CYCLE LENGTH		
Arctictis binturong	82.5 d (or 81.8); se11.8; r18-187; n15 cycles	11591
Arctogalidia trivirgata	6 mo	652
Civettictis civetta	estrus: 1-6 d	3135
Prionodon linsang	estrus: 11 d	3597
GESTATION		
Arctictis binturong	88-89 d; n1	11974
	90-92 d	3707, 3708, 3713
	91.5 d text (or 91.1 abstract); se1; r84-99; n13	11591
	93 d; r88-98; n2	349
	96 d; n1	6125
Arctogalidia trivirgata	~45 d	652
Civettictis civetta	48.75 d; r43-54; n4 last mating to birth	6746, 6753, 6754
	63-68, 66, 79, 80, 81 d; n5	3135
	75.3 d; r72-82; n3; male removed to birth	6753, 6754
Cryptoprocta ferox	~3 mo; few data; field data	116, 120
Fossa fossa	82, 88-89 d; n2; [+ 3 others ~3 mo]	119, 120
Genetta genetta	10-11 wk	11326
	70 d	9216
	3 mo; field data	9215
Genetta tigrina	70 d	9216
Nandinia binotata	64 d	5761
Paradoxurus hermaphroditus	60 d; n1	47
LACTATION		
Arctictis binturong	den emergence: 1 mo	3708
	solid food: 2 wk	3708
	solid food: 8 wk	3707
	solid food: 65 d	11974
	nurse: 5 mo maybe longer	11974
Arctogalidia trivirgata	solid food: 3 wk	652
Civettictis civetta	leave nest: 8 d	11253
	solid food: 21 d	6753, 6754
	solid food: 6 wk	3135
	wean: 84 d	6753
	nurse: 14-20 wk	3135
Cryptoprocta ferox	solid food: 3 mo	116
	wean: 4-4.5 mo	116, 120
Eupleres goudotii	wean: ~9 wk	121
Fossa fossa	solid food: 1.5 mo	119, 120
	wean: 2.5 mo	119, 120
Genetta genetta	solid food: ~6 wk	5761
	wean: 2 mo	11326
	wean: 9 wk	10158
Nandinia binotata	wean: 64 d	5761
Paguma larvata	solid food: 8 wk	7161
Viverricula indica	wean: 4-4.5	120
INTERLITTER INTERVAL		
Arctictis binturong	271.9 d; ±47.4; r107-926; n17 young die	11591
	317.6 d; ±17.6; r111-446; n22 m present	11591
	334.0 d; ±27.2; r107-1168; n53 young live	11591
	468.3 d; ±94.8; r128-1168; n11 m absent	11591
Civettictis civetta	221 d; r193-258; n4 with lactation	6753, 6754

SEASONALITY

Arctictis binturong	mate: Oct-Nov; zoo	3708, 3713
	birth: Apr; n1; captive	2109
	birth: Nov-early Feb; zoo	3713
	birth: yr rd, peak Jan-Mar; Himalayan foothills	11591
Arctogalidia trivirgata	preg f: May; n1; India	8619
Civettictis civetta	mate: Aug-Oct; s Africa	10153
	birth: Feb-Mar, Nov, peak Mar; Nigeria	4354a
	birth: May-Oct; n19/22 lt; zoo	6753, 6754
	birth: end May-June, end Dec; Zaire	11253
	birth: July-Mar; few data; captive, Ghana	3135
	birth: Aug-Jan; S Africa	10158
	birth: Oct-Nov; n12; zoo	1230
	birth: Oct-Nov; n Zimbabwe	331
	lact f: Apr, Aug, Oct; n3; s Cameroun, Benito, Eq Guinea	647
	lact f: Jan, Feb; n2; e Africa	185
Cryptoprocta ferox	mate: Sept-Oct; Madagascar	120
	birth: Dec-Jan; Madagascar	120
Cynogale bennettii	young found: May; Borneo, Indonesia	556
Eupleres goudotii	spermatogenesis: July-Sept; captive	120
	spermatogenesis: Aug-Sept, end winter; Madagascar	121
	birth: Nov; Madagascar	121
Fossa fossa	mate: Aug-Sept; Madagascar	119, 120
	birth: Nov-Dec; Madagascar	119, 120
Genetta spp	lact f: Jan; n1; s Cameroun, Benito, Eq Guinea	647
Genetta felina	young seen: Oct; Angola	11697
Genetta genetta	mate: Jan; captive	9215
	mate: Jan-Sept, peak Feb-Mar; Spain	471
	mate: Oct-Feb; Zimbabwe, Zambia, Malawi	10153
	preg f: Jan; n1; Transvaal, S Africa	8965
	preg f: Oct-Feb; few data; Botswana	10154
	preg f: Nov; n1; Uganda, e Africa	4924
	birth: Mar-June Sept-Nov; captive	9216
	birth: Mar-Nov, peak Apr-June; Spain	471
	birth: Apr; captive	9215
	birth: Aug-Apr; S Africa	10158
	birth: Nov; n1; Tanzania	5761
	birth: spring, fall; zoo, Czechoslovakia	5761
	birth: summer; Transvaal, S Africa	8965
	lact f: Feb, Apr; Transvaal, S Africa	8965
Genetta maculata	preg f: Feb; n1; e Africa	4924
	lact f: Jan-Feb; n1; e Africa	185
Genetta tigrina	preg f: Nov; Transvaal, S Africa	8965
	birth: Aug-Mar; S Africa	10158
	birth: Aug-Nov, Mar; Natal, S Africa	9327
	birth: Feb-May; few data; Zaire	11271
Nandinia binotata	birth: peaks May, Oct; e Africa	5761
	juvenile: Oct; n1; Zambia	331
Osbornictis piscivora	preg f: Dec; n1; Zaire	4434
Paguma larvata	sperm: Mar-July; captive	10930
	testes: large Mar-Aug, small Sept-Feb; captive	10930
	mate: May-June; captive	10930
	lact f: June; n1; central Asia	179
Paradoxurus hermaphroditus	birth: yr rd, peak before monsoon; Sri Lanka	8495, 8501
Poiana richardsoni	lact f: Oct; n1; s Cameroun, Benito, Eq Guinea	647
Prionodon linsang	preg f: May; n2; w Malaysia	6446a
	lact f: May, Oct; n2; w Malaysia	6446a
Prionodon pardicolor	repro: Feb, Aug; Sri Lanka	988a, 8619
Viverra tangalunga	preg f: June; n2; n Borneo, Indonesia	2313
Viverricula indica	birth: Sept-Oct; Madagascar	120
	birth: yr rd; e of Indus, Pakistan	9175
	birth: yr rd; Sri Lanka	8501

Order Carnivora, Family Herpestidae

Mongooses (*Atilax* 2.5-5 kg; *Bdeogale* 1-3 kg; *Crossarchus* 450-2000 g; *Cynictis* 410-900 g; *Dologale* 300-400 g; *Galidia* 700-900 g; *Galidictis* 320-340 mm; *Helogale* 185-680 g; *Herpestes* 225 g-4 kg; *Ichneumia* 2-5 kg; *Liberiictis* 2.3 kg; *Mungos* 1-2 kg; *Mungotictis* 600-900 g; *Paracynictis* 1.0-2.5 kg; *Rhynchogale* 2-3 kg; *Salanoia* 780 g; *Suricata* 620-1000 g) are primarily terrestrial, often diurnal, sometimes colonial carnivores from Africa, Madagascar, central and southeast Asia. Females have an extra chromosome (3492). Male *Herpestes* are often larger than females (5761, 8495, 8501, 9263, 9945).

Induced ovulation may occur in *Herpestes* (4883). *Helogale* family groups are distinctive as usually only the dominant female produces young (2126a, 2126b, 2128, 8930, 9261, 9262), but lactation may occur spontaneously in other females (2127). Placentophagia is reported for *Mungotictis* (120). Females have a bipartite uterus (*Herpestes* 8308). Estrus occurs 1-2 wk (2763, 3129, 9175, 9259) after the birth of usually blind (not *Mungotictis* 120), but furred neonates (757, 4180, 5761, 6215, 9175, 11253). Eyes open at 6-8 d (*Galidia* 120), 8-12 d (*Atilax* 515, 3515; *Mungos* 9984, 10158, 11301), 12 d (*Crossarchus* 3866), 13-14 d (*Helogale* 5761; *Suricata* 3129), or 16-21 d (*Herpestes* 757, 5225, 9175). Females have 2-6 mammae (331, 3129, 3134, 5761, 8308, 9945, 10158). Males have a baculum (8308; cf 9175).

Details of anatomy (7441, 7442, 8308, 9965), endocrinology (4883, 4884, 7838, 10190, 10191, 10192), reproductive energetics (2126a), and light (10193, 10194) are available.

NEONATAL MASS		
Atilax paludinosus	~100 g; from graph	515
Galidia elegans	50 g	117, 120
Herpestes auropunctatus	r19.8-22.1 g; n3 term emb	10863
	r22.3-36.4 g; n18 term emb	7829
	29 g; n1	10863
Mungos mungo	20 g	10158
	24.2 g; n1, 3 d	9984
	73 g; n3, 1 lt [5.4% of 1.35 kg]	11301
Mungotictis decemlineata	50 g	120
Suricata suricatta	25-30 g	9476
	30.5 g; r25-36; n4	3129
NEONATAL SIZE		
Crossarchus obscurus	HB: 9-10 mm [*sic*, est units should be cm]	3866
Galidia elegans	110-130 mm	117, 120
Herpestes auropunctatus	CR: r51-58 mm; n3 term emb	10863
Mungos mungo	HB: 99 mm; n1, 3 d	9984
	HB: 100 mm	5761
	~180 mm	9985
WEANING MASS		
Atilax paludinosus	~550 g; from graph	515
Galidia elegans	183 g; r150-200; n3	117
Herpestes auropunctatus	250-300 g; 10-12 wk	7829
Herpestes ichneumon	800 g; 12 wk [1 mo after weaning]	757
Suricata suricatta	225.5 g; r172-275; n6, 5 wk	3129
LITTER SIZE		
Atilax paludinosus	1 emb; n1 f	331
	1.67; r1-2; n3 f	10158
	2-4 yg	11253
	2 yg; n1 lt	12128
	2.5 yg; sd0.54; r2-3; n6 lt	515
	3 yg ea; n2 lt	3515
Bdeogale crassicauda	1 emb; n1 f	4924
	1 yg; n1 f	4924
Crossarchus alexandri	max 4 yg	5761
Crossarchus obscurus	mode 4 yg; r2-4	3866
	5 emb; n1 f	2955
Cynictis penicillata	1.3 yg; r1-2; n3 lt	12128
	1.67; r1-2	1230
	2.33 emb; r2-3; n3 f	9327
	3.2 emb; r2-5; n6 f	10154, 10158

Galidia elegans	1 yg ea; n3 lt	6215
	1 ea; n11 lt	117, 120
Galidictis fasciata	est 1 yg	120
Helogale parvula	1-6	9260
	2-4; max 7	10153, 10158
	2 emb, n1 f	331
	3 yg, n1 f	331
	3; r2-4; n2 lt, 1 f	4180, 10154
	4 yg; r1-7	8930a
Herpestes auropunctatus	1.7 yg; r1-3; n3 lt	12128
	1.8 emb; se0.35; r1-4	3952
	2-3 yg; n9 f	538
	2.1 emb; r2-4; n52 f	8546
	2.12 ps; r1-3; n154 f	1932
	2.14 emb; r1-3; n69	7829
	2.19 emb; r1-5; n283	7829
	2.25 yg; n27	7829
	2.4 implantations; r2-4; n17 f	8308
	2.4 ps; n35 f	8308
	2.7 cl; r1-4; n16 f	8308
	2.7 emb; r2-4	538
	2.7 emb; r2-4; n9 f	8308
	2.9 emb; r1-3; n68 f	1932
Herpestes edwardsi	1 yg; n1 lt	12128
	mode 2	8501
	2 emb ea; n2 f	7382
	2 ea; n6 lt, 1 f	9175
	3-4	8495
Herpestes fuscus	mode 3-4	8495, 8501
Herpestes ichneumon	2-4	11253
	2-5	8223
	2; n1 lt	2763
	3 emb; n1 f	1149
	3.3 yg; sd0.67; r1-4; n10 lt	757
Herpestes pulverulentus	2-3; 4 not uncommon	9945
Herpestes sanguineus	1.45; mode 1; r1-3; n11 lt	9263
	1.8 emb; r1-2; n5 f	10154, 10158
	2 emb; n1 f	2096
	3 emb; n1 f	11271
	3 yg; r2-4; n3 lt	10714
Herpestes vitticollis	2-3	8501
Ichneumia albicauda	2-4	5277
	2 emb; n1	9327
	2.4 emb, yg; r1-3; n5 f	10153, 10158
Mungos mungo	1 emb; n1 f	6611
	1 yg ea; n2 lt	12128
	2-8	9327
	mode 2; max 3	10153
	2; n1 lt	9441
	2.4 yg	9259
	3-5 yg	9985
	3 yg; 1 lt	11301
Mungotictis decemlineata	mode 1 yg; r1-2	120
Paracynictis selousi	2 yg	4744
	max 2	10153
	2-4	10158
Rhynchogale melleri	3 emb; n1 f; 2 yg ea; n2 nests	10153, 10158
Suricata suricatta	2.9; r2-5; n34 f	10158
	3; n1 lt	4180
	3.75 yg; r3-4; n4 lt	9476
	mode 4; r2-5	3134
	4.0 yg ea; n2 lt	3129
	4.0 yg; r2-5; n4 lt	1230
	4.1 yg; r2-7?; n3 lt	12128

SEXUAL MATURITY		
Atilax paludinosus	m: testes descend: 255.5 d; r204-302	515
Crossarchus obscurus	f/m: ~9 mo	3866
Helogale parvula	f: 1st preg: 15 mo; n1	9260
Herpestes auropunctatus	f: 1st mate: 9 mo	8308
	f: 1st conceive: min 130 d; n1	7829
	m: 1st mate: <6 mo; 400-500 g	7829
Herpestes ichneumon	f/m: 1st mate: 2 yr	2763
Herpestes sanguineus	f: 1st estrus: 13.5 mo	5225
	f: 1st preg: early 2nd yr	9263
Mungos mungo	f: 1st estrus: 9-10 mo	9985
	f: 1st birth: ~1 yr	9985
	f: 1st birth: ~1 yr	9441
Suricata suricatta	f: 1st birth: 1 yr	3129
	m: 1st fertile mating: 1 yr	3129
CYCLE LENGTH		
Galidia elegans	mode 12 d	6215
Helogale parvula	estrus: 1-6 d; n1 f	8930a, 8931
Herpestes auropunctatus	estrus 3 d	7829
	3 wk	7829
Mungos mungo	estrus: 3-6 d	9985
GESTATION		
Crossarchus obscurus	~8 wk	3866
	~10 wk	4354a
Galidia elegans	60.67 d; r52-77; n3	6215
	82.6 d; r74-90; n11	117, 120
Helogale parvula	54-56 d	8930a
Herpestes auropunctatus	6 wk; n1	9175
	46-49 d	3952
	~7 wk	538, 8308
	49.3 d; r49-50; n3	7829
	49.6 d; r47-53; n8	10863
Herpestes ichneumon	60 d	757
	11 wk	5761
Herpestes sanguineus	58-62 d	5225
Mungos mungo	~60 d	9985
	60-62 d	2449
	~2 mo	9259
	~2 mo	9984
Mungotictis decemlineata	100 d; r90-105; n3	120
Suricata suricatta	≤77 d	3129, 3134
LACTATION		
Atilax paludinosus	solid food: 14 d; prechewed; 30 d; not prechewed	3515
	wean: 35.9 d; r30-46	515
Crossarchus obscurus	solid food: ≤3 wk	3866
Galidia elegans	solid food: 1 mo	117
	wean: 2 mo	117, 120
Helogale parvula	solid food: 10-12 d	8930a
	wean: after ~3 wk; den emergence: 13 d	5761
Herpestes auropunctatus	nurse: 3-5 wk	3952
	nurse: est 4 wk	8308
	wean: 4-8 wk	7829
	independent: 12 wk	10863
Herpestes ichneumon	in nest: 6 wk	757
	solid food: 32 d	757
	nurse: 2 mo	757
Herpestes sanguineus	wean: ~50-65 d	9263
	wean: 52 d	5225
Mungos mungo	den emergence: 3 wk	9258
Mungotictis decemlineata	solid food: 15 d	120
	wean: 2 mo	120
Suricata suricatta	solid food: 4-6 wk	3129
INTERLITTER INTERVAL		
Herpestes auropunctatus	4 mo	7829

LITTERS/YEAR

Atilax paludinosus	2	11253
Crossarchus obscurus	2-3	3866
Galidia elegans	1	117
Helogale parvula	2	11253
	2-3	9260
Herpestes auropunctatus	est 2	538, 8308
	2	7829
	2	8546
	2	3952
Herpestes edwardsi	3.3; 5 lt/18 mo; n1 f	9175
Herpestes ichneumon	mode 1; 2 with good conditions	757
	2?	11253
Herpestes sanguineus	2	11253
Mungos mungo	1	11253
	max 4	9259
Suricata suricatta	1	3134

SEASONALITY

Atilax paludinosus	preg f: Oct; n1; n Zimbabwe	331
	birth: July, & rainy season; Zaire	11253
	birth: Aug-Dec; S Africa	10158
Bdeogale crassicauda	preg f: Dec; n1; Mt Kilimanjaro, e Africa	4924
	lact f: Dec; n1; Mt Kilimanjaro, e Africa	4924
Crossarchus obscurus	preg f: Mar; n1; Cameroun	2955
	birth: yr rd; captive	3866
Cynictis penicillata	preg f: Sept; n3; Natal, S Africa	9327
	birth: peak Oct-Jan; S Africa	10158
	lact f: Nov; Natal, S Africa	9327
Galidia elegans	mate: Apr, July-Oct, Dec; captive	120
	mate: July-Nov; Madagascar	117, 120
	birth: Apr, Aug-Sept; n3; zoo	6215
Galidictis fasciata	birth: summer; Madagascar	120
Helogale parvula	estrus: end Mar-early Apr; zoo	8931
	mate: Oct-Dec; Zambia, w Zimbabwe	10153
	preg f: Feb; n2; Botswana	10154
	birth: Oct-Mar; S Africa	10158
	birth: Nov-May; Tanzania	9260
	birth: twice/yr; Zaire	11253
Herpestes auropunctatus	spermatogenesis: yr rd; Fiji I	3952
	testes: large Feb-Mar; HI	8308
	testes: large Sept-Dec, small Feb-July; Fiji I	3952
	mate: Jan-Oct; Puerto Rico	8546
	mate: Feb-Aug; Oahu, HI	10190, 10194
	mate: end Feb-July; HI	538
	mate: end summer, with monsoon; Pakistan	9175
	preg f: Feb-Aug; Virgin I	1932
	preg f: Feb, Aug-Sept; Oahu, HI	4884
	preg f: end Feb-July; HI	8308
	preg f: Aug-Feb, peak Oct-Dec; Fiji I	3952
	preg f: yr rd, low Nov-Dec; Grenada	7829
	birth: peaks Mar-Apr, July-Aug; Puerto Rico	8546
	lact f: Mar-Oct; Oahu, HI	4884
Herpestes edwardsi	preg f: Apr; n2; Nepal	7382
	birth: Apr, July; zoo	44
	not seasonal; Sri Lanka	8495, 8501
Herpestes fuscus	birth: yr rd, peak 1st few mo of yr; Sri Lanka	8495, 8501
Herpestes ichneumon	mate: end Feb-Mar; Israel	757
	mate: Mar-Apr; n1 pr; zoo	2763
	birth: Feb-Sept; s Uganda	5761
	birth: Apr-May; Israel	757
	birth: July; n1 pr; zoo	2763
	birth: July-Aug; s Spain	5761
	birth: twice/yr, rainy season; Zaire	11253

Herpestes sanguineus	mate: Aug-Sept, sometimes Apr; captive, S Africa	5225
	preg f: Feb-Apr, Oct-Nov; e Africa	10715
	preg f: Dec; n1; Botswana	10154
	birth: Oct-Mar, peak Oct-Dec; captive, S Africa	5225
	birth: mid dry season, rainy season; Zaire	11253
	lact f: Feb; n2; Botswana	10154
	lact f: June; n1; Kenya	4924
	repro: yr rd; Serengeti, Tanzania	9263
Herpestes vitticollis	lact f: May; n1; Sri Lanka	8501
Ichneumia albicauda	birth: Sept-Dec; S Africa	10158
	birth: Oct-Jan; s Africa	10153
Liberiictis kuhni	lact f: Aug; n1; Liberia	3867
Mungos mungo	mate: end Aug, Oct; Natal, S Africa	9327
	mate: Oct-Jan; Zambia	10153
	preg f: Oct-Jan; captive	9984
	birth: Oct-Feb; S Africa	10158
	birth: Oct, Dec, Feb; Natal, S Africa	9327
	birth: Nov-Dec; Zaire	11253
	birth: peak Dec; captive	9984
	repro: yr rd; Uganda	9259
Mungotictis decemlineata	mate: June-Dec; captive, Madagascar	120
	birth: Sept-Mar; captive, Madagascar	120
Paracynictis selousi	birth: Aug-Mar; S Africa	10158
	birth: Oct; s Zambia, Zimbabwe	10153
Rhynchogale melleri	birth: ~Nov; Zimbabwe	10153
Suricata suricatta	mate: Sept-Oct; captive	3129, 3134
	birth: Jan, Nov, Sept; n4; zoo	1230
	birth: Oct-Mar; S Africa	10158
	birth: Oct-Apr; via natives; sw Africa	4560
	birth: Nov-Dec; captive	3129, 3134

Order Carnivora, Family Protelidae, Genus *Proteles*

The 5-14 kg aardwolf, *Proteles cristatus*, from the arid plains and woodlands of Africa eats termites by night and rests in burrows by day. Monogamous pairs copulate 1-4.5 hr (9104). Neonates are born blind and helpless (5761), and reach adult size by 9 mo (5761). Females have 4 teats (9945). Details of anatomy (11584) are available.

LITTER SIZE		
Proteles cristatus	mode 2-3; r1-5	5761
	most 2-4 yg; max 5	5907
	2.36 yg; r1-4; n11 lt, 5 f	9103, 9104
	3 yg; n1 nest	9945
	3 yg; r2-4	10153, 10158
CYCLE LENGTH		
Proteles cristatus	estrus: 1-3 d	5907, 9104
GESTATION		
Proteles cristatus	55 d; r54-56; n2; field data	9103
	~90 d	5907
	~3 mo; r90-110 d	5761
LACTATION		
Proteles cristatus	den emergence: 1 mo	5907
	den emergence: 2 mo	9103
	solid food: 9-12 wk	5907
	solid food: 105-120 d	9103
	wean: 120-145 d	9103, 9104
SEASONALITY		
Proteles cristatus	mate: Oct-Nov; Zimbabwe	10153
	mate: mid May-mid Aug; S Africa	9103, 9104
	birth: May; n1; s Kenya	5761
	birth: yr rd, peak Oct-Feb; S Africa	10158
	lact f: Nov; n1; Namibia	9945

Order Carnivora, Family Hyaenidae

Hyaenas (*Crocuta* 35-85 kg; *Hyaena* 25-55 kg) are terrestrial, nocturnal carnivores from Africa and southwest Asia. *Crocuta* females are larger than males and have male-like genitalia (1716, 2315, 4075, 4265, 6074, 6963, 7802, 7858, 11173, 11481; cf 11615), while *Hyaena* males are larger than females (5761, 10033). The labyrinthine, hemochorial, villous, zonary placenta (232, 2479, 6963, 6969, 7555, 8031, 11967, 11968) implants in a bipartite uterus and is probably eaten after birth (3861). A postpartum estrus occurs 2-3 wk (*Hyaena* 9621) after the birth of furred neonates that either have their eyes open (*Crocuta* 2427, 5761, 6964, 8709, 9621, 10158) or whose eyes open 1-2 wk later (*Hyaena* 5761, 7979, 9128, 9670, 10158). Females have 4-6 mammae (9175, 9621, 9945, 11482). Males have penial spines (11584). Details of anatomy (6963, 11482, 11584) and endocrinology (*Crocuta* 3471, 3880, 6471, 6472, 8838; *Hyaena* 3046) are available.

NEONATAL MASS

Crocuta crocuta	m: 1.1 kg; n1	9621
	1.46 kg; n3	8709
	1.5 kg; r1.35-1.6; n3	3861
Hyaena brunnea	f: 635 g; n1	6182
	m: 635 g; n1	6182
	f: 657.5 g; r645-680; n4	9670
	693.2 g; se17.6; n13	7349
	710.5 g; r635-812; n4	2879
	m: 727.5 g; r690-805; n4	9670
	760 g; n1	6182
Hyaena hyaena	650 g	9444
	f: 660 g; n1	10033
	m: 668 g; r661-675; n2	10033
	700 g	9130

WEANING MASS

Crocuta crocuta	14.5 kg; n1, 100 d	8709
Hyaena brunnea	27.4 kg; r24.9-29.9; n2, ~10 mo	7349

LITTER SIZE

Crocuta crocuta	1-2	6964
	1 emb; n1 f	6612
	1.5 yg; r1-2; n2 lt	3861
	1.5 yg; r1-2; n9 lt	12128
	1.83 yg; se0.56; mode 2; r1-3; n24 lt	3470
	2-4	8514
	mode 2; r1-3	2427
	2; r1-4	10158
	2 emb, 2 yg; n2 f	4924
	2 yg ea; n2 lt	9026
	2; r1-3; n9 lt, 2 f	7555
	2.2 yg; r1-3; n5	8709
Hyaena brunnea	1; n1 lt	1230
	1.8; mode 4; r1-5; n6 lt	11616
	2.0; r1-4; n6 lt	8170
	2.3 yg; r1-4; n9 lt	2879
	2.3; r1-5; n61 lt (literature compilation)	7348
	2.8 yg; r1-5; n5	10024
	3.0 yg; r1-4; n3 lt, 1 f	9670
	3.0 yg; r2-4; n5 lt, 2 f	7350
Hyaena hyaena	r1-4; n21 lt	9621
	1.33; n3 lt	1230
	mode 2-4; r1-6	5761
	2.3; r2-3; n3 lt	10033
	2.4 yg; r1-5; n43 lt	9128, 9129
	2.5 yg; r2-3; n2 lt	12128
	4 yg; n1 lt	9444

SEXUAL MATURITY		
Crocuta crocuta	f: 3 yr	9621
	m: 1-2 yr	5761, 10158
Hyaena brunnea	f: 1st birth: 3 yr; from graph	7350
Hyaena hyaena	f/m: 2-3 yr, rarely 1.5	9128
CYCLE LENGTH		
Crocuta crocuta	estrus: 1 d	9627
	14 d; +14 d anestrus	4075
Hyaena brunnea	estrus: 1-4 d	2879
	estrus: 1 wk	7352
	30 d; n3-4 cycles/f, 2 f	2879
Hyaena hyaena	estrus: mode 1 d; r1-2	9130
	40-50 d	9130
GESTATION		
Crocuta crocuta	110 d; r97-132	9621
Hyaena brunnea	74-93 d; n1	6182
	~3 mo	7345
	95.8 d; r93-100; n5	2879
	98-99 d; n1	4075
Hyaena hyaena	89.3 d; r86-92; n6 lt	9128, 9129
	90 d	7979
	90-91 d	4582
	90-92 d	8798
LACTATION		
Crocuta crocuta	den emergence: 5 wk	11253
	solid food: 2-4 wk	9026
	solid food: 6 mo	6074
	solid food: 7-8 mo	5761, 10158
	wean: 8 wk	9026
	wean: ~6 mo	6963
	wean: 12-14 mo	6074
	wean: ~1 yr-16 mo	5761
Hyaena brunnea	solid food: 3 mo	7352
	wean: 9 mo	7352
	min 10 mo	8169
	wean: 12-15 mo	10158
Hyaena hyaena	solid food: 30 d	9128
	solid food: >2 mo	7979
	nurse: 4-5 mo	9130
INTERLITTER INTERVAL		
Crocuta crocuta	~12 mo	6963
	1 yr	11253
	15.9-19.3 mo	10158
	17.4 mo; r9-26; n7 intervals	6074
Hyaena brunnea	1 yr	7352
	1.8 yr; r0.9-3.5; n5 intervals; from graph	7350
Hyaena hyaena	est 0.3-1 yr	7979
SEASONALITY		
Crocuta crocuta	mate: yr rd; central, s Africa	5761
	preg f: May; n1; e Africa, Eritrea, Somalia	4924
	preg f: yr rd; Serengeti, Tanzania	6963, 6964
	birth: Mar-Apr; Africa	5774
	birth: early rainy season; Zaire	11253
	birth: yr rd, peak June-Aug; Kruger Nat Pk, S Africa	3145
	birth: yr rd, peak end summer; S Africa	10158
	lact f: May; n1; e Africa, Eritrea, Somalia	4924
	repro: Oct-Apr; Elosha Game Reserve, Namibia	2427
	repro: yr rd, Jan-Mar, peak end summer; S Africa	6470
Hyaena brunnea	estrogen: high Sept; zoo	3046
	mate: end Nov; zoo	2879
	birth: May; n1; zoo	1230
Hyaena hyaena	birth: Jan-Feb; n3; zoo	1230
	birth: Apr-May; central Asia	7979
	birth: yr rd; captive	9129

Order Carnivora, Family Felidae

Cats (*Acinonyx* 22-72 kg; *Felis* 1-100 kg; *Lynx* 4-32 kg; *Neofelis* 16-37 kg; *Panthera* 17-320 kg) are terrestrial, primarily nocturnal, usually solitary carnivores ranging over most of the earth. Males are larger than females (833, 852, 3571, 4924, 5761, 6468, 7979, 8238, 9175, 9360, 9546, 9559, 9567, 9675, 9945, 10172, 10499, 10760).

Ovulation is probably triggered by coitus (7496, 9706, 9708, 11687; cf 9704). The 25 g (small *Felis* spp), labyrinthine, zonary or annular placenta (6367, 6965, 7563, 7839, 9563) implants into a bicornuate uterus and is eaten after birth (6268, 10158, 10788, 10870). Postpartum estrus may occur (2875, 2877, 2878, 5400, 5592, 5838, 9358, 10158, 12045; cf 1018, 1230). Cats are born with fur and closed eyes (2494, 3202, 3625, 5761, 7088, 7624, 9175, 9546, 9563, 10157). Their eyes open at 1-15 d (*Acinonyx* 2883, 3339, 5761, 10020, 10158, 10869, 11284; *Felis* 41, 300, 370, 2375, 3571, 4462, 5761, 6422, 6423, 6605, 7259, 7781, 7979, 8407, 8639, 8822, 9560, 9564, 9565, 9573, 9574, 9674, 9675, 10870, 10871, 11038, 11331, 11378, 11936, 12045; *Panthera* 29, 833, 2109, 2375, 3537, 5465, 5761, 5930, 6829, 7979, 9546, 10308, 10499) or 8-17 d (*Lynx* 1474, 3571, 3982, 4314, 5761, 5995, 7979, 9175, 10499). Females have 4-8 mammae (331, 1474, 3134, 3571, 6965, 9546, 10760, 11633). Males have a baculum (6926, 9559, 9568, 9569).

Details of anatomy (2770, 6428, 6965), endocrinology (*Acinonyx* 9796, 11684, 11686, 11688; *Felis* 1096; *Panthera* 968, 1357, 1358, 9604, 9605, 9704, 9706, 11687), semen (11680, 11684, 11686), capicitation (1515), and ovulation (*Felis* 7495, 7496, 7822; *Panthera* 2718, 9348) are available. Information on the reproduction of domestic cats is reviewed in supplement 39 of the Journal of Reproduction and Fertility published in 1989.

NEONATAL MASS

Acinonyx jubatus	250-280 g; n3	3339
	~300 g; n1	3338
	m: 520 g; n1	11333
	m: 666.7 g; r600-700; n3	11284
	f: 695.0 g; r690-700; n2	11284
Felis aurata	215.0 g; r195-235; n2	10870
Felis bengalensis	82.5 g; r75-95; n4	2282
	92.5 g; r90-95; n2	8639
	113.14 g; r93-120; n7, 4 lt	26
Felis chaus	106.21 g; r83-125; n24	35, 41
	156.25 g; r150-161; n4	9565
Felis concolor	225-450 g	12045
	390-450 g	5210
	450 g	7088
Felis geoffroyi	65 g ea; n2	9574
	91.7 g; n1	6268
	122.9 g; n1 with placenta	6268
Felis manul	89 g; n1	9563
Felis margarita	39 g	9573
	39 g; se2	4646
	105 g; n1	4644
Felis marmorata	70-100 g	9564
Felis nigripes	72.0 g; r60-84; n2, 1 d	6423
	m: 60 g; n1	9674
	85.2 g; r82.5-87.9; n2	370
Felis pardalis	200-276 g	9567
	206-340 g	8407
Felis serval	246.0 g; r230-260; n5	11378
	280 g; r270-290; n2	3168
Felis silvestris	~40 g	3571
	65 g; n1 stillborn	11331
	82-149 g; median 116; n71	813
	83-120 g	9560
	97.0991 g; 95%ci 23.1204; n78	7839
	m: 97.75 g; r82.7-107.4; n6	6240
	100 g	10871
	~100-163 g live; r65-163; n54, with stillborns	9561
	f: 103.92 g; r97.0-119.5; n6	6240

	106.4 g; sd13.5; se0.81; r70-144; n128, 33 lt	4237
	106.5 g; r103-110; n2 term emb	2490
	119 g; r97-141; n6; 5 stillborn, 1 alive	6437
	119.44 g; r80-135; n9	11331
	134.67 g; r100-163; n24	7259
Felis temmincki	110 g; n1 black fur, did not survive	24
	250 g; n1	6605
Felis viverrina	170 g; without tail	11038
Felis wiedii	m: 163 g; n1	8407
	f: 170 g; n1	8407
Lynx canadensis	204 g; r197-211; n2	9510a
Lynx caracal	165 g; n1	5592
Lynx rufus	112-226 g	1474
	~300 g	1871a
Lynx lynx	289 g; r250-360	10942
Neofelis nebulosa	142-170 g; n4	3378
	200 g; n1	7648
Panthera leo	1000-2000 g	5761
	f: 1152 g	10308
	1199 g; r1100-1350; n5	2278
	1200-2100 g	9546
	f: 1266 g; n571	9730
	1284.5 g; r1210-1359; n2	2375
	m: 1288 g	10308
	1306 g; r540-2190; n1173	9730
	m: 1345 g; n601	9730
	1466.3 g; r1243-1800; n3	2109
	~1500 g	10158
Panthera onca	680 g; n1, 2 d	5098
	800 g; r700-900	9811
	840 g; r510-1070; n29	9730
	m: 847 g; n1	2109
	m: 950 g; n1, 1 d	10332
Panthera pardus	43-60 g	5761
	50-60 g	10158
	710 g; r500-1000; n7	2494
	560 g; n1	2109
	498.5 g; r430-567; n2	2375
	553.4 g; r338-858; n174	9730
Panthera tigris	f: 678-1243 g	2109
	785-1043 g	75
	785-1500 g	10499
	1100-1430 g	29
	f: 1300.0 g; r950-1600; n6	11283
	f: 1341.4 g; r990-1575; n24	9729
	m: 1386-1700 g	2109
	m: 1387.5 g; r1100-1610; n4	11283
	1400 g	7979
	1496 g; r808-1893; n128	9730
	m: 1519.75 g; r1210-1760; n20	9729
Panthera uncia	300-380 g	6829
	f: 320 g; n1	5465
	r320-567 g; n7	4617
	m: 350 g; n1	5465
	540.9 g; r368-708; n10	3537
	581.1 g; r510-652; n10, 1 d	5838
	623 g; n1	3496
	666 g; r623-709; n2	3499
NEONATAL SIZE		
Acinonyx jubatus	HB: ~250 mm	3338, 3339
Felis bengalensis	178.5 mm; r175-182; n2	8639
	TL: 241.4 mm; r220-250; n7, 4 lt	26
Felis chaus	TL: 231.7 mm; r220-226; n24	35, 41

Felis concolor	200-300 mm	12045
	300 mm	7088
Felis geoffroyi	HB: 135 mm; n2	9574
Felis manul	HB: 152 mm; n1	9563
Felis margarita	143 mm	7979
Felis nigripes	150 mm	9675
Felis serval	f: 190 mm; n1	8856
	m: 195 mm; n1	8856
Felis silvestris	m: 152 mm; n1	185
	f: 155 mm; n1	185
	CR: 126.7 mm; 95%ci 9.2; n78	7839
Felis temmincki	HB: 200 mm; n1 black fur, did not survive	24
Lynx canadensis	CR: 160.5 mm; r158-163; n2	9510a
Panthera onca	400 mm	9811
Panthera pardus	150 mm	5761
Panthera tigris	315-400 mm	75
	364 mm	29
Panthera uncia	HB: 233.5 mm; r230-237; n2	4617
WEANING MASS		
Felis chaus	2212.5 g; r2100-2300; n4, 90 d	9565
Felis concolor	3-4 kg; 1-2 mo	2186
Felis silvestris	m: 362.73 g; r266.4-486.7; n6, 4 wk	6240
	398 g; sd70; r264-521; n70, 30 d	4237
	f: 402.42 g; r330.2-475.4; n6, 4 wk	6240
	604 g; sd102; r390-876; n70, 50 d	4237
	600 g	2020
	f: 683.8 g; r644.8-760.0; n6, 8 wk	6240
	m: 713.92 g; r558.7-820.4; n6, 8 wk	6240
	968.0 g; r890-1230; n5, 2 mo	7259
	1055 g; r1030-1080; n2, 62 d	11331
Neofelis nebulosa	f: 1055 g; r1050-1060; n2, 6 wk	3202
	7500 g ea; n2, 32 wk	3202
	m: 1155 g; n1, 6 wk	3202
	12000 g; n1, 32 wk	3202
Panthera tigris	11003 g; r10760-11850; n3, 88 d	11283
LITTER SIZE		
Acinonyx jubatus	1 yg; n1	3338
	1 yg; n1 lt	6794
	2 yg; n1 lt	9504
	2 yg; n1 lt	6793
	2.5 yg; r2-3; n2 lt	10788
	3; n1 lt	3339
	3; n1 lt	4659
	3 yg; n1 lt	11065
	3 yg; n1 lt	334
	3 yg; n2 lt, 1 f	6792
	3; r1-5; max 6	10158
	3.1 yg; n14 lt	6795
	3.48; n217 lt	7992
	3.7; r1-8; n15 lt	2873, 2880, 2883
	4 yg; n1 lt	10020
	4 yg; n1 lt	3001, 3002
	4; r3-5; n2 lt	11284
	5 yg; n1 lt	10869
	5 *ea*; n2 lt	11333
Felis aurata	2 yg *ea*; n2 lt	10870
Felis bengalensis	1.75 yg; r1-2; n4 lt	26
	2	6448a
	2-3	7781
	2.0; r1-3; n3 lt	3514
	2.4; n18 lt	8639
	3-4	8619
	3; n1 lt	179
	3.0; n5 lt, 1 f	9175

	3 yg ea; n5 lt	2282
Felis chaus	2.89; mode 3; r1-6	9565
	3; n1 lt	8822
	3 emb; n1 f	7382
	3-5; max 10	7979
	3.5; mode 3; r3-5; n8 lt	35, 41
Felis concolor	r1-4 yg; n14 f	9385
	2 emb; n1 f	7324
	2.4; r1-4	7088
	2.4; mode 3; r1-4; n35	1230
	2.5 yg; r2-3; n2 lt, 1 f	8822a
	2.7; n11 lt	2881
	2.7 yg; n11 lt, 9 f	6545
	2.76; mode 3; r1-4	11338
	2.9; r1-4; n7 lt, 1 f	7234
	2.9 yg; r1-5; n258 lt	9185
	3 emb; n1 f	6723a
	3 yg; n1 f	8468
	3.0 yg; r2-4; n4 lt	9066
	3.4 emb; r1-6; n66 f	9185
	4 yg; n1 lt	502
Felis geoffroyi	1	5476
	1.67 yg; r1-2; n3	6268
	2.0 yg; r1-3; n2	9574
	2.57 yg; r2-3; n7 lt	300
	3 emb; n1 f	11977
Felis manul	1; n1 lt	9563
	4-6 yg	9558
	5-6	10499
Felis margarita	mode 3; r2-5	9559
	4	4618
	6.0; r4-8; n2 lt	9573
	8; n1 lt	4646
Felis marmorata	2 yg	9564
Felis nigripes	1.6; r1-2; n5 lt	370
	1.7 yg; 1-2; n3	6423
	2 emb; n1 f	11311
	2; r1-3; n3 lt	9675
	2; r1-3; n5 lt	9674
	2.5 yg; r2-3; n2	11311
Felis pardalis	1-2 yg	9567
	1.3; r1-2; n3 lt	12128
	2 yg; n1 lt	5900
	2 emb; n1 f	8422
Felis rubiginosa	2 yg; n1 lt	8502
	2-3 yg	8501
Felis serval	1 yg; n1 lt	1018
	2.17 yg; n6 lt, 1 f	1014
	2.35 yg; r1-4; n20 lt, 3 f	11378
	2.4 yg; r1-4; n11 lt, 1 f	3168
	2.5 emb; r1-3; n7 f	10153, 10155, 10157
	3 yg; n1 lt	334
	4 cl; n1 f	9327
Felis silvestris	2.0; r1-4; n4 lt	10871
	2.4; r2-3; n5 lt	348
	3-5; max 10	7979
	mode 3-5; r3-10	4645
	3; r1-5; n16 lt	11331
	3.3 emb; r1-6; n39 f	2490
	3.4; r2-5; n7 lt	10154
	3.42 yg; r1-8; n61 lt	2019, 9560
	3.59 yg; sd1.55; mode 3; r1-8; n51 lt	6362
	3.6; mode 3; r2-6; n25 lt	2020
	3.63 yg; sd1.30; mode 3-4; r1-8; n119 lt	6362

	3.88 yg; mode 3,4; r1-8; n33 lt	4237
	4 yg; n1 lt	6395
	4; n6 lt	1973
	mode 4; r3-5	9843
	mode 4-5; r2-6	6468
	4; r1-8; n16 lt, 4 f	7259
	4.20 yg; se0.05; r1-10; n973 lt	9197
	4.35 emb; sd1.27; r2-7; 5.07 cl; sd1.28; r3-7; n17 f	5400
	4.59 emb; sd1.12; r2-7; 5.88 cl; sd1.12; r4-8; 2.66 weaned; sd0.76; n17	11072, 11074
	4.7 emb; r1-9; n14	2044
Felis temmincki	1 yg	24
	1 yg	6605
	2 yg	4813b
Felis tigrina	1-2	6422
Felis viverrina	1.5 yg; r1-2; n2	11038
	2; n1 lt	5275
	2; n1 lt	9368
Felis wiedii	1 emb; n1	7624
	2 yg; n1 lt	8207
Felis yagouaroundi	mode 2	5090
	2-3	1474
Lynx canadensis	1 ps, 4 ca; n1 f	10839
	2; r1-4	1474
	2 yg; n1 lt	9510a
	4.3-11.9 cl; across age	1229
	mode 4; r1-6	5620
	max 4 yg	5146
Lynx caracal	r1-4 emb; n7 f	8758
	1-6	5761
	max 2	10153
	2-4; max 5	10158
	2-4	9945
	est 2-4	8223
	2-5	10156
	2-6	5995
	2-6	9175
	2; r1-3	5592
	2; r1-2; n6	1230
	2 yg; n1 f	9327
	2.2 yg; r1-4; n37 lt	833
	2.3; r2-3; n3 lt	12128
	3-4	9496
Lynx lynx	1 yg; 1 lt	11501
	mode 2; max 4 rare	11226
	mode 2-3; r1-5	3571
	mode 2-3; r1-5	7979
	mode 2-3; r1-5	10499
	2 yg; n1 lt	4314
	2 yg; n1 lt	9066
	2.0; r1-3; n6 lt	8772
	2.2; r1-3; n5 lt, 4 f	6102
	2.26 yg; mode 2; r1-4; n19 lt	736
	2.5 emb; n4 f	3527
	2.50 emb; sd0.53; r2-3; 2.88 cl; sd0.64; r2-4; n8 f	6137b
	3.0; n3 lt	6103
	3.10 cl; se0.16; r1-10; n104 f	6137a, 6137b
	4.18 ps; n95 f	8813
	4.2 cl; r2-10; 2.5 ps; r1-4; n64 f	3527
Lynx pardinus	2-3 yg	10809
	2.5; mode 2; r1-4; n14 lt	11070
Lynx rufus	1 emb ea; n2 f	11228a
	2 emb; n1 f	8316
	2 yg; n1 f	4249

	2.02 ps; n664 f	5355
	2.22-2.74 ps; r1-6; n3 ages	8239
	2.3 yg; r1-3; n9 lt	5884a
	2.54 ps; ±1.09; n157	5885
	2.71 ps; ±0.79; n53	5885
	2.8 yg	1871a
	3.0 emb; r2-4; n3 f	502
	3.0 yg ea; n3 lt	9066
	3.0 emb; r2-4; n8 f	12044
	3.2 emb; mode 3; r1-8; n356 f	3625
	3.3; r2-5; n6 lt, n1 f	11815
	3.5 yg; mode 4; r1-6; n47 f	3625
	3.9 ps; mode 4; r3-5; n12 f	3625
	4.8 cl; mode 5; r2-9; n21 f	3625
Neofelis nebulosa	1.5 yg; r1-2; n2 lt	3660
	1.82; r1-2; n11 lt	7648
	3 yg; 1 lt	3202
	4 yg; 1 lt	3378
Panthera leo	1 yg; n1 lt	30
	1.7-2.3	9546
	mode 2-4; r1-9	5761
	2.1; r1-4; n7 lt	2278
	2.4; n41 lt	2875, 2882
	2.48 yg; r1-5; n64 lt	2039
	2.5; mode 3; r1-5	12128
	2.5 yg; r2-3; n59 lt, 4-6 wk	852
	2.57 yg; r1-6; n112 lt	9360
	2.6; r1-4; max 6; n19	10158
	2.75; r1-4; n4 lt	2583
	2.9 yg; r2-4; n15 lt	9066
	3; mode 3 52%; r1-4; n19 lt, 4 wk	9358
	3.01; r1-8; mode 3; n707 lt	9730
	3.02 yg; mode 3; r1-4; n47 lt	10170
	3.04; r1-6; mode 3; n323 lt	10412
	3.08 cl; mode 3; r1-5; n25 f	10170
	3.08 emb; mode 3; r1-5; n12 f	10170
	3.1; mode 3; r1-5; n84 lt	1230
	3.3; r3-4; n6 lt	9360
	7; n1 lt	11167
Panthera onca	1-4	10809
	1 emb; n1 f	7324
	1.67; r1-2; n3	10332
	1.93; mode 2; n15 lt	9730
	2-4	1474
	2; n5	1230
	3 yg; n1 lt	2109
Panthera pardus	1-2	9546
	1-4	2439
	1-4	8223
	1.5 yg; r1-2; n2 lt	2109
	1.6 yg; mode 1-2; r1-3; n39 lt	2494
	1.7 yg; r1-3; n60 lt, 8 f	2878
	1.70; se0.06; r1-3; mode 2; n159 lt	9196
	1.75; mode 1; r1-3	12128
	1.77 yg; n39 lt	2494
	1.78; r1-3	1230
	mode 2; r2-4	9175
	2-3; r2-5	10499
	2-3; max 6	331
	mode 2-3; max 6	10158
	2-4	6654
	2-4	9945
	2-5	7979
	2; n1 lt	6712

	2; r1-3; n17 lt	2936
	2.09; se0.06; mode 2, 4 rare; r1-4; n171 lt	9196
	2.2 yg; r1-3; n10 lt	2651
	2.32; mode 2; n75 lt	9730
	2.4 yg; r2-3	9066
	2.5 yg; r2-3; n2 lt	9726
	3 yg; n1 f	9725
	mode 3 born; mode 2 weaned	8501
	mode 3-4; r2-6	5761
	4	11253
Panthera tigris	1.53; mode 2; r1-4; n30 lt	1230
	mode 2-3; r1-5	19
	2.3; mode 3	12128
	2.3; n530 lt	9706
	2.33 yg; r1-3; n12 lt	29
	2.5 yg; r2-4; n15 lt	9066
	2.53 yg; r1-5; n652 lt	7407
	2.55 yg; r1-4; n27 lt	11089, 11090
	2.6; mode 2,4; r1-5; n22 lt	2272
	2.65; n148 lt	7004
	2.79; mode 2-3; r1-7; n107 lt	9474
	2.8 yg; r1-5; n15 lt	11283
	3 yg; n1 lt	2333
	3.63; n49 lt	9729, 9730
	5 emb; n2 f	3425
Panthera uncia	mode 2-3; r2-5	10499
	2; n1 lt	5465
	2 yg; n1 lt	1557
	mode 2; r1-4; n6 lt	6829
	2.04; r1-4; n47 lt	3499
	2.2; mode 2; r1-3; 19 lt	5930
SEXUAL MATURITY		
Acinonyx jubatus	f: 1st conceive: 21-22 mo	5761
	f: 1st birth: 3.67 yr	3338
	f: 21-24 mo	10158
	f/m: 1st birth: 2.5-3.5 yr	6795
	f/m: 14-16 mo	2873
	f/m: ~2 yr	8798
Felis bengalensis	f: 1st birth: 13-14, 31, 36 mo; n3	9566
Felis chaus	f: 11 mo	9565
Felis concolor	f: 1st mate: 2-3 yr	12045
	f: 1st birth: 20-21 mo	2881
	f: repro: 3-4 yr	6545
	m: 1st mate: 16-29 mo	2881
Felis nigripes	f: 1st estrus: 251 d	370
	m: spraying: 228 d	370
Felis pardalis	f/m: 1st mate: 20-24 mo	9567
	f/m: 16-18 mo	9567
Felis serval	f: 1st estrus: 26 mo	3168
	f: 1st mate: 28 mo	1018
	f: 1st birth: 28 mo	3168
	m: 1st mate: 17 mo	1018
Felis silvestris	f: 1st cl: ~8 mo	11072
	f: 1st birth: 11-13 mo; n5	2019, 9560
	f: 1st birth: 1 yr	7259
	f: 10 mo	11331
	f: 10 mo	3571, 9175
	f: 10-12 mo; 2500 g	5400
	f: < 12 mo	6965
	m: spermatogenesis: ~7 mo; 2.4 kg; ±0.17; min 2.2	11072
	m: spermatogenesis: 2600 g	5400
	m: 9-10 mo	6965
	m: 1 yr	2019, 9560
	m: 12-14 mo; 3200-3800 g	5400

	f/m: 1st season	10760
	f/m: ~1 yr	5761
Lynx canadensis	f: 1st conceive: 9 mo	8238
	f/m: 1st mate: 1 yr	5620
	f/m: 1st mate: 22 mo	8238
	f/m: 22-23 mo	1871a
Lynx caracal	f: 1st birth: 15 mo	5592
	f: 21 mo	9175
	m: 1st sire: 12.5-14 mo	833
Lynx lynx	f: large ovary: 7.5-11.5 mo 50%	6137a
	f: 1st repro: 1-2 yr	3527
	f: 1.75 yr	3571
	m: large testes: 19.5-23.5 mo 50%	6137b
	m: spermatogenesis: 1.0-1.5 yr	3527
	m: 2.75 yr	3571
	f/m: 1st mate: end 2nd yr	11226
	f/m: 2 yr	10499
Lynx rufus	f: 1st cl: >12 mo	8765
	f: 1st mate: 1-2 yr	1871a
	f: 1st birth: 1 yr	8239
	f: 1st birth: 2 yr; n1	11815
Neofelis nebulosa	f: 1st estrus: 21.8 mo; sd2.9; r17-28; n9	11991a
	f: 1st conceive: 26.3 mo; sd7.9; r13-38; n13	11991a
	m: 25.5 mo; sd6.6; r16-38; n16	11991a
Panthera leo	f: 1st estrus: 3.5-4.5 yr	9546
	f: 1st mate: 2-3 yr; n21	9730
	f: 1st conceive: 2.5 yr	9358
	f: 1st conceive: 48 mo; r43-55	10170
	f: 1st birth: 24 mo; n1	2874
	f: 1st birth: 2.87 yr	30
	f: 1st birth: 4.75 yr	852
	m: 1st sperm: median 30 mo; r26-34	10170
	m: 1st mate: 2.17-3.5 yr; n8	9730
	f/m: 1st mate: 20-24 mo	9704
	f/m: 1st mate: <2 yr	10412
Panthera pardus	f: 1st mate: 28 mo; r23-32; n8	2878
	f: 1st mate: <3 yr	9730
	f: 1st birth: 2.25-4.17 yr; n6	2494
	f: 1st birth: 36.9 mo; r27-52; n8	2878
	m: 1st sire: 1.5 yr; n1	9730
	f/m: 22 mo	9546
Panthera tigris	f: 1st estrus: 3.5-4.83 yr; n6	2272
	f: 1st mate: <3 yr	9730
	f: 1st mate: 3 yr	38
	f: 1st birth: 3.25 yg; n2	11090
	f: 1st birth: 40 mo; n1	10555
	f: 1st birth: 4.08-5.17 yr; n2	2272
	f: 4 yr	19
	m: 1st mate: <3 yr, 2 in 1 case	9730
	m: 1st mate: 4 yr	38
	m: 4-5 yr	7004
CYCLE LENGTH		
Acinonyx jubatus	27 d; n1 cycle, 1 f	4659
Felis bengalensis	4-7 d	7781
Felis concolor	estrus: 6.6 d; r4-9; n8, 3 f	9422
	estrus: 8-11 d; n9	8822a
	estrus: ~9 d	12045
	22.8 d; interestrous interval	8822a
Felis pardalis	estrus: 5.35 d; r1-22; n20 cycles, 8 f	2877
Felis silvestris	estrus: 2-3 d	11271
	estrus: 4.39 d; sd1.46; n18	7776
	estrus: 5-6 d; max 8	2019
	27.2 d; ±1.1; 12-21 49.8%; r6-120	11683
	32.9 d; r13-53	11271

Felis wiedii	estrus: 10.13 d; r7-13; n8	8406
	26.5 d	8406
	32 d; n1 f, 5 mo of obs	8207
Lynx caracal	estrus: 4.5 d; r3-6; n20	833
	14 d; n15 cycles	833
Lynx rufus	~44 d	1871a
Neofelis nebulosa	estrus: 6.2 d; sd4.1; r1-17; via behavior	11991a
	estrus: 6.1 d; sd4.2; r1-16; via mating	11991a
	29.9 d; sd13.8; median 27; r10-55; n72 cycles, 28 f	11991a
Panthera leo	estrus: 3 d	1017
	estrus: 4-8 d	9730
	estrus: 6.76 d	2875
	estrus: 7.2 d; r4-16; n14, 3 f	9422
	19-66 d	2882
	r3-8 wk; n8 cycles, 3 f	9605
	28 d	1017
	55.4 d; n9 cycles, 3 f	9422
	76.45 d; n192 cycles, 24 f, n3 prides	2875
Panthera onca	estrus: 7.4 d; r5-11; n13, 2 f	9811, 10322
	estrus: 12.0 d; ±1.0; n7	11687
	estrus: 12.9 d; r6-17; n10, 1 f	9422
	27.4 d; r13-58; n11, 2 f	9811, 10322
	42.6 d; r25-60; n8 cycles, 1 f	9422
	47.2 d; se5.4; n6 cycles, 1 f	11687
Panthera pardus	estrus: 6.7 d; r4-12; n34, 3 f	9422
	estrus: 7.5 d; r6-9	6712
	3.4 wk; n33 cycles no luteal phase, 3 f	9604
	45.8 d; r19-115; n27 cycles, 3 f	9422
	47 d; r27-58; n5 cycles, 1 f	6712
	7.3 wk; n31 cycles with luteal phase, 3 f	9604
	54.96 d; r6-210; n24 cycles, 7 f	2878
Panthera tigris	estrus: 1 wk	7979
	estrus: 3-13 d	75
	estrus: 5.3 d; se0.2; n12, 4 f	9706, 9708
	estrus: 7.1 d; r3-10; n14, 1 f	9422
	estrus: 7.2 d; r2-22	5848
	estrus: 7.6 d; r5-10	29
	estrus: 10-15 d	11283
	20 d	9474
	25.0 d; se1.3; n12 cycles, 4 f	9706, 9708
	25; r20-30; n5 cycles, 1 f	10555
	41 d; r33-53; n12 cycles, 2 f	29
	49.12 d; r12-86 d; n69 cycles, 2 f	5848
	r45-55 d; n3, 1 f, from graph	9422
	65.9 d; r25-138; n15 cycles, 2 f	43
Panthera uncia	estrus: 4-5 d	9131
	estrus: 6.2 d; r5-7; n5, 1 f	9422
	55-65 d; n3, 1 f, from graph	9422
GESTATION		
Acinonyx jubatus	90 d	11697
	~3 mo; n1	10788
	91-92 d; n1	3338, 3339
	91-95 d	6792
	94 d	10020
Felis bengalensis	~2 mo	7979
	~2 mo	10499
	63-69 d; 1st, last mating to birth	7781
Felis chaus	56 d	9175
	~2 mo	4645
	64-66 d; n1	9565
Felis concolor	82-98 d	1871a
	87 d	4582
	88-90 d; n2	8822a
	88-97 d	1474

	~90 d	11936
	3 mo	1355
	91 d	7234
	~92 d; r88-96	11338
	91-97 d	7088
	92-104 d	10809
	92.3 d; r84-106; n12	2881
	93.2 d; r86-98; n10	9185
	95 d; n1; artificial insemination	7495, 7496
Felis geoffroyi	65.3 d; r62-70; n9	300
	73.3 d; r72-74; n3	6268
	74-75 d	5476
	74, 75-76 d; n2	9574
Felis manul	74-75 d; n1	9563
Felis margarita	59-63 d	4646
	61 d; r59-63; n2	9573
	77-82 d	9564
Felis nigripes	~67 d	9675
	67 d; n1	9674
	67-68 d; n3	6423
Felis pardalis	79.7 d; r77-82; n23	2877
	82 d; r77-90; n32	9567
Felis serval	56 d	11697
	64-78 d	5761
	71-76 d; n2	1014
	74 d	3168
	74 d; r67-77; n3 f, 14 births	11378
Felis silvestris	56 d	8223
	~56 d	2439
	63-67 d	11331
	65 d	7839
	65-69 d	10871
	9 wk; median 66 d; r60-70; n75	813
	66 d; 63-69.5	2019
	~68 d	6965
	69 d; n1	7259
Felis tigrina	74-75, 76, 76 d; n3	6422
Felis viverrina	63 d; n1	11038
	80 d; n1	9368
Felis wiedii	81 d; n1	8207
Felis yagouaroundi	72-75 d; n3	5090
Lynx canadensis	min 60 d	5146
	60-65 d	5620
	~62 d	1474
	est 63-70 d	10942
Lynx caracal	66 d; r64-68; n2	10156, 10158
	69 d *ea*; n2	3982
	69-70 d	9175
	69-78 d	5761
	79 d; r78-81; n3	5592
	79 d; r78-81; n5	833
Lynx lynx	~9-10 wk	3571, 4645, 7979, 10499
	67, 68-70, 70, 70 d; n4	1451
	10-11 wk	11226
	73-74 d	6102
Lynx pardinus	63-73 d	10809
Lynx rufus	~50 d	2088
	~50 d	6387
	50-60 d	1474
	50-60 d; hunting records	12044
	50-70 d	1871a
Neofelis nebulosa	86-94 d	3203
	90-97, ~95 d	3660
	mode 91 d 56%; r88-109; n9	7648

	93.4 d; sd6.3; r85-121	11991a
Panthera leo	100-113 d	10308
	100-119 d	5761
	100.48 d; n42 lt, 3 prides	2875
	102.1 d; sd3.02; n7	2882
	105 d; r102-112	4582
	~3.5 mo	852
	106.2 d; n398; last mating to birth	9730
	est 108 d	10170
	~108 d	1230
	109.74 d; r100-114; n51	2039
	110 d; sd10	9704
	111 d	1355
	111.9; n419; 1st mating to birth	9730
	112-115 d	1017
	114.75 d; r111-119; n4 lt, 3 f	9422
	115 d; n1	2583
Panthera onca	93-105 d	10809
	95-108 d	1230
	98-109 d	2109
	3.33 mo	1355
	101, 103, 103-105 d; n3	10332
	3.5 mo	4582
	109.5 d; r108-111; n2, 1 f	9422
Panthera pardus	~69 d; stillborn after artificial insemination	2718
	90 d; n1 stillborn	9422
	90-100 d	11697
	3 mo	1355
	3 mo; r87-99 d	4582
	~3 mo	10499
	~3 mo	7979
	91.9 d, r84-98	2494
	92-95 d	2439
	92-95 d	8223
	92-105 d	5761
	93-103 d	1230
	96 d	6712
	98-105 d	9175
	100 d ea; n2	9422
	112 d	2651
Panthera tigris	95-107 d	19
	102 d; n1	5848
	102-106 d; last mating to birth; 108-111 d; 1st mating to birth	9730
	mode 103; r95-107	75
	103 d; r102-104; n5 f, n5 births	2272
	103-104 d; r98-110	4582
	104 d; n4	9708
	~15 wk	6654
	105 d	7979
	105.4 d; n7, 1 f	9422
	106.27 d; r93-114; n11 lt, 5 f	9474
	109 d	1355
	111.7 d; r93-105; n9 lt	11283
	~113 d	2333
Panthera uncia	54 d; n1 premature birth	9422
	~3 mo	10499
	90-100 d	7979
	98 d	5465
	98-103 d; n5	6829
	98-103 d	3498
	98-105 d	5838
	99 d; n1	1557
	99-100 d; n1	3496

LACTATION

Acinonyx jubatus	solid food, den emergence: ~4 wk	8798
	solid food: 1 mo; n1 lt	3339
	solid food: 5-6 wk	10158
	solid food: 47 d	6793
	wean: 3 wk	9546
	wean: ~3 mo	10158
	wean: 14-24 wk	5761
	wean: 5 mo	3339
	wean: 4-5 mo	8798
Felis bengalensis	solid food: 4th wk	3514
Felis chaus	solid food: 49 d	9565
	wean: 3 mo, 102 d	9565
Felis concolor	wean: 4-5 wk	11936
	wean: 4-5 wk	12045
Felis geoffroyi	wean: 8-10 wk	300
Felis nigripes	solid food: 1 mo	9675
	solid food: 34-35 d	6423
Felis pardalis	wean: 6 wk	9567
Felis serval	solid food: 39 d	1018
Felis silvestris	solid food: 25-30 d	4237
	solid food: 1 mo	7259
	solid food: 35-36 d	10871
	solid food: 1.5 mo	3571
	wean: 50-60 d	11331
	wean: 3 mo	2490
	wean: ~4 mo	4252
	driven off: 3 mo	11331
	wean: 6-8 wk	7259
	wean: ~4 mo	3571
Felis tigrina	solid food: 55 d	6422
Felis viverrina	solid food: 53 d	9175
	wean: 6 mo	9175
	independent: 4 wk	9368
Felis wiedii	solid food: 53.6 d; r52-57; n3	8407
Lynx caracal	solid food: 1.5 mo	3982
	solid food: 2 mo	5761
	wean: 6 mo	5995
	wean: 2.5 mo	3982
	wean: 15-20 wk	833
Lynx lynx	solid food: 6 wk; n1 yg, 1 lt	11501
	wean: ~4 mo	11226
	wean: 2 mo	3571
	wean: 2-3 mo	10499
	wean: 3 mo	4645
Lynx rufus	f took food to den: 7-8 wk	11815
	wean: 60-70 d	5210
	nurse: 16th wk	11815
Neofelis nebulosa	solid food: 6 wk	3202
	nurse: 93 d	3660
Panthera leo	solid food: 5 wk	9546
	solid food: ~2 mo	852
	solid food: ~2 mo	5761
	solid food: 2.5 mo	1017
	nurse: 5-8 mo	5761
	nurse: 6-7 mo	10158
	wean: 5.5 mo	1017
	wean: 7-8 mo	9546
	wean: max 15 mo	9360
	wean: 6-8 mo	9704
	wean: 6-8 mo	852
	wean: 8-9 mo	10170
Panthera onca	solid food: 10-11 wk	9811
	wean:5-6 mo	9811

Panthera pardus	solid food, den emergence: 42 d	5761
	wean: 3 mo	5761, 10158
	wean: max 129 d; r114-130	2494
Panthera tigris	solid food: ~1 mo	7979
	solid food: 2 mo	7004
	solid food: 43 d	75
	wean: 3 mo	11283
	nurse: 3 mo	9474
	nurse: 3-6 mo	7004
	independent: 18 mo	10555
	wean: ~6 mo	10499
Panthera uncia	wean: > 2 mo	5465
	wean: 48 d	3499
INTERLITTER INTERVAL		
Felis bengalensis	163.3 d; r81-305; n3 intervals, 1 f	26
Felis chaus	93-131 d	9565
Felis concolor	17.5 mo; r16-19; n2	6545
	2-3 yr	1474
	2-3 yr	12045
Felis pardalis	174.3; n15	2877
	2 yr; n2 f; circumstantial evidence	2999
Felis serval	5-6 mo	3168
	184 d w/normal lact	11378
Felis silvestris	114 d; r112-116; n2	11072
Panthera leo	1 yr 3 mo 23 d; n7	36
	19 mo	852
	24 mo	9546
	2 yr	10158
	2 yr	9358
Panthera pardus	8.44 mo; r 3.5-40; n51 intervals, 7 f	2878
	1 yr 2 mo 27 d; n10	36
	20-21 mo	9726
Panthera tigris	1 yr 6 mo 5 d; n5	36
	1.71-2.21 yr	29
	2-2.5 yr	10555
	2-3 yr	6654
LITTERS/YEAR		
Felis chaus	2	9565
Felis concolor	1/2.5 yr	7848
	est 0.5	9185
Felis geoffroyi	1-2	300
Felis nigripes	4	9675
Felis pardalis	1; 2 if yg die	2877
Felis serval	2	5761
	3?	11253
Felis silvestris	1	7259
	est 1	10760
	2	2044
	2	5400
	2, occasionally 3	6965
	3.11; mode 4; r1-6	2019
Felis yagouaroundi	est 2	1474
Lynx lynx	1	3571
Panthera leo	1	11253
	max 3; yg removed ~6 wk	1230
Panthera pardus	1	10499
	1	8223
	1; occasionally 2	9196
	2?	11253
Panthera tigris	1/2-3 yr	7979
Panthera uncia	1	10499

SEASONALITY

Acinonyx jubatus	mate: Jan; n1; captive	10788
	mate: peak July-Sept; captive	2880
	birth: Jan-Aug, peak Mar-June; e Africa	5761
	birth: Jan-Aug; Serengeti, Tanzania	9546
	birth: Apr; n1; captive	10788
	birth: Nov, Mar-Apr; few data; Mumbwa, n Zimbabwe	331
	birth: fall-winter; Kruger Nat Park, S Africa	8514
	birth: yr rd; n67; captive	11684
	yg seen: yr rd; Natal	11616
	repro: yr rd; n14; zoo	6795
Felis bengalensis	estrus: Mar; Amur Basin, Siberia, Russia	7979
	mate: Feb-Mar, Aug-Sept; Malaysia	6448a
	mate: Mar; e USSR	10499
	mate: irregular; captive	7781
	birth: Feb-Aug; zoo	8639
	birth: Mar; central Asia	179
	birth: May; Amur Basin, Siberia, Russia	7979
	birth: mid-end May; e USSR	10499
Felis chaus	estrus: Feb-Mar; USSR	7979
	mate: spring; Europe, USSR	4645
	preg f: Apr; n1; Nepal	7382
	birth: May; USSR	7979
	birth: Dec-May; captive	9565
	birth: winter-early spring; Pakistan	9175
	lact f: Dec; n2; Nepal	7382
Felis concolor	mate: Dec-Mar; WI	5210
	mate: yr rd; zoo	1230
	birth: May-June; n2; FL	6723a
	birth: yr rd; most Apr; w US	5146
	birth: yr rd, 60% June-Sept, peak July; UT, NV	9185
	birth: yr rd, except Nov, peak May-Oct; zoo	2881
	birth: yr rd, peak spring-summer; Canada	11936
	birth: yr rd; w US	12045
	repro: yr rd; MAN, Canada	7848
Felis geoffroyi	birth: Dec-May; Uruguay	11977
Felis manul	birth: Apr-May; Transbaikalia, Russia	9558
	birth: end Apr-May; Transbaikalia, Russia	10499
Felis margarita	mate: Jan-Feb; n Africa, Asia, Arabia	9559
	birth: early-mid Apr; USSR	7979
	birth: Apr, Sept; zoo	4618
	birth: Apr-May; n Africa, Asia, Arabia	9559
	birth: spring, end summer; Pakistan	9175
Felis marmorata	repro: not seasonal; Asia, Indonesia	9564
Felis nigripes	birth: Nov-Dec; S Africa	11311
Felis pardalis	birth: Nov, Apr; TX	9567
	birth: Dec-Feb; S America	9567
	birth: Jan; Yucatan, Mexico	9567
Felis serval	mate: Jan; n1; zoo	1018
	preg f: Nov, Jan-Mar; S Africa	10157
	birth: Jan-Apr; s Africa	10155
	birth: Jan-June: n part of range	10155
	birth: Mar-Apr, Sept-Nov; n8; Uganda, e Zaire	5761
	birth: Apr; n1; zoo	1018
	birth: Aug-Feb; Zimbabwe	10153
	birth: peak Sept-Apr; Zimbabwe	10157
	birth: Sept-Apr; S Africa	10158
	birth: twice/yr, start/end rainy season; Zaire	11253
	repro: est yr rd; zoo	11378
Felis silvestris	estrus: end Jan-Mar; USSR	7979
	estrus: Feb-Mar; central Europe	4252
	mate: Feb-Mar; Europe, USSR	4645
	mate: Feb-Mar, May-early June; Germany	3571
	mate: Feb-Apr, Oct; India	9843

	mate: end Feb-early Mar; Caucasus, Asia	9370
	mate: Mar; Zambia	10153
	mate: Apr, Aug, Oct; Zimbabwe	10153
	mate: Dec-June; captive	2019
	mate: spring, fall; Scotland, UK	10760
	preg f: Oct-Apr; Macquarie I, Australia	2044
	birth: Mar-June; zoo	11331
	birth: Mar-Aug; zoo	7259
	birth: Mar-Aug, peak May; n60; captive	2019
	birth: Mar-Oct, peak Apr; ne France	2020
	birth: end Mar-Apr; USSR	7979
	birth: Apr-May; central Europe	4252
	birth: peak Apr-May; Europe	2019, 9560
	birth: Apr-June, end July-Aug; Germany	3571
	birth: Apr-June; Europe, USSR	4645
	birth: early May, Aug, end Nov-early Dec; Scotland, UK	6965
	birth: May-June; Caucasus, Asia	7979
	birth: Aug-Apr, peaks Oct-Nov, Jan; S Africa	348
	birth: peak Sept-Mar; S Africa	10158
	birth: mid Sept-mid Mar; subantarctic	11074
	birth: peak Sept-Mar; se Australia	5400
	birth: Oct-Dec; Angola	11697
	birth: peak Oct-Jan, none May-July; Kerguelen, subantarctic	2490
	birth: Nov-Mar; Macquarie I, Australia	5399
	birth: yr rd; captive	9197
	wean: June; USSR	7979
	yg found: Mar-Oct; France	2020
	repro: Oct-May, peaks Nov, Apr; subantarctic	2491
Felis wiedii	estrus: Dec-Apr; captive	8406
Felis yagouaroundi	mate: Sept-Nov; Paraguay	10809
	mate: Nov-Dec; Mexico	10809
Lynx canadensis	mate: Jan-Feb; Canada	1474
	mate: Jan-Feb; w US	5146
	mate: Jan-Feb; WI	5210
	mate: Feb-Mar; Canada	5620
	mate: Mar-early Apr; NEW, Canada	10942
	mate: Mar-Apr; AK	10942
	mate: Apr-May; ALB, Canada	10942
	birth: Mar-Apr; Canada	1474
	birth: Mar-Apr; w US	5146
	birth: Mar-Apr; WI	5210
	birth: Apr-May; Canada	5620
	birth: end May-early June; NEW, Canada	10942
	birth: end May-early June; WY	1871a
	birth: spring; Cape Breton, NS, Canada	8238
Lynx caracal	mate: Sept-Dec; Zambia	10153
	preg f: Apr, Sept-Nov; n8; Bedford District, S Africa	8758
	birth: Apr; Karakum, USSR	9496
	birth: Sept; n1; Natal, S Africa	9327
	birth: peak Oct-Jan; S Africa	833
	lact f: Nov; few data; n Zimbabwe	331
	repro: yr rd; Morocco	8223
Lynx lynx	estrus: Feb-early Mar; Romania	11226
	mate: Jan-Mar; Europe	3571
	mate: end Jan-Apr; Europe, USSR	4645
	mate: Feb-Mar; zoo	4058
	mate: Feb-early Apr; Norway	6137b
	mate: mid Feb-Mar; USSR	10499
	mate: Mar; some regions earlier; USSR	7979
	birth: Apr-May; Europe	3571
	birth: May-June; USSR	10499
	birth: mid-end May; zoo	4058
	birth: end May; Russia	10942
	birth: end May-early June; Norway	6137b

Lynx pardinus	mate: Jan; Spain	11070
Lynx rufus	follicular size: large Jan-Mar; SC	8765
	CL present: early Mar; ne US	8647
	mate: mid Jan-July; WY	1871a
	mate: peak Feb-May; UT	12044
	mate: end Feb-Mar; WI	5210
	preg f: Jan-Sept, peak Mar-Apr; UT	3625
	preg f: Feb-June; N America	12044
	preg f: Mar; n2; CA	11228a
	birth: Apr-May; BC, Canada	2088
	birth: Apr-May for 5 yr; Apr, Oct 1 yr; FL	11815
	birth: mid Apr-early Sept; WY	1871a
	birth: May; ne US	8647
	birth: est Aug; n1 lt; MA	9886
	birth: spring; Cape Breton, NS, Canada	8239
	birth: spring-summer; w US	5146
	birth: yr rd; UT	12044
Neofelis nebulosa	estrus: peak Oct-Feb; captive	11991a
	birth: peak Mar; captive	11991a
Panthera leo	mate: yr rd; Kenya	9358
	birth: Feb-Mar, June; n3; zoo	44
	birth: Mar-Apr; S Africa	5780
	birth: peak Dec-Jan; Kenya	9358
	birth: dry season; Zaire	11253
	birth: yr rd; S Africa	10170
	birth: yr rd; S Africa	8514
	birth: yr rd; n Zimbabwe	331
	birth: yr rd; n38; zoo	12128
	birth: yr rd; S Africa	3145
	birth: yr rd; Serengeti, Tanzania	852
	birth: yr rd; zoo	1230
	birth: yr rd; zoo	10412
Panthera onca	mate: Jan; n Mexico	1474
	birth: Jan-Feb, June, Sept; few data; zoo	1230
	birth: peak Mar-July; Argentina	9811
	birth: Apr-May; n Mexico	1474
	birth: peak June-Aug; Belize	9811
	birth: peak: July-Sept; Mexico	9811
	birth: peak Nov-Dec; Paraguay	9811
	birth: peak Dec-May; Brazil	9811
Panthera pardus	estrus: Jan; e Asia	7979
	estrus: peak May, Oct; zoo	2878
	estrus: yr rd; captive	2494
	mate: Jan-Feb; Amur, Siberia, Russia	10499
	mate: Nov-Dec; Nepal	7382
	mate: Dec-Jan?; Wilpatti Nat Park, Sri Lanka	2936
	mate: est not seasonal; few data; n Zimbabwe	331
	birth: Feb-Mar; India	988a
	birth: Feb-Mar; India	6654
	birth: Feb-Mar; Nepal	7382
	birth: Apr-May; USSR	7979
	birth: Apr-May; Amur, USSR	10499
	birth: Apr-June, Dec; n5; zoo	44
	birth: peak Aug, low Oct; zoo	2878
	birth: spring-early summer; Pakistan	9175
	birth: rainy season; Angola	11697
	birth: start rainy & start dry seasons; Zaire	11253
	birth: not seasonal; Namibia	9945
	birth: yr rd; S Africa	10158
	birth: yr rd, peak May, low Dec; n146 lt; zoo	3497
	birth: yr rd; S Africa	3145
	birth: yr rd; n27; zoo	12128
	birth: yr rd; S Africa	3145
	repro: yr rd; zoos	1230, 9196

Panthera tigris	estrus: end Jan-early June; zoo	9706, 9708
	estrua: Dec-Jan; USSR	7979
	estrus: not seasonal; e Asia	7979
	mate: end Dec-early Jan; Khabaroush, Siberia, Russia	10499
	mate: Mar-Apr, July-Aug; n12; zoo	11283
	mate: not seasonal; zoo	1230
	birth: Mar-July, Nov-Dec; zoo	29
	birth: Apr; Khabaroush Terr, Siberia, Russia	10499
	birth: peak May-June; n1239; zoo	9706
	birth: July, Dec; n2; zoo	44
	birth: low Oct-Mar, peak Apr-June; zoo	9706
	birth: yr rd, peak May; zoo	9706
	birth: yr rd, peak Mar-June; Rajasthan, India	9474
	birth: yr rd; n17; zoo	12128
Panthera uncia	estrus: winter-early spring; USSR	7979
	mate: Feb; zoo	9131
	mate: Mar; n1; zoo	5465
	mate: end winter-early spring; USSR	10499
	birth: Apr; USSR	7979
	birth: Apr-May; USSR	10499
	birth: May 62%; zoo	3499
	birth: peak May; zoo	3497
	birth: end May-early June; n Chitral, Pakistan	9175
	birth: none Oct-Jan; zoo	3497

Order Carnivora, Family Otariidae

Eared or fur seals (*Arctocephalus* 20-700 kg; *Callorhinus* 30-280 kg; *Eumetopias* 270-1000 kg; *Neophoca* 60-300 kg; *Otaria* 140-340 kg; *Phocarctos* 230-400 kg; *Zalophus* 40-450 kg) are polygynous, semiaquatic, marine carnivores from the southern hemisphere and north Pacific. Males are larger than females (359, 1083, 1086, 1090, 1474, 2121, 3717, 3720, 4257, 6170, 6828, 6939, 6940, 6977, 8024, 8278, 8279, 8417, 8908, 8910, 8911, 9459, 9570, 9677, 11237, 11238, 11239, 11403, 11404).

The 500-1000 g, zonary, deciduous placenta implants into a bicorunate uterus (359, 2062, 2102, 2666, 3716, 3717, 4258, 4845) and is not usually eaten after birth (1083, 1605, 3716, 4257, 8024, 8908, 11404). Implantation can be delayed (*Arctocephalus* 9570, 11461; *Callorhinus* 2252, 2254, 2255, 3032, 10736). Estrus occurs 0-3 wk after birth (359, 621, 1083, 1090, 2102, 2299, 2667, 3716, 3717, 3719, 5683, 5752, 6170, 6287, 6485, 6827, 6940, 7318, 7319, 8024, 8417, 8568, 8908, 8911, 9458, 9459, 9658, 10158, 10736, 10809, 10898, 10899, 11461). The precocial neonates are furred with open eyes (1604, 1605, 3716, 7104, 8908, 8909, 11239). Females have 2-6 mammae (1604, 3716, 5755). Males have a baculum (1473, 9587, 11239).

Details of anatomy (236, 2102, 4258, 9586, 9969, 10728, 12029), endocrinology (4845), birth (11220), suckling (1604, 4730), milk composition (420, 1083, 2062, 3718, 8035, 8040, 8542, 8697, 8908, 9608, 10902, 10904), and light (10276) are available.

NEONATAL MASS

Arctocephalus australis	3-5 kg	5755
	3.5-5.5 kg	11238
Arctocephalus forsteri	3.5 kg	5755
	3.7 kg; r2.90-4.40; n5	6977
	4.3 kg	2121
Arctocephalus galapagoensis	3.0-4.5 kg	10898
	f: 3.1-3.4 kg; n36	9509
	m: 3.3-3.9 kg; n34	9509
Arctocephalus gazella	5-6 kg	1086
	f: ~5.6 kg; n12	8279
	6 kg	5755
	m: 6 kg; n9	8279
	7 kg	2666
Arctocephalus pusillus	3.6-5.4 kg	1605
	4.5-6.4 kg	8908, 8911
	4.5-7 kg	9570, 9864

	f: 7.1 kg; r4.5-10	11461
	m: 8.1 kg; r5-12.5	11461
Arctocephalus tropicalis	4-5 kg	5755
	~4.2 kg	10158
	4.9 kg	1090
Callorhinus ursinus	4-6 kg	359
	f: 4.8 kg; sd0.7; r3.3-6.0; n16	9591
	5 kg	5755
	f: 5.10 kg; se0.11; n11	2062, 3718
	m: 5.4 kg; sd0.9; r4.1-7.1; n23	9591
	5.5 kg	5683
	f: 5.55 kg; r5.3-5.8; n2	8419
	m: 6.23 kg; se0.27; n8	2062, 3718
Eumetopias jubatus	15.9 kg	5683
	16-23 kg	6940
	18 kg	5146
	18-22 kg	5755
	18.25 kg; r17-22; n4	9585
	19-22 kg	6944
Neophoca cinerea	6-8 kg	5755
	6.4-7.9 kg	6481a, 6483, 11403
Otaria byronia	f: 11.43 kg; r10.28-13.40; n6	11237, 11239
	f: 12.3 kg; sd1.4; n38	1591a
	m: 13.7 kg; sd1.6; n37	1591a
	m: 14.15 kg; r13.26-14.77; n4	11237, 11239
Zalophus californianus	5-6 kg	6939
	~6.0 kg	3718
	6.2 kg; n1	2109
	f: 7.7 kg; n28	8024
	m: 7.82 kg; n1	9623
	m: 9 kg; n33	8024
NEONATAL SIZE		
Arctocephalus forsteri	40-45 cm	2121
	55 cm	2122
Arctocephalus gazella	60-66 cm	1086
	65 cm	5755
Arctocephalus pusillus	60-70 cm	9570, 9864
	65 cm	5755
	m: 69.7 cm; r62-79	11461
	80 cm	8911
Arctocephalus tropicalis	60 cm	5755
	63 cm	1090
Callorhinus ursinus	f: 63.1 cm; sd4.45; r54.4-68.7; n16	9591
	m: 65.9 cm; sd4.58; r54.4-75.4; n23	9591
Eumetopias jubatus	100 cm	6940
Neophoca cinerea	62-68 cm	6481a, 11403
	70 cm	5755
	75 cm	5752
Otaria byronia	f: 78.5 cm; r73.0-82.0; n6	11237, 11239
	f: 79 cm; sd3; n29	1591a
	m: 81.7 cm; r79.0-85.0; n4	11237, 11239
	m: 82 cm; sd4; n29	1591a
	m: 83.8 cm; n1	4257
Phocarctos hookeri	~75 cm	6828
	75-80 cm	5755
Zalophus californianus	f: 70.5 cm	8024
	m: 74 cm; n33	8024
	75 cm	6939
	m: 78 cm; n1	9623
WEANING MASS		
Arctocephalus galapagoensis	f: 14 kg; 18-36 mo	3718
	m: 16 kg; 18-36 mo	3718
Arctocephalus gazella	f: 14.5 kg; 4 mo	3718
	m: 17.1 kg; 4 mo	3718

Arctocephalus pusillus	m: 25.0 kg; 9-11 mo	3718
	27.3 kg	8911
Callorhinus ursinus	f: 11.7 kg; 4 mo	3718
	f: 12.0 kg; sd2.2; r7.0-16.6; n91; 3 mo	9591
	m: 13.9 kg; sd2.2; r8.9-18.8; n82; 3 mo	9591
	m: 14.1 kg; 4 mo	3718
Zalophus californianus	m: ~25 kg; 10-12 mo	3718
WEANING SIZE		
Eumetopias jubatus	179 cm; yearlings	6599
LITTER SIZE		
Arctocephalus pusillus	1	1605
	1; 2 rare	8908, 8911
Arctocephalus tropicalis	1 *ea*; n2 lt	1428, 1429
	2 wean; n1 f	891
Callorhinus ursinus	1	3032
	1.04; r1-2; n84 lt	2102
	2 emb; n3 f	7914
Eumetopias jubatus	1	6944
	1 emb; n1 f	566
	1; 2 rare; 1 case in 3 yr	3716
Neophoca cinerea	1	10698
	2 aborted; n1 f	6481
Otaria byronia	1	4257
Zalophus californianus	1	2914
	1 *ea*; n12 lt	12128
	1 yg	5146
	2; n1 f	11020
SEXUAL MATURITY		
Arctocephalus australis	f: 3 yr	11238
	m: 7 yr	11238
Arctocephalus forsteri	f: 4-6 yr	2121
	m: 4-6 yr	2121
	m: socially mature: 10-12 yr	2121
Arctocephalus galapagoensis	f: 1st mate: 3 yr	10898
Arctocephalus gazella	f: 1st birth: 3-4 yr	1086
	f: 1st birth: 3-4 yr	8279
	m: 1st territory: ~8 yr	8279
	m: 3-4 yr	8279
	m: 6-7 yr	1086
Arctocephalus pusillus	f: 1st mate: 6 yr	9570
	f: 1st mate: after 2nd winter	8908
	f: 1st birth: 3rd yr	8911
	f: 1st birth: 4 yr	9864
	f: 3-6 yr	11461
	m: 4-5 yr	11461
	m: socially mature: 9-12+ yr	11461
Callorhinus ursinus	f: 1st estrus: mode 4 yr; min 2 yr	1705
	f: 1st mate: 3 yr; Asia; 4 yr; N America	6170
	f: 1st ovulate: 70% 4 yr	2102
	f: 1st cl: 3-7 yr	3717
	m: sperm: 3 yr; mode 4-5 yr	9591
	m: spermatogenesis: 3 yr	359
	m: 1st mate: 6 yr	359
	m: 1st mate: 7-9 yr	3717
	m: 4-5 yr	6170
	m: 5 yr	3717
Eumetopias jubatus	f: 1st ovulate: 4.6 yr; 95%ci 0.8; n ≥ 38	8568
	f: 3-8 yr	6599
	f: 4-5 yr	6940
	m: 1st mate: 9-13 yr	8568
	m: socially and physically competitive: 7-9 yr	6940
	m: 3-7 yr	6599
	m: 5-7 yr	6940
	m: *est* 5-7 yr	8568

	f/m: 1st repro: 4.9 yr; 95%ci 1.2	8568
Neophoca cinerea	f: 1st mate: 3 yr	5752
	m: 1st mate: 6+ yr	5752
Otaria byronia	f: 1st birth: 4 yr	4257, 4258
	m: 6 yr	4257
Phocarctos hookeri	m: 6 yr	6828
Zalophus californianus	f: 1st mate: 3 yr	1474
	m: 5 yr	1474
CYCLE LENGTH		
Arctocephalus forsteri	birth to estrus: 8 d	5755
Arctocephalus gazella	birth to estrus: 8 d	5755
Arctocephalus pusillus	birth to estrus: 6 d	5755
Arctocephalus tropicalis	birth to estrus: 8-12 d	5755
Callorhinus ursinus	birth to estrus: 7 d	5755
	behavioral estrus: 48 h	3717
	physiological estrus: ~1 mo	3717
Eumetopias jubatus	birth to estrus: 10-14 d	5755
	estrus: short	9459
Neophoca cinerea	birth to estrus: 4-9 d	5755
Zalophus californianus	birth to estrus: ~14 d	5755
GESTATION		
Arctocephalus galapagoensis	emb development: 7 mo	10898
Arctocephalus gazella	51 wk	1086
	4 mo; 1 yr; 8 mo	6287
	~1 yr	8278
Arctocephalus pusillus	implant: ~3 mo	11461
	implant: 4 mo	8908, 8910, 8911
	359 d with delay; primiparous longer up to 15 mo	8908, 8910, 8911, 8912
	51 wk	11461
	est 365-414 d; n10	1605
Callorhinus ursinus	implant: 3-5 mo	2254
	implant: 3.5-4 mo	6170
	implant: 4 mo	2102
	implant: min 4? mo	3032
	implant: 125 d	10736
	51 wk	6170
	360 d	2254
	1 yr	2102
	~1 yr	621
Eumetopias jubatus	implant: 3 mo	8568
	implant: 3 mo	6940
	11 mo	8128
	11.5 mo total	8568
	1 yr	1474, 5146
	~1 yr	3716
	~1 yr	5683
	~1 yr	6940
Neophoca cinerea	~12 mo	5752
	18 mo	11403
Otaria byronia	~1 yr; field data	11236
Phocarctos hookeri	~12 mo	6828
Zalophus californianus	11 mo; field data	8023, 8024
	~11 mo	4582
	50 wk	6939
	11.5-12 mo	1355
	~1 yr	8417
LACTATION		
Arctocephalus australis	wean: 6-12 mo	11238
Arctocephalus forsteri	solid food: 1 d (?)	7104
	wean: 10-11 mo	2121
	wean: ~1 yr; field data	7318
Arctocephalus galapagoensis	wean: 18-36 mo	3718
	wean: 2 yr	10898, 10899

Arctocephalus gazella	wean: 117 d; sd8.0; n8	2667
	nurse: 4 mo; r110-115 d	8278, 8279
Arctocephalus pusillus	solid food: 4 mo	8911
	solid food: 7 mo	9864
	wean: 8-9 mo	1605
	wean: most 9-11 mo; r8-18 mo	2299, 8908, 8911
	wean: 11-12 mo	11461
	wean: 12+ mo	9570
	wean: 1-2 yr	10903
Arctocephalus tropicalis	nurse: ~100 d	1083
	wean: 2 yr; but author states lactation 7 mo	1090
Callorhinus ursinus	wean: 2-3 mo	8416
	wean: 3 mo	5683, 9591
	wean: 4 mo	6170
	wean: 125 d	3719
	nurse: 3-4 mo	10276
Eumetopias jubatus	wean: min 3 mo	5683
	wean: 3 mo	5146
	wean: 8-11 mo	6940
	wean: 1 yr	9658
	wean: 12-15 mo	3716
	wean: 12-16 mo	9458
	nurse: est 1 yr	8568
Neophoca cinerea	wean: >1 yr	6827
	wean: 2 yr	10429
Otaria byronia	wean: ~1 yr	11239
Phocarctos hookeri	wean: est 6-8 mo	11404
Zalophus californianus	solid food: 4-5 mo	10900
	wean: 5-12 mo	6939
	wean: 10 mo-1 yr, exceptionally 2 yr	10900
	nurse: 1 yr	8417
INTERLITTER INTERVAL		
Arctocephalus galapagoensis	1 yr	10898
Arctocephalus gazella	1 yr	1086
Arctocephalus pusillus	1 yr	8911
	1 yr	9570
	1 yr	11461
Arctocephalus tropicalis	1 yr	1090
Eumetopias jubatus	1 yr	8568
	1 yr	3716
Neophoca cinerea	17-18 mo	6484, 6485, 11403
	est 18 mo	10698
Otaria byronia	1 yr	4257
Zalophus californianus	2 yr	2914
SEASONALITY		
Arctocephalus australis	mate/birth: end Nov-Dec; Falkland I	11238
	implant: Mar-Apr; Falkland I	11238
	birth: Nov-Dec; Uruguayan I	11236
	birth: Oct-early Dec; s hemisphere	5755
Arctocephalus forsteri	mate: mid Nov-mid Jan, peak mid Dec; South I, New Zealand	7318, 7319
	mate: end Nov-mid Jan, peak mid Dec; New Zealand	2122
	birth: Nov-mid Jan, peak mid Dec; New Zealand	2122
	birth: mid Nov-mid Dec; s hemisphere	5755
	birth: end Nov-mid Dec, peak early-mid Dec; South I, New Zealand	7318
Arctocephalus galapagoensis	birth: Aug-Nov, peak Oct; Galapagos, Ecuador	10898
	birth: mid Aug-mid Nov; Galapagos, Ecuador	10899
	repro: Aug-Nov; Galapagos, Ecuador	10809
Arctocephalus gazella	mate/birth: Oct-Nov; S Georgia I, Antarctic	8278
	mate/birth: Nov-Dec; S Georgia Sea, Antarctica	1086
	mate/birth: Nov-Dec; n7; S Georgia I, Antarctic	2667
	mate/birth: Dec; Antarctic	6287
	ovulate: before Jan; Marion I, Antarctic	8909
	birth: end Nov; S Africa	10158

	birth: early Dec; S Georgia I, Antarctic	8279
	birth: Dec; S Georgia I, Antarctic	2666
	wean: end Feb-Mar; Bird I, Antarctic	6287
	wean: Apr; S Georgia I, Antarctic	2667
Arctocephalus philippii	mate: Nov-Jan; Juan Fernandez Archipelago, Chile	10809
Arctocephalus pusillus	mate: Oct-Jan, median early Dec; Namibia, S Africa	9570
	mate: Nov-Dec; sw Africa	2299, 8908, 8912
	mate/birth: Nov-Dec; NSW, s TAS, Australia	11461
	birth: Mar-May; captive, ONT, Canada	1604
	birth: June; n hemisphere	5755
	birth: Oct-Dec; Punta San Juan, Peru	10903
	birth: Oct-Jan, median early Dec; Namibia, S Africa	9570
	birth: mid Oct-Dec; sw Africa	8908, 8911, 8912
	birth: Nov-Dec; s hemisphere	5755
	wean: Sept-Oct; S Africa	8908, 8911
Arctocephalus townsendi	mate/birth: May-July; Guadalupe, Mexico	8418
Arctocephalus tropicalis	mate/birth: Nov-Dec; Tristan da lunha Gough, Prince Edward, New Amsterdam, St Paul I, Indian Ocean	1090
	mate/birth: Dec; S Georgia, Antarctica	1083
	birth: est end Jan; Falkland I	8013
	birth: Dec; Gough I, s Atlantic Ocean	890
	birth: mid Dec; S Africa	10158
Callorhinus ursinus	mate: June, primiparous until July; Pribilofs, US	359
	mate: June-Aug; Pribilofs, US	3032
	mate/birth: June-July; Pribilofs, US	3719
	mate/birth: end June-July; Bering Sea, Okhotsk Sea	5683
	ovulate: mid July, primiparous Aug; e Pacific, Pribilofs, US	2102
	implant: early Nov; Bering Sea, n Pacific	6170
	implant: Nov; e Pacific, Pribilofs, US	2102
	birth: end May-July; n831; San Miguel I, CA	10736
	birth: June-July; St Paul I, Pribilofs, US	8416
	birth: June-early Aug; St Paul I, Pribilofs, US	621
	birth: mid June; Pribilof I, US	359
	birth: mid June-July, peak mid July; n433; St George I, Pribilofs, US	10736
	birth: end June-early Aug; Bering Sea, n Pacific	6170
	birth: mid July; e Pacific, Pribilofs, US	2102
	nurse: mid July-mid Oct; AK	9591
	wean: Nov; Pribilofs, US	3719
Eumetopias jubatus	mate: end May-mid July, peak June; n215; AK	8568
	mate: June; Año Nuevo I, CA	3716
	mate: June-July; Año Nuevo I, CA	9459
	mate: mid June-mid July; OR	6938
	mate: mid June-mid July; Año Nuevo I, CA	8128
	mate: mid June-mid July; Año Nuevo I, CA	9350b
	mate/birth: mid May-June; n Pacific	6940
	mate/birth: end May-early July; peak mid June; AK	9458
	mate/birth: June-July; CA, Bering Sea	5683
	implant: end Sept-Oct; AK	8568
	implant: Sept-Oct; n Pacific	6940
	birth: mid May-mid June; peak early June; AK	8568
	birth: mid May-mid July, peak June; Año Nuevo I, CA	3716
	birth: end May-early July; Año Nuevo I, CA	8128
	birth: end May-June; nw Pacific	6944
	birth: June; San Miguel, CA	9658
Neophoca cinerea	birth: est July; WA, Australia	6480
	birth: Oct; Dangerous Reef, Australia	6827
	birth: Oct-Dec; Australia	5752
	birth: Dec-Jan; s hemisphere	5755
	birth: yr rd; Australia	6484, 11403
Otaria byronia	mate: end Dec-Feb, Aug-Sept; Uruguayan I	11236
	mate/birth: Dec-Jan; Falkland I	4258
	birth: Dec-Jan; Falkland I	4257
	birth: mid Dec-early Feb, peak Jan; Uruguayan I	11236, 11237, 11239

	birth: mid Dec-mid Feb; Argentina	11239
Phocarctos hookeri	birth: Nov-Dec; s hemisphere	5755
	birth: Dec; Auckland I	3160
	birth: Dec-early Jan; Auckland I	6828
Zalophus californianus	m: territorial: May-Aug, peak June; San Nicolas I, CA	8417
	mate: May-June; CA, Mexico	5683
	mate: May-Aug; w US	5146
	mate: June-July; San Nicolas I, CA	8023, 8024
	mate/birth: June-July; San Nicolas I, CA	8417
	birth: Jan; Galapagos I, Ecuador	2924
	birth: mid May/June-July/Aug; Santiago I, Galapagos, Ecuador	10900
	birth: end May-early June; n12; zoo	12128
	birth: end May-June; CA, Mexico	6939
	birth: mid May-June; San Nicolas I, CA	8023, 8024
	birth: June 90%; n38; zoo	10635
	birth: June; n5 same f; zoo	2914
	birth: June-July; Pacific coast	1474
	birth: peak Aug-Oct, low Apr-May; Galapagos, Ecuador	2922
	birth: Oct-Dec; Galapagos, Ecuador	6939
	birth: mid? Dec-mid Jan; Galapagos, Ecuador	5755

Order Carnivora, Family Odobenidae, Genus *Odobenus*

Walruses (*Odobenus rosmarus* 600-1900 kg) are circumpolar, gregarious carnivores from arctic waters of the northern hemisphere. Males are larger than females and have larger tusks (1282, 1474, 3181, 3182, 6600). Implantation occurs 3-3.5 mo post coitus into a bicornuate uterus (1282, 3182). Gulls eat the placenta after birth (3182). Neonates can can swim at birth (1457). Males have inguinal testes and a baculum (1332, 1473, 3182, 5755). Details of anatomy (3182, 6786, 7627) and milk composition (3181, 3182) are available.

NEONATAL MASS		
Odobenus rosmarus	f: 33.2 kg; n1	1332
	41 kg	1457
	55 kg	6789
	f: 63.3 kg; r33.8-85; n15	3181, 3182
	m: 63.8 kg; r45-77; n5	3181
NEONATAL SIZE		
Odobenus rosmarus	f: 100 cm	1332
	105 cm	1457
	m: 112 cm; r104-122; n6	3182
	f: 113 cm; r95-123; n11	3182
	m: 119.4 cm; n1, term emb	6786
	f: 121.4 cm; r102-133; n30	3181
	m: 122.4 cm; r112-137; n18	3181
WEANING MASS		
Odobenus rosmarus	200 kg; 1 yr	6789
LITTER SIZE		
Odobenus rosmarus	1	1514
	1	6789
	1-2	5171
	1; 2 occasional	753
	1; 2 rare	3181, 3182
SEXUAL MATURITY		
Odobenus rosmarus	f: 1st cl: 5-6 yr	3181, 3182
	f: 1st cl: 6.25 yr; r5-9; n4	6786
	f: 1st cl: 6-7 yr	3183
	f: 1st mate: 10 mo	5171
	f: 1st mate: 4 yr	6789
	f: 1st birth: 5 yr; few data	1332
	f: repro: 5 yr 20%, 7 yr 50%	6077
	f: 3-4 yr	1514
	f: 4-5 yr	753
	f: 5-7 yr	1282

	m: spermatogenesis: 7-10 yr	3181, 3182
	m: testes large: 7 yr; few data	1332
	m: fertile: 8-10 yr	3183
	m: 5 yr	1514
	m: 5-6 yr	753
	m: 6 yr	6786, 6789
	m: 5-10 yr	1282
GESTATION		
Odobenus rosmarus	implant: 3-3.5 mo	1282
	implant: 4 mo	3181
	implant: 4-5 mo	5755
	implant: ~5 mo	3182
	postimplantation: 10 mo	3182
	postimplantation: 11.5-12 mo	1282
	est 11 mo	1984
	11 mo; field data	753
	~1 yr	1332
	~1 yr	5171
	15 mo with delay	1282
	~15 mo	6789
	15-16 mo with delay	3181, 3182, 3183
LACTATION		
Odobenus rosmarus	solid food: 5-6 mo	3182
	wean: 1-2 yr	1332
	wean: 1-2 yr; 2 yr more common	1282
	wean: >1 yr; perhaps 2 yr	753
	wean: >1 yr; probably 2 yr	6789
	wean: ≥ 16 mo	6600
	wean: 18-24 mo	1457
	wean: ≥2 yr	1514
	wean: ~2 yr; some 3 yr	3182, 3181
	wean: 2.5 yr	5171
INTERLITTER INTERVAL		
Odobenus rosmarus	2 yr	1514
	2 yr	6600
	~2 yr	3182
	≥2 yr	3181
	2-3 yr	1332
	2-3 yr	1474
	2-4 yr	6789
SEASONALITY		
Odobenus rosmarus	mate: Jan-Mar; circumpolar	3181
	mate: peak Feb-Mar; Greenland	6789
	mate: Feb-Mar; Greenland, Canadian arctic, Chukchi/Laptev seas	1282
	mate: early-mid June; Bering Sea, Pacific	753
	mate: Nov-May, peak Apr-May; e Canadian Arctic	6786
	mate: Dec-Mar; Bering Sea	3182
	ovulate: Jan-Feb; Bering Sea	3181, 3182
	implant: June; Bering Sea	3182
	birth/mate: Apr-June; Novaja Zeml'a, Russia	1514
	birth: Apr-early May; Bering Sea	1332
	birth: Apr-June; Eskimo account; Hudson Bay, Canada	6600
	birth: Apr-June; AK	1474
	birth: mid Apr-early June, peak May; Bering Sea	3181, 3182
	birth: early May; Bering Sea, Pacific	753
	birth: Apr-June, peak May; Greenland, w Canadian Arctic	6786, 6789

Order Carnivora, Family Phocidae

Earless seals (*Cystophora* 140-410 kg; *Erignathus* 110-370 kg; *Halichoerus* 100-310 kg; *Hydrurga* 115-590 kg; *Leptonychotes* 325-550 kg; *Lobodon* 200-240 kg; *Mirounga* 250-3700 kg; *Monachus* 200-400 kg; *Ommatophoca* 130-300 kg; *Phoca* 30-250 kg) are marine carnivores with a worldwide distribution. Males are larger than females in *Cystophora* (9005, 9783), *Halichoerus* (1979, 3571), *Mirounga* (2097, 6275, 6286, 6329, 6482, 7017, 7018, 7019), and *Phoca* (925, 927, 1088, 1142, 4511, 7087, 7753, 10145; cf 6655, 10430), while females are larger than males in *Erignathus* (1458; cf 772), *Hydrurga* (4904), *Leptonychotes* (854), *Monachus* (1281; cf 189, 5682), and *Ommatophoca* (4903, 6289).

The hemoendothelial, endotheliochorial, or epitheliochorial, 500 g-2 kg, labyrinthine, zonary placenta has an antimesometrial or central implantation into a bipartite or bicornuate uterus (928, 1456, 4427, 4431, 4708, 4845, 5470, 5755, 6277, 7563, 7864, 9965, 10001, 11099) and is not usually eaten after birth (854, 5166, 5875, 6276, 6333, 10426). Implantation can be delayed (*Halichoerus* 478, 1172, 4708; *Hydrurga* 6971; *Leptonychotes* 4427, 6971; *Lobodon* 4427, 6971; *Mirounga* 3763; *Phoca* 3275, 4515). Estrus usually does not occur immediately after birth, but at the end of the short lactation period (*Cystophora* 9005; *Erignathus* 1458; *Halichoerus* 1079, 1171, 4416; cf 1979, 2187; *Leptonychotes* 1906; cf 6287; *Mirounga* 1622, 1623, 2023, 5055, 6276, 6277, 6286, 6329, 7018, 7019, 7062; cf 622, 6328, 6482; *Phoca* 1088, 10014; cf 3188, 3275, 7087, 7786).

Neonates are furred with open eyes (234, 622, 854, 970, 1142, 1359, 1374, 1457, 1622, 1979, 2023, 2024, 4415, 5682, 6275, 6276, 6277, 6333, 6787, 7853, 8703, 9244, 10158, 10781, 11831). Females have 2-4 mammae (854, 5682, 5755, 6788, 9243). Males have inguinal testes and a baculum (1395, 1473, 4259, 5755).

Details of anatomy (233, 236, 926, 1146, 1171, 4062, 4415, 4427, 5754, 5755, 6277, 6971, 7087, 9586, 10727), endocrinology (1169, 1170, 1177, 4060, 4843, 4845, 8849, 9473), birth (1479, 5875, 9246), suckling (2142, 3365, 5552a, 5552b, 5552c, 8130), milk composition (234, 240, 1034, 2034, 5959, 6265, 6331, 7244, 8036, 8285, 9126, 9244, 10407, 10525, 10729, 11168, 11505, 11698), reproductive energetics (285, 2061, 3186, 10402, 10405), and light (929) are available.

NEONATAL MASS

Cystophora cristata	4-5.5 kg	625
	10-15 kg	9783
	22.0 kg; se0.61; n21	1156, 1158, 8038
	30 kg	9005
Erignathus barbatus	29-43 kg	8691
	m: 32 kg; r30-34; n2; 1 term emb, 1 pup	5353
	33.6 kg; n13	1456, 1457, 1458
	43 kg	10431
Halichoerus grypus	4.5-6 kg	3571
	9.1-17.2 kg	1092, 4725
	9.2 kg	2187
	10-15 kg; from graph	3186
	13.6 kg; r11.8-15	240
	f: 14 kg; r10-19; n34	2067
	m: 14.24 kg; r13.63-14.54; n3	233, 234
	14.5 kg	1087
	f: 14.6 kg	5980
	f: 14.8 kg; r10.9-19.5; n26	1180
	m: 15 kg; r9-20; n61	2067
	m: 15.8 kg; r12.3-19.9; n25	1180
	m: 16.3 kg	5980
Hydrurga leptonyx	29.0 kg; 1 stillborn	10158
	29.5 kg; n1	1359
	35 ? kg	4904
Leptonychotes weddelli	22-25 kg	2400
	24 kg; n9	10729
	27.24 kg	854
	f: 27.7 kg; n8	1399
	m: 29.0 kg; n8	1399
	29 kg; r23-35; n17	6479
	30-40 kg	5755

Lobodon carcinophagus	20 kg	6285
	22 kg	854
Mirounga angustirostris	30-34 kg	7062
	30 kg; r22-45.5	6333
	f: 31.5 kg	6329
	34 kg	6330, 6331
	m: 36.0 kg	6329
	40 kg	2061, 9024
Mirounga leonina	29.1 kg; r26.7-30.5; n3	7853
	30.6 kg; r30-31; n4	5881
	m: 31 kg; n1	5881
	f: 35.0 kg; n19	6507
	m: 37 kg; r30-50; n13, Sept birth	1624
	37.7 kg; n46	6507
	f: 38.1 kg; se0.8; n7	2024
	38.5 kg; ±7.0; n29	4137
	38.5 kg; ±6.5; n56	4137
	m: 38.9 kg; se2.1; n8	2024
	f: 39 kg; n3	1397
	m: 39.09 kg; n27	6507
	f: 39.3 kg; sd7.29; n3	6507
	f: 39.4 kg; sd4.9; n28	6507
	40.9 kg; sd5.1; n6	6507
	41 kg; coefficient of variation 36.6	1396
	m: 41 kg; r28-54; n14, Oct birth	1624
	41.02 kg; sd5.3; n61	6507
	41.8 kg; stillborn, 38.8 kg; dead within 24 hr	5879
	m: 42.4 kg; sd5.3; n33	6507
	m: 42.5 kg; sd2.14; n3	6507
	m: 43 kg; n3	1397
	45-50 kg	6482
	f: 45 kg; r39-51; n8	6275
	45.2 kg; sd5.7; n12	7019
	m: 49 kg; r43-56; n7	6275
Monachus albiventer	17 kg; term emb	10910
Monachus monachus	17-26 kg	1145
	20 kg	9788
Monachus schauinslandi	10.65 kg; n10	11831
	15.2 kg stillborn, 18.6, 19.3 kg	9090
	~16 kg	5679
	16-17 kg	1281
Ommatophoca rossi	16.8 kg; n1	10781
	est 27 kg	6289
Phoca caspica	5 kg	8681
Phoca fasciata	8.5 kg; near term	1457
	10 kg	10430
	10.5 kg	1459
Phoca groenlandica	2.7-3.6 kg	3571
	3-4 kg	625
	6.2-14 kg	10402
	9.30 kg; se0.62; n6	8038
	9.9 kg; sd1.7; n32	5978, 6264
	10-14 kg	4012
	10.7 kg; sd1.6; n48	5978, 6264
	10.8 kg; 95%ci 0.65; n40	5152, 10404
Phoca hispida	3.5-4 kg	5353
	4.0 kg	1457
	4-5 kg	8679
	7.5 kg	3188
	mode 8-12; r5.5-19.5 kg	3571
	8.1 kg	3534
Phoca largha	7.1 kg; r6.9-7.3	1089
Phoca sibirica	1.5-4.5 kg	10782
	3 kg	8680

Phoca vitulina	6.9-15 kg	3571
	mode 8-9 kg; r7-10.5	11829
	m: 8.9 kg; n1	233
	m: 9 kg; n1	9246
	9-11 kg	4415
	f: 9.49 kg	1143
	9.5-11.81 kg	1457
	m: 9.53 kg	1143
	10.2 kg; 95%ci 1.5; n9	925
	f: 10.4 kg	1143
	m: 10.59 kg	1143
	f: 10.71 kg; sd2.76; n13	7864
	f: 11 kg	1142
	m: 11 kg	1142
	m: 14.81 kg; sd2.74; n5	7864
	15 kg	4511
	est ~15-18 kg; n1	5166
NEONATAL SIZE		
Cystophora cristata	m: 87-115 cm	9005
	100 cm	9783
Erignathus barbatus	87 cm	10431
	120 cm	7086
	m: 129.25 cm; r127.5-131.0; n2; 1 term emb, 1 pup	5353
	130 cm	1457
	131.3 cm; n13	1456
Halichoerus grypus	60-110 cm	3571
	76 cm	5755
	83.5 cm; r87-100; n2	2187
	m: 87-105 cm	1087
	95-105 cm	1091
Hydrurga leptonyx	150-160 cm	4904
	est 150-160 cm	8121
	157.5 cm; n1	1359
Leptonychotes weddelli	120 cm	854
	120-130 cm	5755
Lobodon carcinophagus	nose-tail: 113.03 cm; r102-120; term emb	6278a
	114-127 cm; n19	6278a
	120 cm	6285
	130 cm; r115-145; n35 term emb	8121
	132 cm; n1	8121
	150 cm	5755
Mirounga angustirostris	120 cm	5755
	~127 cm	7062
	141.8 cm; se10	6333
	f: 147 cm	6329
	m: 153 cm	6329
Mirounga leonina	f: 116.0 cm; se2.6; n7	2024
	m: 116.0 cm; se9.0; n8	2024
	120-127 cm	6275, 6277, 6286
	130 cm	6482
	136.5 cm; r133-142; n4	5881
Monachus albiventer	120 cm; term emb	10910
Monachus monachus	80 cm	9788
	98.8 cm; r87-120	1145
Monachus schauinslandi	100 cm	1281
	100 cm	5679
Ommatophoca rossi	96 cm	5755
	est ~105 cm	8121
	est 105-120 cm	6289
Phoca caspica	65-79 cm	8681
Phoca fasciata	80 cm	10430
	85 cm; near term	1457, 1459
Phoca groenlandica	75-90 cm	10402
	90-105 cm	6264

	95 cm	3571
	105 cm	5152
Phoca hispida	50-60 cm	8679
	54 cm	3534
	58 cm	1457
	64-72 cm	3188
	65 cm; n5; n3 term emb, 2 new born	7087
Phoca largha	76-85 cm	1089
	~85 cm; r78.0-92.5; term emb	7752, 7753
Phoca sibirica	70 cm	8680
	226 mm; emb	10782
Phoca vitulina	m: 75 cm; n1	233
	f: 75.76-77.26 cm; n2 samples	1143
	m: 75.94-78.16 cm; n2 samples	1143
	m: HB: 76 cm; n1	9246
	76.4 cm; r75.9-77.7 for 3 yr data	1146
	mode 80 cm; r71-90	11829
	81.6 cm; 95% ci 6.2; n9	925
	84 cm	1457
	85 cm	5166
	95-105 cm	4511
	98 cm; n20	7752, 7753
	98.2 cm; se3.2	1088
WEANING MASS		
Cystophora cristata	42.6 kg; se1.37; n11, 4 d	1158
	43.6 kg; se1.78; n5	8038
	43.7 kg; se1.9; n12	1156
Erignathus barbatus	43-90 kg	8691
	85 kg; n11	1456, 1458
Halichoerus grypus	f: 35.8 kg	5980
	m: 40.0 kg	5980
	41-45 kg; 17 d	2067
	41.8 kg	240
	42.5 kg	2187
	45.1 kg; sd7.7; n17	11933
	50 kg; from graph	3186
Leptonychotes weddelli	m: 93.75 kg; n8, 5 wk, 2 sites	1399
	f: 101.67 kg; n8, 5 wk, 2 sites	1399
	110 kg	6787
	120-140 kg	10729
	125 kg	2400
	135 kg; n1, 52 d	6479
Mirounga angustirostris	f: 129.73 kg; se10.55; n4	9025
	136 kg; r113-238	6331, 9024, 9126
	m: 137.53 kg; se19.30; n5	9025
	158 kg	2061, 6329, 6330
Mirounga leonina	m: 70.5 kg; r63-78; n2	2024
	106.5 kg; ±26.5; n36, 3 wk	4137
	109.0 kg; ±21.0; n22, 3 wk	4137
	~110-160 kg	6482
	119.5 kg; ±13.0; n20, 3 wk	4137
	127.0 kg; ±22.5; n40, 3 wk	4137
	m: 129 kg; n3, 3 wk	1397
	f: 141 kg; n3, 3 wk	1397
	182 kg	6277
Monachus schauinslandi	~64 kg	5679
Phoca groenlandica	m: 20-55 kg	10402
	22-40 kg; 12-15 d	5978
	31.4 kg; sd4.8; n15	11933
	34.4 kg; 95%ci 2.6; n29	10404
	38.6 kg; se1.45; n5	8038
Phoca hispida	14 kg	3534
Phoca vitulina	~17 kg; from graph; 3 wk	11829
	m: 20.9 kg; r19.1-22.7; n2	925
	f: 25.3 kg; r20.9-28.6; n8	925

	f: 26.2 kg; n6	1146
	m: 27 kg; n1	9246
	m: 27.27 kg; n11	1146
WEANING SIZE		
Mirounga leonina	f: 123.0 cm; se5.3; n7	2024
	m: 125.4 cm; se4.8; n7	2024
Erignathus barbatus	147.3 cm; n48	1456
Phoca groenlandica	87-110 cm	10402
Phoca vitulina	f: 90.0 cm; n5, 30-36 d	1146
	m: 90.7 cm; n11, 30-36 d	1146
LITTER SIZE		
Cystophora cristata	1	1474
Erignathus barbatus	1	7086
Halichoerus grypus	1	1979
	1	1079
	1 ea; n3 lt	6396
	1 emb ea; n36 f	4843
Hydrrurga leptonyx	1 emb ea; n19 f	8121
Leptonychotes weddelli	1	10846
	1	1374
	1; n~81	6637
	1; 2 pr twins found dead	6479
	1; twins 1%	854
Lobodon carcinophagus	1 emb; 0.4% twins; r1-2; n226 f	8121
Mirounga angustirostris	1	2097, 6330
	1	5055
Mirounga leonina	1	4062
	1; 2 rare	6275, 6276, 6277
Monachus schauinslandi	1	9090
Ommatophoca rossi	1 emb ea; n7 f	8121
Phoca fasciata	1	1474
Phoca groenlandica	mode 1; occasionally 2	1474
	1; r1-2	3571
Phoca hispida	1; 2 uncommon	3534
Phoca sibirica	~1.02 yg; 2.2-3.9% twins; r1-2	8252
Phoca vitulina	1	925
	1 yg; 1 report of twin emb	9590
	1 yg; n1 lt	9246
	1 yg ea; n2 lt	7391
	1.01; 1.26% twins; r1-2;; n556 lt	11829
SEXUAL MATURITY		
Cystophora cristata	f: 1st ovulate: median 2.8 yr; r2-9; n114	8122
	f: 1st birth: median 3.8 yr; r3-11; n114	8122
	m: 4-6 yr	9005
Erignathus barbatus	f: 1st cl: 3 yr	1456
	f: 1st cl: 3 yr; 4 yr 20% have cl	8691
	f: 1st cl: 6 yr	7086
	f: 1st conceive: 5 yr	1456
	f: 1st birth: 6 yr	1456
	f: 3 yr (8%), 4 yr (70%)	10833
	f: 5-6 yr	10431
	m: increase testes: 4-5 yr	8691
	m: testes large: 7 yr	7086
	m: 5 yr (50%), 6 yr (66%), 7 yr (100%)	10833
	m: 6-7 yr	10431
Halichoerus grypus	f: 1st mate: 3-4 yr	1979
	f: 1st preg: 4 yr 50%	1174
	f: 1st birth: 5 yr 16%; 6 yr 45%; 7 yr 39%	4465
	f: 5 yr; few data	2187
	m: 1st mate: 3-4 yr	1979
	m: 6 yr; few data	2187
Hydrurga leptonyx	f: 1st ovulate: est 2 yr; n1	8121
	f: 1st cl: 2nd yr	4259
	f: 1st cl: 3-6 yr	10834

	f: 1st conceive: 3 to just over 4 yr	6278
	f: 1st conceive: 5 yr; n1	8121
	m: epididymal sperm: 3-6 yr	8121
	m: histology testes: 3rd yr	4259
	m: testes large: 4-5 yr	10834
Leptonychotes weddelli	f: 1st conceive: 26 mo	854
	f: 1st preg: 4 yr 50%; n8/16	10428
	f: 1st preg: 5 yr	6787
	f: 1st birth: 4-5 yr; r2-8	2161
	f: 1st birth: 4-6 yr 26%; n45/170	2886
	f: 1st birth: mode 6 yr; mean 7.1 yr	10759
	f: 2 yr	6479
	f: 3-6 yr	2400
	m: 3-6 yr; social maturity: 7-8 yr	2400
	f/m: 1st mate: 6-8 yr	10427
Lobodon carcinophagus	f: 1st ovulate: 3 yr; median 3.5 yr	8121
	f: 1st ovulate: 3.8 yr	761
	f: 1st cl: 3-5 yr	10834
	f: 1st birth: 5 yr	761
	f: 2-6 yr	6285
	m: epididymal sperm: 3-6 yr	8121
	m: testes large: 4-7 yr	10834
Mirounga angustirostris	f: 1st birth: 3-5 yr	6329, 9024
	f: 1st birth: 3-5 yr; mode 4 yr	5055
	f: 1st repro: 3.9-4.6 yr	5056
	m: 3 yr n1	10278
	m: 4-5 yr; social maturity: 9-10 yr	6329
Mirounga leonina	f: 1st mate: 2 yr	6276, 6277, 6286
	f: 1st mate: 3 yr; r2-5	6287
	f: 1st birth: 3 yr	6276, 6286
	f: 1st birth: 4-7 yr	1622
	f: 1st birth: mode 4 yr; r3-5	7017, 7019
	m: 1st sperm: 3-4 yr	6275, 6276, 6277, 6286
	m: 1st mate: 4-6 yr	6287
	m: 1st mate: 6-7 yr	6276, 6286
	m: 1st mate: 10 yr	1622
	m: 1st mate: 10 yr	5401
	m: 5-6 yr	1622
Monachus monachus	f: 1st mate: 4 yr	9243
	f/m: 4 yr	4417
	f/m: est 4 yr; few data	1145
Monachus schauinslandi	f: 5 yr	5332
Ommatophoca rossi	f: 1st ovulate: 3.6 yr; r2-7; n7	8121
	f: 1st cl: 3-5 yr	10834
	f: 3-4 yr; few data	6289
	m: testes large: 3-5 yr	10834
	m: 2-7 yr; few data	6289
Phoca caspica	f: 1st birth: 5-7 yr	8681
	m: 7 yr	8681
Phoca fasciata	f: 1st birth: 2 yr; 50% 3 yr	1459
	f: 2 yr 44%; 3 yr 86%	10833
	f: 2-4 yr	10430
	m: 2 yr 15%; 3 yr 69%	10833
	m: 3 yr 22%; 4 yr 75%	1459
	m: 3-5 yr	10430
Phoca groenlandica	f: 1st cl: 3 yr	7786
	f: 1st cl: 4.3-5.5 yr	9782
	f: 1st ovulate: 3-6 yr	9245
	f: 1st ovulate: 4-5 yr	9780
	f: median 5 yr	6264
	m: testes large: 50% 3-5 yr	9782
	m: median 6 yr	6264
	f/m: 3-5 yr	7786
	f/m: 4-5 yr	9244

Phoca hispida	f: 1st cl: 5 yr n1/12; 6 yr n10/10	5353
	f: 1st cl: 5-8 yr	7087
	f: 1st conceive 4-7 yr	3534
	f: 1st ovulate: 4 yr	3261
	f: 1st ovulate: 6 yr	10432
	f: 1st cl: 3 yr	3534
	f: 1st cl: 4 yr, 12.5% in Cumberland, 40.9% Home Bay	10145
	f: 1st preg: 7-8 yr 52%	4594
	f: 4 yr (18%), 5 yr (74%)	10833
	f: 5-7 yr	3188
	m: spermatogenesis: 7th yr	5353
	m: testes size, sperm in ejaculate: 7 yr	7087
	m: 5 yr (40%), 6 yr (43%), 7 yr (100%)	10833
	m: 5-7 yr	3534
	m: 6-8 yr	3188
	m: 7 yr	10145
	f/m: 3 yr 20%; 4 yr 60%; 5 yr 80%	6655
Phoca largha	f: 3-4 yr	1089
	m: mating attempts: 4 yr	739
	m: 4-5 yr	1089
Phoca sibirica	f: 1st mate: 5-6 yr	8680
	f: 1st mate: mode 6 yr; r4-6	10782
	m: 1st mate: ≤7 yr	10782
	m: 1st mate: 7-8 yr	8680
Phoca vitulina	f: 1st ovulate: 3.7 yr	4515
	f: 1st cl: 2 yr 20%; 3 yr 38%; 4 yr 34%; 5 yr 8%	925
	f: 1st cl: 3 yr 20%; 4 yr 100%	7752
	f: 1st conceive: end 3rd yr	4511
	f: 1st preg: 4.6 yr	4515
	f: 1st birth: 3 yr	9590
	f: 2-5 yr	1088
	f: 3 yr 10%; 4 yr 96%	10833
	f: 4 yr	11829
	f: 6 yr	9246
	m: large baculum: 3 yr	4511
	m: spermatogenesis 3 yr 20%, 4 yr 100%	7752
	m: end 3rd yr	11829
	m: 3 yr 33%; 4 yr 100%	10833
	m: 3-6 yr	925
	f/m: 3-7 yr	1146
CYCLE LENGTH		
Mirounga leonina	estrus: 4 d	7021
	estrus: ~5 d	6276
Phoca vitulina	estrus: 1-9 wk; proestrus: 3-4 wk	926
GESTATION		
Cystophora cristata	implant: 16 wk; field data	9005
	implant: 4 mo	9783
	postimplantation: 7.7 mo	9783
	postimplantation: 240-250 d; field data	9005
	11+ mo	1991
	11.5 mo	5755
	50-51 wk; field data	9005
	11.7 mo	9783
Erignathus barbatus	implant: 1.5-3.5 mo	8691
	implant: 2 mo	10833
	implant: 2 mo; field data	1458
	implant: 2.3-3 mo	10431
	implant: 2.5 mo	1456
	implant: 2.5 mo	7086
	postimplantation: 7 mo	7086
	postimplantation: ~8 mo	10431
	postimplantation: 8.5-10.5 mo	8691
	postimplantation: 9 mo	10833
	postimplantation: 9 mo; field data	1458

	9.5 mo	7086
	10.5-11 mo	10431
	11 mo	10833
	~11 mo; field data	1458
	12 mo	8691
Halichoerus grypus	implant: 3 mo	1087
	implant: 130 d	4708
	implant: 135.69 d; sd14	1171, 1172
	334 d; n1	5552b
	343 d; n1	5552c
	355 d	6396
	11.5 mo	1087
	1 yr	1171
	1 yr	4708
	1 yr	2187
	373 d; n1	5552a
Hydrurga leptonyx	implant: 1 mo	6287
	implant: 2.5-3.5 mo	5755
	postimplantation: 11 mo	6287
	~8 mo	4259
	est 8-9 mo	1374
	11 mo with 2.5-3 mo delay	10834
	~1 yr	6287
Leptonychotes weddelli	implant: 1 mo	6287
	implant: 1 mo	854
	implant: 2 mo	2400
	postimplantation: 9 mo	2400
	postimplantation: 11 mo	6287
	est > 9 mo and < 10 mo 10 d	6479
	est 9.5-10 mo	854
	~11 mo	2400
	11.5 mo	5755
	~1 yr	6287
Lobodon carcinophagus	implant: 2-3 mo	6285, 6287
	postimplantation: 8-9 mo	6285, 6287
	9 mo	854
	11 mo	6285, 6287
Mirounga angustirostris	implant: 3 mo	6329
	implant: 4 mo	5755
	postimplantation: 8.3 mo	6329
	11 mo; field data	7062
	11.3 mo	6329
Mirounga leonina	implant: 16 wk	6482
	implant: 4 mo	6275, 6276, 6277, 6286, 6287
	implant: ~4 mo	3763
	postimplantation: ~7 mo	6482
	postimplantation: 7.5 mo	6275, 6276, 6277, 6286, 6287
	postimplantation: ~8 mo	3763
	~11 mo	6482
	11.5 mo	6275, 6276, 6277, 6286, 6287
	1 yr	3763
	12 mo; n2	5879
Monachus monachus	10 mo	4417
	est 11 mo	1145
Ommatophoca rossi	implant: 3.5-4.5 mo	5755
	~11.5 mo	5755
	almost 1 yr	10698
Phoca caspica	11 mo	5755, 8681
Phoca fasciata	implant: 2 mo	10833
	postimplantation: 9 mo	10833
	9 mo	10062

	10.5-11 mo	10430
	11 mo	10833
Phoca groenlandica	implant: 2-3 mo	9244
	implant: 11 wk	9245
	implant: 4.5 mo	6264, 10406
	postimplantation: 7 mo	6264, 10406
	postimplantation: 7.5 mo	9245
	postimplantation: 8 mo	9244
	11 mo	9245
	11+ mo	1991
	11.5 mo	6264, 10406
	11.5 mo	9244
Phoca hispida	implant: 2 mo	10833
	implant: 3.5 mo	10432
	implant: 3.5 mo	7087
	postimplantation: 7-7.5 mo	10432
	postimplantation: 8.5 mo	7087
	postimplantation: 9 mo	10833
	8 mo	10062
	~9 mo	1474
	10.5-11 mo	10432
	11 mo	10833
	~11 mo	3571
	~1 yr	7087
Phoca largha	10.5 mo	1089
Phoca sibirica	~9 mo with delay	10782
Phoca vitulina	implant: 1-3 mo	926
	implant: 2 mo	10833
	implant: ~2 mo	566
	implant: est 2-3 mo	4415
	implant: 3 mo	1146
	implant: ~3 mo	4515
	postimplantation: 7-9 mo	926
	postimplantation: 8 mo	1146
	postimplantation: 9 mo	10833
	9 mo	10062
	9.5 mo; field data	11245
	~10 mo	4511
	10.5-11 mo	1088
	~11 mo	8680
	11 mo	7752
	11 mo	5470
	11 mo	10833
	11 mo	1146
	11 mo with 2-2.5 mo delay	925, 928, 3275
	341 d	4582
	341 d; r10-11 mo	3571
LACTATION		
Cystophora cristata	wean: 4 d; r3-5	1156, 1158, 8038
	wean: 7-12 d	9005
	wean: 10 d	9783
Erignathus barbatus	solid food: 13-14 d	1458
	wean: 12-18 d	1456, 1457, 1458
	wean: 3-4 wk	10833
	wean: 6-7 wk	5755
Halichoerus grypus	wean: 14 d	4708
	wean: 16-21 d	1084, 1085, 1092, 2067
	wean: 17 d	5980
	abandoned: 17-18 d	234
	wean: 18 d	3186
	wean: 20 d	2187
	wean: 20.3 d; 18-24 d; n3	5552a, 5552b, 5552c
	wean: 23 d	6396
Hydrurga leptonyx	wean: ~1 mo	10834

Leptonychotes weddelli	wean: 12-18 d	5755
	wean: 4-5 wk	8682
	wean: 5-6 wk	6637
	abandoned: 33-44 d	5578
	wean: 6 wk	2400
	wean: 6-8 wk	1906
	wean: 45 d	10729
	wean: 7 wk	854
	wean: 7 wk	6787
	permanent separation: 50.3 d; r40-55; n18	6479
Lobodon carcinophagus	wean: est 4 wk	10008
Mirounga angustirostris	wean: 22-28 d; varies with age of mother	2061, 9024
	wean: 22-32 d	5055
	wean f: 26.6 d; ±1.42	2097, 6331, 6332, 6333, 9025
	wean m: 27.8 d; ±1.54	2097, 6331, 6332, 6333, 9025
	independent: 3-4 mo	622
Mirounga leonina	wean: 14-22 d	10158
	wean: 20-34 d	4578
	wean: 3 wk	6482
	wean: 3 wk	1397
	wean: 3 wk	10698
	wean: 21.8 d; ±2.4; n62	4137
	wean: 22.5 d; ±3.5; n18	2023, 2024
	wean: 23 d	6275, 6276, 6277, 6286, 6287
	wean: 23 d	889
	wean: 23 d	7021
	wean: 23-25 d	7019
	wean: 24.2 d; ±2.3; n10	2024
Monachus monachus	wean: 5 wk	4417
	wean: 6-7 wk	9243
	wean: 6-7 wk	1145
Monachus schauinslandi	wean: ~1 mo	11831
	wean: 5-6 wk	5679, 5680, 5682
Phoca fasciata	wean: 3-4 wk	1457, 1459
	wean: 3-4 wk	10833
Phoca groenlandica	wean: 8-12 d	6264, 10404
	wean: 10 d	10402
	wean: 10-12 d	9245
	wean: 12-15 d	5978
	wean: ~13 d	8038
Phoca hispida	wean: ~3 wk	3571
	wean: 3-4 wk	10833
	wean: 4-6 wk	10432
	wean: ~4.5-7 wk	3534
	wean: 5-7 wk	1457
	wean: 2 mo	5755
	wean: 5 mo	8679
Phoca largha	wean: 2-3 wk	7752
Phoca sibirica	wean: 1.5-2.5 mo	10782
	wean: 2-2.5 mo	8680
Phoca vitulina	wean: 2-3 wk	11829
	wean: 2-6 wk	926
	wean: 3 wk	4415
	wean: 28 d	9246
	wean: 30 d	1146
	wean: 3-4 wk	4416
	wean: 4-5 wk	5470
	wean: 4-6 wk	1084
	wean: 4-6 wk	9590
	wean: 4-6 wk	7752, 7753
	wean: 4-6 wk	7864
	wean: 4-6 wk; then 8-14 d fast before solid food	3571

	wean: 5-6 wk	925, 928
	wean: 2 mo	10195
INTERLITTER INTERVAL		
Erignathus barbatus	1 yr	10833
	2 yr	7086
	2 yr	10062
Halichoerus grypus	1 yr	1079
	1 yr	1979
	1 yr	2187
Leptonychotes weddelli	1 yr	6787
	est 2 yr	10007
Lobodon carcinophagus	2 yr	854
Mirounga angustirostris	1 yr	2097, 6329
	1 yr	7062
Mirounga leonina	1 yr	1622
	1 yr	6277, 6286
Monachus monachus	2 yr	9243
Monachus schauinslandi	382 d; r361-393	11831
	1.5 yr	5332
Phoca fasciata	1 yr	10833
Phoca groenlandica	1 yr	3571
Phoca hispida	1 yr	3571
	1 yr	10062
	1 yr	10833
Phoca vitulina	1 yr	925
	1 yr	10062
	1 yr	10833
SEASONALITY		
Cystophora cristata	mate: mid-end Mar; N Atlantic	9005
	birth: end Feb; N Atlantic	1474
	birth: Mar-Apr; n hemisphere	5755
	birth: mid Mar; ne NEW, Canada	1156
	birth: mid-end Mar; Greenland	9783
	birth: mid-end Mar; N Atlantic	9005
Erignathus barbatus	spermatogenesis: Mar-Apr; Barents Sea	8691
	mate: end Mar-mid May; Barents Sea	8691
	mate: 1st 3 wk May; Bering Strait/Sea, Okhotsk Sea	1458
	mate: May; N Atlantic, Pacific	10431
	mate: June-July; Okhotsk Sea	10062
	ovulate; Mar-June; Barents Sea	8691
	ovulate; May; e Canadian Arctic	7086
	implant: mid July-early Aug; Bering Strait & Sea, Okhotsk Sea	1458
	implant: end July-Aug; e Canadian Arctic	7086
	birth: mid Mar-early May, peak Apr; Bering Strait/Sea, Okhotsk Sea	1456, 1458
	birth: end Mar-mid May; Barents Sea	8691
	birth: Apr; Bering, Chukchi Seas	1457
	birth: Apr-May; n hemisphere	5755
	birth: Apr-May; N America	1474
	birth: mid Apr; Okhotsk, Bering Seas	10833
	birth: mid-end Apr; AK	5353
	birth: mid Apr-May; N Atlantic & Pacific	10431
	birth: Apr-May; e Canadian Arctic	7086
Halichoerus grypus	mate: Feb-Mar; Baltic Sea	2187
	mate: Feb-Mar; Baltic Sea	3571
	mate: Feb-Mar; Baltic Sea	1085
	mate: Mar; Baltic Sea	4961
	mate: Mar-Apr; E Anglia, England, UK	4416
	mate: Sept-Oct; Pembrokeshire, UK	478
	mate: Oct; UK	233
	mate: peak end Oct; Orkney, Shetland I, UK	4708
	mate: end Oct; Fame I, UK	1084
	mate: end Oct-Nov; Fame I, Northumberland, UK	2066
	mate: end Dec-mid Feb; Sable I, Canada	1079

	mate: fall; Fro I, Norway	1979
	ovulate: Mar-Apr; E Anglia, England, UK	4416
	conceive: end Nov; Farne I, UK	1172
	implant: after early Jan; n1; Pembrokeshire, UK	478
	implant: end Jan; Farne I, UK	1084
	implant: Mar-May; Farne I, UK	1172
	birth: Jan; e Canada	4961
	birth: Jan; n1; UK	1288
	birth: Jan-Feb; w Atlantic	1087
	birth: Jan-Feb; Netherlands	1
	birth: Jan-Mar; Baltic Sea	3571
	birth: Feb-Mar; Baltic Sea	1087
	birth: Feb-Mar; Baltic Sea, St Lawrence	5755
	birth: end Feb-early Mar; Baltic Sea	2187
	birth: Mar; E Anglia, England, UK	4416
	birth: Mar; n2; Rumsey I, Pembrokeshire, UK	479
	birth: end Aug-early Dec; UK	5755
	birth: Sept-Nov; UK	4961
	birth: Sept-Oct; Norway, Sweden	4961
	birth: Sept-Jan; e Atlantic	1087
	birth: mid Sept-early Nov; Orkney, Shetland I, UK	4708
	birth: end Sept-mid Oct; Fro I, Norway	1979
	birth: Oct; Iceland	4961
	birth: Oct; n Rona, UK	285
	birth: mid Oct-Dec, peak Nov; Farne I, Northumberland, UK	1091, 1092, 1093, 2066, 2067, 4725, 4726, 4727
	birth: Nov-Dec; N Atlantic	1171
	birth: Dec; Kola Peninsula, Baltic Sea	4961
	birth: Dec-Feb, peak Jan; Netherlands	11240a
Hydrurga leptonyx	mate: Jan-Mar; Falkland I	4259
	mate: end Dec; Antarctic Peninsula	6287
	mate: autumn; Australia	10698
	birth: Oct-Dec; S Africa	10158
	birth: mid Oct-mid Nov; Pacific Antarctic	10834
	birth: est Nov; sw Atlantic	8121
	birth: Nov-Dec; Antarctica	4904
	birth: Nov-Dec; s hemisphere	5755
	birth: mid Nov; n1; Heard I, Southern Ocean	1359
	birth: end Nov; Antarctic Peninsula	6287
	wean: end Dec; Antarctic Peninsula	6287
Leptonychotes weddelli	spermatogenesis: Nov; Graham Land, Antarctica	854
	testes: large Sept-Nov, small June; Antarctic	4427
	mate: ear-mid Nov; King George I, Antarctica	8682
	mate: Nov-Dec; Ross I, Antarctica	10007
	mate: mid Nov-mid Dec; Graham Land, Antarctica	854
	mate: mid Dec; McMurdo Sound, Antarctica	6287
	mate: summer; Australia	10698
	ovulate: end Nov; Graham Land, Antarctica	854
	ovulate: mid Dec; McMurdo Sound, Antarctica	10428
	implant: mid Jan-mid Feb; McMurdo Sound, Antarctica	10428
	implant: mid Jan-mid Feb; Ross I, Antarctica	10007
	birth: Aug-Sept; Signy I, S Orkneys, UK	6787
	birth: end Aug-early Oct; King George I, Antarctica	8682
	birth: end Aug-Sept; Graham Land, Antarctica	1374
	birth: Sept-Oct; S Africa	10158
	birth: Sept-Oct; s hemisphere	5755
	birth: Sept-early Nov; Antarctica	2400
	birth: 3rd wk Sept; Antarctic	854
	birth: Oct; Antarctica	6637
	birth: mid Oct-Nov; Hutton Cliffs, Antarctica	5578
	birth: Nov; McMurdo Sound, Antarctica	6287
	birth: Nov-Dec; Ross I, Antarctica	6444
Lobodon carcinophagus	epididymal sperm: Oct; sw Atlantic	8121
	mate: Sept-Oct; Antarctica	10008

	mate: Oct-Dec; Graham Land, Antarctica	854
	mate: mid Nov; Antarctic Peninsula	6287
	mate: end spring; Australia	10698
	ovulate: Oct; sw Atlantic	8121
	birth: Sept; Graham Land, Antarctica	854
	birth: Sept-Oct; Antarctica	10008
	birth: Sept-Oct; Antarctica	6285
	birth: Sept-early Nov; s hemisphere	5755
	birth: mid Sept-early Nov; S Africa	10158
	birth: Oct; sw Atlantic	8121
	birth: Oct-early Nov; s hemisphere	5755
	birth: end Oct; Antarctic Peninsula	6287
Mirounga angustirostris	estrus: Feb; CA	622
	mate: Dec-Feb; Baja CA Mexico, n CA US	6329
	mate: Jan; Año Nuevo I, CA	9024
	mate: mid Jan; w US, Mexico	7062
	mate: mid Jan-mid Mar; Año Nuevo I, CA	6328
	birth: Dec-Feb; CA	622
	birth: Dec-Feb; Baja CA Mexico, n CA US	6329
	birth: mid Dec; w US, Mexico	7062
	birth: end Dec-early Feb; Farallon I, CA	5055
	birth: Jan; Año Nuevo I, CA	6332, 9024
	birth: peak Jan-Feb; CA	6330
Mirounga leonina	spermatogenesis: Aug-Nov; Macquarie I, Heard I, Southern Ocean	4060, 6482
	testes: large Aug-Nov; Falkland I	6277
	testes: large Nov, small May; Macquarie I, Southern Ocean	4060
	mate: Aug-Oct, peak Sept; S Georgia, s Atlantic Ocean	7021
	mate: Sept-Nov; Macquarie I, Southern Ocean	4060, 4061, 4062
	mate: Oct-mid Nov; Macquarie I, Heard I, Southern Ocean	6482
	mate: mid Oct-mid Nov; Heard I, Southern Ocean	3763
	mate: peak end Oct-early Nov; Marion I, S Africa	2023
	mate: Nov; Cumberland E Bay, S Georgia, s Atlantic Ocean	7018
	mate: Nov; S Georgia, s Atlantic Ocean	6287
	implant: end Feb; Macquarie I, Heard I, Southern Ocean	6482
	birth: Sept; Macquarie I, S Georgia, s Atlantic Ocean	7017
	birth: Sept-Oct; Australia	10698
	birth: Sept-Oct; S Georgia, s Atlantic Ocean	7021
	birth: Sept-Oct, few Nov; Signey I, S Orkney, S Georgia, s Atlantic Ocean	6276
	birth: Sept-Oct; S Georgia, s Atlantic Ocean, Macquarie I, Heard I, Southern Ocean	1622, 1623
	birth: Sept-Oct, peak mid Oct; Macquarie I, Heard I, Southern Ocean	6482
	birth: Sept-Nov; Gough I, s Atlantic Ocean	889
	birth: Sept-Nov; S Georgia, s Atlantic Ocean, Macquarie I, Southern Ocean	1622
	birth: Sept-Nov; S Orkney, S Georgia, s Atlantic Ocean, Macquarie I, Southern Ocean	6286
	birth: Sept-Nov; Marion I, S Africa	2023, 2024
	birth: mid Sept-Oct; Kergeulen I, Indian Ocean	8269
	birth: end Sept-Oct; s hemisphere	5755
	birth: end Sept-early Nov; S Georgia, s Atlantic Ocean	7020
	birth: Oct; Cumberland E Bay, S Georgia, s Atlantic Ocean	7018, 7019
	birth: peak Oct; S Africa	10158
	birth: Oct-Nov; Antarctica	4578
	birth: Oct-Nov; Heard I, Southern Ocean	3763
	birth: mid Oct; S Georgia, s Atlantic Ocean	6287
	wean: mid Nov; S Georgia, s Atlantic Ocean	6287
Monachus monachus	mate: Mar-July; Mediterranean	4417
	mate: Aug; n1; Mediterranean, n Atlantic	9788
	birth: Mar-May; Mediterranean	4417
	birth: May-Nov, peak Sept; Mediterranean, n Atlantic	9243, 9788
	birth: Sept-Oct; n hemisphere	5755

	yg found: yr rd; Mediterranean	1145
Monachus schauinslandi	birth: Dec-July, peak Apr-May; Fr Frigate Shoals, Lisianske, Laysen, Pearl, Hermes Rf, Kure Atoll, N Pacific	1281
	birth: Dec-Aug, peak Mar-May; Kure, Pearl, Hermes reef, Lisianski, Laysan, French Frigate shoals, N Pacific	5679
	birth: Jan-June; Leeward I, Pacific	5682
	birth: Jan-Aug, peak Mar-Apr; HI	5332
	birth: Feb-June, peak Mar-May; n47; HI	11831
	birth: end Mar-early June; n hemisphere	5755
Monachus tropicalis	preg f: Dec; n1; W Indies	189
	birth: Nov-Dec; n hemisphere	5755
	birth: Dec; Caribbean	5753
	lact f: Dec; n1; W Indies	189
Ommatophoca rossi	mate: est Nov; sw Atlantic	8121
	birth: early-mid Nov; Pacific Antarctic	10834
	birth: early-mid Nov ?; Antarctic	10781
	birth: Nov-Dec; s oceans	6289
	birth: Nov-Dec; Australia	10698
Phoca caspica	birth: Jan-Feb; n hemisphere	5755
	birth: end Jan-Feb; Caspian sea	8681
Phoca fasciata	mate: end Apr-early May; Bering, Okhotsk Seas	10430
	mate: end Apr-early May; Bering, Okhotsk, Chukchi Seas	1459
	mate: end July-Aug; Far East	10062
	ovulate: ?-early May; Bering, Okhotsk, Chukchi Seas	1459
	birth: Mar-Apr; n hemisphere	5755
	birth: end Mar-mid Apr; Bering, Okhotsk Seas	10430
	birth: Apr-early May; Bering, Okhotsk, Chukchi Seas	1459
	birth: mid Apr; Okhotsk, Bering Seas	10833
Phoca groenlandica	mate: Mar-Apr; Europe	3571
	mate: mid-end Mar; White Sea	7786
	mate: est mid-end Mar; Gulf of St Lawrence	9244, 9245
	conceive: mid Mar; nw Atlantic	10406
	delay: Mar-June; nw Atlantic	10406
	implant: Aug; nw Atlantic	10406
	birth: Jan-Apr, peak end Feb-early Mar; n Atlantic	9245
	birth: Feb-early Mar; NEW, Canada	1991
	birth: end Feb-early Mar; White Sea	6264
	birth: end Feb-early Apr; Europe	3571
	birth: Mar; NEW, Canada	6264, 10403, 10406
	birth: Mar; White Sea	10014
Phoca hispida	spermatogenesis: Mar-mid May; e Canadian Arctic	7087
	spermatogenesis: Mar-May; AK	5353
	mate: Mar-early May; arctic	10432
	mate: Mar-mid May; Okhotsk Sea	3188
	mate: Mar-mid May, some June; NWT, Canada	10145
	mate: Mar-May; e Canadian Arctic	7087
	mate: Apr-June; Europe	3571
	mate: end Apr-early May; n Atlantic	3534
	mate: Aug-mid Sept; June-Aug; Far East	10062
	ovulate: Apr; AK	5353
	birth: mid Feb-mid Mar; Lake Ladoga, Russia	8679
	birth: Mar-Apr; Europe	3571
	birth: end Mar?; NWT, Canada	10145
	birth: end Mar-Apr; arctic	10432
	birth: end Mar-May; Okhotsk Sea	3188
	birth: Apr; N Atlantic	3534
	birth: mid Apr; Okhotsk, Bering Seas	10833
Phoca largha	mate: Apr-May; captive	739
	birth: end Jan-early May, peak mid Mar-mid Apr; Bering, Okhotsk Seas	1089
Phoca sibirica	birth: Feb-Mar; Lake Baikal, Russia	5755
	birth: peak mid Mar; Lake Baikal, Russia	8680
Phoca vitulina	epididymal sperm: Mar-Nov; se Vancouver I, BC, Canada	925
	estrus Aug-Sept; se Vancouver I, BC, Canada	928

mate: Apr-July; NEW, NB, Canada 1146
mate: June; Baltic Sea 4961
mate: June; Hokkaido, Japan 4515
mate: June-July; Okhotsk Sea 10062
mate: end June; Hokkaido 7752
mate: end July-early Aug; E Anglia, England, UK 233, 4416
mate: Aug; zoo 5470
mate: Aug-early Sept; Europe 3571
mate: Aug-Sept; Netherlands 4511
mate: Aug-Sept; se Vancouver I, BC, Canada 925, 928
mate: Sept; e Pacific 5683
mate: Sept; w US 5146
mate: Sept-early Oct; Shetland I, UK 11245
mate: mid Sept-mid Oct; Europe 1084
ovulate: mid June; NEW, NB, Canada 1146
ovulate: end July-Aug, a few May; E Anglia, England, UK 4416
ovulate: Aug-Oct; se Vancouver I, BC, Canada 928
conceive: June; NEW, NB, Canada 1146
conceive: Aug-Sept; se Vancouver I, BC, Canada 928
implant: Sept; Nova Scotia, NB, Canada 3275
implant: mid Sept; NEW, NB, Canada 1146
implant: end Sept; Hokkaido, Japan 4515
implant: end Oct-mid Dec; Europe 1084
implant: Nov-early Dec; E Anglia, England, UK 4416
implant: mid Nov; se Vancouver I, BC, Canada 925, 928
birth: mid-end Mar; Hokkaido, Japan 7752
birth: end Mar-early May; CA 926
birth: Apr-July?; Kurile I, Russia 10195
birth: mid Apr; Okhotsk, Bering Seas 10833
birth: May; Hokkoido, Japan 7753
birth: May; NEW, NB, Canada 1146
birth: May-early June, peak end May; Sable I, NS, Canada 1143
birth: May-early June; Salile I, NS, Canada 1142
birth: May-June; Kurile I, Russia, AK 7753
birth: May-June; NS, NB, Canada 3275
birth: May-June; n hemisphere 5755
birth: May-June; Yaquina estuary, OR 674
birth: May-Aug; e Pacific 5683
birth: mid-end May; Hokkaido, Japan 7752
birth: mid-end May; Pacific 7864
birth: mid May-mid June; southern range 6788
birth: end May-early July; Hokkaido I, Japan 7753
birth: June; E Anglia, England, UK 4415, 4416
birth: June; Baltic Sea 4961
birth: June; n1; captive 9246
birth: June-mid July; Europe 1084
birth: June-July; Netherlands 4511
birth: June-early Sept; BC, Canada 926
birth: mid-end June; UK 233
birth: mid June-early July; Shetland I, UK 11245
birth: mid June-mid July; Europe 3571
birth: mid June-mid July; N Sea, Germany 11829
birth: end June-early July; Bering, Chukchi Seas 1457
birth: end June-early Sept, peak July;
se Vancouver I, BC, Canada 925, 928
birth: July; zoo 5470
birth: mid Aug-mid Sept; Gertrude I, WA 7864
birth: spring; w US 5146
birth: early summer; w Atlantic 1474
lactation: ?-Sept; Kurile I, Russia 10195